THE AREA FUNCTION OF THE EXCURSION SKELETON OF THE MÖBIUS SUM AND HORIZON-LOCALIZED CRITERIA FOR THE RIEMANN HYPOTHESIS

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ABSTRACT. Let $M(x) = \sum_{n \le x} \mu(n)$ be the Möbius sum, also called the Mertens function. Its integer zeros cut the walk $n \mapsto M(n)$ into excursions of constant sign, and this paper studies the Riemann hypothesis through the geometry of these excursions. To every completed excursion of length ℓ and height H we attach a triangular skeleton of height $\sqrt{H}$, and we compute exactly its area, its energy and its quadratic mass; the quadratic mass of an excursion equals minus a lag-weighted sum of the Möbius correlations internal to that excursion.

Four results carry the main weight. (i) A counting lemma for disjoint intervals, combined with the known frequency of large oscillations of M, shows that RH is equivalent to Littlewood's bound $M(t) \ll_\varepsilon t^{1/2+\varepsilon}$ imposed only on the initial part $x_0 < t \le \min\{x_1, \lambda x_0\}$ of every excursion $(x_0, x_1]$, for any fixed $\lambda > e^{\pi/\gamma_1} = 1.2489\ldots$; this constant is the sharp threshold of the interval-counting comparison used here. (ii) With $\Theta = \sup \Re\rho$, the absolute area of the skeleton up to x is $x^{1+\Theta/2+o(1)}$, and the Dirichlet series of $1/\zeta$ grouped by excursions has absolute block-convergence abscissa Θ. (iii) Inside the RH-equivalent window the values of M are explicit linear functions of μ on the prefix up to χu for every $\chi > \frac{1}{2}e^{\pi/\gamma_1}$; we split them into scale channels and into a smooth part and a large-prime part which, in the range of the split considered, have size $(1+o(1))t/\log^2 t$ and opposite signs. (iv) At every zero v of M, $\sum_{n<v} M(n)^2 = -\sum_{h<v} hC_h^{[v]}$, and the corresponding signed weighted contribution lies at lags $h > v^\alpha$ for every $\alpha < \Theta$. We also show that divisor equations reduce the 2^J orientations of J excursions to at most $2N$ candidates, that no sublinear set of divisor rows controls a fixed-shift correlation, and that several natural strategies (termwise contraction, relaxed divisor constraints, fixed multiplicative depth) cannot by themselves reach the RH scale.

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1. Introduction

The first part of this introduction (Sections 1.1–1.8) recalls the objects that the paper uses: the zeta function and the Riemann hypothesis in Riemann's form, the distribution of primes, the Möbius function and its summatory function, and the relations between them. The second part (Sections 1.9–1.11) describes the method and the results.

1.1. **The zeta function and Riemann's hypothesis.** For $\Re s > 1$ the Riemann zeta function is

$$\zeta(s) = \sum_{n\geq 1} \frac{1}{n^s} = \prod_p \Big(1 - \frac{1}{p^s}\Big)^{-1}, \tag{1.1}$$

the product running over the primes. The Euler product (1.1) is the analytic form of unique factorization, and it shows that $\zeta(s) \neq 0$ for $\Re s > 1$. Riemann [45] continued ζ meromorphically to $\mathbb{C}$, with a single simple pole at $s = 1$ of residue 1, and proved the functional equation

$$\xi(s) = \xi(1-s), \qquad \xi(s) := \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\tfrac{s}{2})\zeta(s), \tag{1.2}$$

where ξ is entire. The zeros of ζ at $s = -2, -4, -6, \dots$ come from the poles of $\Gamma(s/2)$ and are called trivial; all other zeros, the nontrivial zeros $\rho = \beta + i\gamma$, are the zeros of ξ and lie in the critical strip $0 < \beta < 1$. Writing $s = \frac{1}{2} + it$, the function $\xi(\frac{1}{2} + it)$ is real for real t, and Riemann stated that it is very likely that all its zeros are real. This is the Riemann hypothesis (RH): every nontrivial zero satisfies $\beta = \frac{1}{2}$. Infinitely many zeros lie on the critical line (Hardy [12]); all zeros with $0 < \gamma \leq 3 \cdot 10^{12}$ lie on it and are simple [42]; the lowest positive ordinate is $\gamma_1 = 14.134725\ldots$. Because the zero set is symmetric under $\rho \mapsto \overline{\rho}$ and $\rho \mapsto 1 - \rho$, the quantity

$$\Theta := \sup\{\Re\rho : \ \zeta(\rho) = 0, \ 0 < \Re\rho < 1\} \tag{1.3}$$

satisfies $\frac{1}{2} \leq \Theta \leq 1$, and RH is the statement $\Theta = \frac{1}{2}$. General accounts of the problem are [53, 5]; a spectral interpretation of the zeros through a trace formula on the adèle class space, which reduces RH to the validity of that trace formula, is due to Connes [4].

1.2. **Primes.** Let $\pi(x)$ be the number of primes up to x, $\mathrm{li}(x) = \int_2^x dt/\log t$, and let $\psi(x) = \sum_{n \leq x} \Lambda(n)$, where $\Lambda(n) = \log p$ if $n = p^k$ and $\Lambda(n) = 0$ otherwise. Taking the logarithmic derivative of (1.1),

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n \geq 1} \frac{\Lambda(n)}{n^s} \qquad (\Re s > 1). \tag{1.4}$$

Riemann's paper contains the explicit formula for $J(x) = \sum_{k \geq 1} \pi(x^{1/k})/k$,

$$J(x) = \mathrm{li}(x) - \sum_{\rho} \mathrm{li}(x^{\rho}) - \log 2 + \int_x^{\infty} \frac{dt}{t(t^2-1)\log t} \qquad (x > 1), \tag{1.5}$$

with the sum over ρ taken symmetrically; the analogous formula

$$\psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \log 2\pi - \tfrac{1}{2}\log(1 - x^{-2}) \qquad (x > 1, \ x \neq p^k) \tag{1.6}$$

shows directly that every zero contributes an oscillation of size x^{β} [53, 35]. The prime number theorem $\pi(x) \sim x/\log x$, proved by Hadamard [11] and de la Vallée Poussin [54] from $\zeta(1+it) \neq 0$, is equivalent to $\psi(x) \sim x$. By (1.6), $\psi(x) = x + O(x^{\Theta}\log^2 x)$, and RH is equivalent to $\pi(x) = \mathrm{li}(x) + O(\sqrt{x}\log x)$ [35]. The best unconditional error term comes from the Vinogradov–Korobov zero-free region [55, 9].

1.3. **The Möbius function and the Möbius sum.** The Möbius function, introduced by Möbius [34] in the inversion formula that bears his name, is $\mu(1) = 1$, $\mu(n) = (-1)^k$ if n is a product of k distinct primes, and $\mu(n) = 0$ if $p^2 \mid n$ for some prime p. It is multiplicative and is characterized by

$$\sum_{d \mid n} \mu(d) = \mathbf{1}_{n=1} \qquad (\mu * 1 = \delta), \tag{1.7}$$

equivalently by the Dirichlet series

$$\frac{1}{\zeta(s)} = \prod_p \Big(1 - \frac{1}{p^s}\Big) = \sum_{n \geq 1} \frac{\mu(n)}{n^s} = s\int_1^{\infty} M(x)x^{-s-1}\,dx \qquad (\Re s > 1), \tag{1.8}$$

where

$$M(x) = \sum_{n \leq x} \mu(n) \tag{1.9}$$

is the *Möbius sum*. It is usually called the Mertens function, after Mertens [32], who tabulated it and conjectured $|M(x)| \le \sqrt{x}$; Stieltjes had claimed $M(x) = O(\sqrt{x})$ in 1885 [17]. Every recursion used below comes from (1.7): for $n \ge 2$, $\mu(n) = -\sum_{d|n,\, d<n} \mu(d)$, and, summing,

$$\sum_{k \le x} M(x/k) = 1 \qquad (x \ge 1) \tag{1.10}$$

[35, §3.2]. Squarefree integers have density $6/\pi^2$: $\sum_{n\le x} \mu(n)^2 = \frac{6}{\pi^2} x + O(\sqrt{x})$. The value $M(n)$ is the determinant of the $n \times n$ Redheffer matrix [43] and, up to sign, the Euler characteristic of a cell complex constructed by Björner [1]. Combinatorial algorithms based on (1.10) evaluate M quickly [6, 16], and Hurst [18] computed $M(x)$ for all $x \le 10^{16}$.

1.4. **Relations between the Möbius sum and the primes.** The Dirichlet series (1.4) and (1.8) give the convolution identities $\Lambda = \mu * \log$ and $\mu \log = -\mu * \Lambda$. Summed, they read

$$\psi(x) = \sum_{d \le x} \mu(d)\, T(x/d), \qquad \sum_{n \le x} \mu(n) \log n = -\sum_{d \le x} \mu(d)\, \psi(x/d), \tag{1.11}$$

with $T(y) = \sum_{m \le y} \log m = y \log y - y + O(\log y)$, and by partial summation

$$M(x) \log x = \sum_{n \le x} \mu(n) \log n + \int_1^x \frac{M(t)}{t}\, dt. \tag{1.12}$$

Thus information on ψ passes to M and back, with losses. At the qualitative level, the prime number theorem is equivalent to $M(x) = o(x)$ [24] and to $\sum_n \mu(n)/n = 0$ [27], and the Vinogradov–Korobov region gives $M(x) \ll x \exp\bigl(-c(\log x)^{3/5}(\log\log x)^{-1/5}\bigr)$ for some absolute $c > 0$ [55, 9]. We also use $\sum_n \mu(n) \log n/n = -1$, which is the value at $s = 1$ of $-\frac{d}{ds}\zeta(s)^{-1} = \sum_n \mu(n)(\log n) n^{-s}$, the convergence at $s = 1$ being a consequence of the prime number theorem with error term.

At the quantitative level, both $\psi(x) - x$ and $M(x)$ are governed by Θ. For M the analogue of (1.6) is, if RH holds and all zeros are simple, for non-integer $x > 0$ [53, §14.27],

$$M(x) = \lim_{k\to\infty} \sum_{|\gamma| < T_k} \frac{x^\rho}{\rho \zeta'(\rho)} - 2 + \sum_{n \ge 1} \frac{(-1)^{n-1} (2\pi/x)^{2n}}{(2n)!\, n\, \zeta(2n+1)}, \tag{1.13}$$

along a sequence $T_k \to \infty$ on whose horizontal segments $1/\zeta(\varsigma + iT_k) \ll T_k^{\varepsilon}$; such a sequence exists under RH [53, §14.16]. The last series collects the residues at the trivial zeros, and the ρ-sum is not known to converge absolutely. In place of $\log x$ weights, the coefficients $1/\zeta'(\rho)$ now appear, which is why M is more sensitive than ψ to the zeros individually. For every $\varepsilon > 0$,

$$M(x) \ll_\varepsilon x^{\Theta+\varepsilon} \qquad \text{and} \qquad |M(x)| \ge x^{\Theta-\varepsilon} \text{ for arbitrarily large } x; \tag{1.14}$$

the upper bound follows from Perron's formula and the bound $1/\zeta(s) \ll |t|^{\varepsilon}$ to the right of $\Re s = \Theta$ (Littlewood [25], [53, §14.25], [41, §4]); the lower bound holds because otherwise $\int_1^\infty M(x) x^{-s-1} dx$ would continue $1/(s\zeta(s))$ analytically beyond $\Re s = \Theta$. This gives Littlewood's criterion: RH holds if and only if $M(x) \ll_\varepsilon x^{1/2+\varepsilon}$ for every $\varepsilon > 0$ [25]. If moreover $|M(x)| \le C\sqrt{x}$ for all x, then $|1/(s\zeta(s))| \le C/(\varsigma - \frac12)$ for $\varsigma > \frac12$, so every zero on the critical line is simple [53, §14.29]. Under RH the best known bound is $M(x) \ll \sqrt{x} \exp\bigl((\log x)^{1/2}(\log\log x)^{14}\bigr)$ (Soundararajan [47], improving Maier–Montgomery [26]); under RH and a hypothesis on $\sum_{\gamma \le T} |\zeta'(\rho)|^{-2}$, Ng [36] obtained a limiting distribution for $e^{-y/2} M(e^y)$.

1.5. **Oscillation.** The Mertens conjecture is false (Odlyzko–te Riele [38]; effective form by Pintz [40], see also [23]); currently $\limsup M(x)/\sqrt{x} > 1.826$ and $\liminf M(x)/\sqrt{x} < -1.837$ [18]. Hence M changes sign, and vanishes, at arbitrarily large integers. Ingham [20] showed that $\limsup M(x)/\sqrt{x} = +\infty$ follows from RH and the linear independence of the ordinates. Kaczorowski and Pintz [22] proved that M has at least $(\gamma_1/\pi - o(1)) \log Y$ sign changes in $[1, Y]$ with

oscillations of size $x^{\Theta-\varepsilon}$, and that every range $[Y^{1-\varepsilon}, Y]$ contains large values of both signs (Theorem 9.5). Pintz proved that the mean $\frac{1}{Y}\int_1^Y |M|$ tends to infinity [39], and recently that this mean and the late maximal order of $|M|$ are both $Y^{\Theta+o(1)}$ [41] (Theorem 8.10). These two inputs, the logarithmic frequency of large oscillations and their exact exponent, are the only deep information on M used in the paper.

1.6. **Liouville's function.** In this subsection only, $\lambda(n)$ and $L(x)$ denote Liouville's function and its summatory function; everywhere else λ is a horizon factor and L a block length. Liouville's function $\lambda(n) = (-1)^{\Omega(n)}$ has $\sum \lambda(n)n^{-s} = \zeta(2s)/\zeta(s)$; hence $\lambda = \mathbf{1}_\square * \mu$ and $\mu(n) = \sum_{d^2|n} \mu(d)\lambda(n/d^2)$, so with $L(x) = \sum_{n\le x} \lambda(n)$,

$$L(x) = \sum_{d\le\sqrt{x}} M(x/d^2), \qquad M(x) = \sum_{d\le\sqrt{x}} \mu(d)L(x/d^2), \tag{1.15}$$

and RH is equivalent to $L(x) \ll_\varepsilon x^{1/2+\varepsilon}$. Pólya's conjecture $L(x) \le 0$ is false [14]. Most correlation results below are proved for λ and μ by the same arguments; we state them for μ.

1.7. **Short intervals and Chowla's conjecture.** Chowla's conjecture [3] asserts, for distinct shifts $h_1, \dots, h_k$, $\sum_{n\le x} \mu(n+h_1)\cdots\mu(n+h_k) = o(x)$ (Cesàro form); the logarithmic form replaces $\sum_{n\le x}$ by $\sum_{n\le x} \frac{1}{n}$ and $o(x)$ by $o(\log x)$, and is implied by the Cesàro form. Sarnak [46] observed that Chowla implies Möbius disjointness from zero-entropy dynamics, and Tao proved the equivalence of the logarithmic Chowla and logarithmic Sarnak conjectures [50]. The known cases are the logarithmic two-point case (Tao [49]), the logarithmic odd-order cases (Tao–Teräväinen [51, 52]), and Cesàro statements averaged over the shifts: Matomäki–Radziwiłł–Tao [29] prove $\sum_{h\le K} |\sum_{n\le x} \mu(n)\mu(n+h)| = o(Kx)$ for every $K = K(x) \to \infty$. The two-point Cesàro case for a fixed shift remains open. These results rest on the Matomäki–Radziwiłł theorem [28] on multiplicative functions in almost all short intervals. For *all* short intervals, Matomäki–Teräväinen [31] prove $\sum_{x<n\le x+x^\theta} \mu(n) \ll x^\theta(\log x)^{-1/3+\varepsilon}$ for $\theta > 0.55$ and Fourier uniformity for $H \ge x^{3/5+\varepsilon}$, extended by Matomäki–Shao–Tao–Teräväinen [30]; primes in all short intervals are due to Huxley [19] and, with the Guth–Maynard large-value estimate, reach length $x^{17/30+o(1)}$ [10]. The distinction between all intervals and almost all intervals is essential here, because the endpoints of excursions are selected by M itself.

1.8. **The walk viewpoint.** Denjoy [8] observed that, if the signs $\mu(n)$ on the squarefree integers behaved like independent coin tosses, $M(x)$ would be of size $x^{1/2+o(1)}$, which by Littlewood's criterion is exactly RH; this is a heuristic, not an equivalence. Wintner [56] introduced random multiplicative functions, whose partial sums have better than square-root cancellation (Harper [13]). Areas and quadratic functionals of lattice excursions are classical [48, 37, 44, 21, 7, 2]; here they are applied to the deterministic walk $n \mapsto M(n)$, whose steps are $\mu(n) \in \{-1, 0, 1\}$.

1.9. **The method.** The integer zeros of M cut $[0,\infty)$ into zero runs and excursions $I_\nu = (x_{0,\nu}, x_{1,\nu}]$ on which M has constant sign σ_ν and does not vanish; we write ℓ_ν for the length and H_ν for the height of I_ν. Every analytic statement about M is then read through three kinds of excursion data.

Geometry. Each completed excursion carries a triangular skeleton of height $\sqrt{H_\nu}$ whose signed area is $\sigma_\nu\sqrt{H_\nu}\ell_\nu/2$. Its quadratic mass $Q_\nu = \sum_{n\in I_\nu} M(n)^2$ satisfies the exact bridge identity $Q_\nu = -\sum_h hC_h^{(\nu)}$, where $C_h^{(\nu)}$ are the Möbius correlations inside I_ν, and H_ν, ℓ_ν bound Q_ν and the linear mass sharply (Sections 3–5).

Counting. The excursions are disjoint intervals. Any family of disjoint intervals whose members satisfy $x_1 > \lambda x_0$ has at most $\log X/\log\lambda + O(1)$ members starting below X, while the Kaczorowski–Pintz theorem produces at least $(\gamma_1/\pi - o(1))\log X$ excursions that carry values of size $x^{\Theta-\eta}$. Comparing the two counts localizes the large values of M (Section 9).

Arithmetic inside an excursion. Starting from a zero u of M, the values of M on $(u, 2u)$ are exact linear functions of μ on $[1, u]$, and on $(u, u^2]$ exact bilinear functions; these identities, the recursion (1.10) and the factorization $n = ap$ with a large prime p describe how an excursion is built (Section 11).

1.10. **Results.** *Horizon localization* (Theorem 9.12, Corollaries 9.14 and 9.15). If a horizon function $\mathfrak{h}$ is crossed by at most $(\kappa + o(1)) \log X$ excursions starting below X, with $\kappa < \gamma_1/\pi$, then RH is equivalent to Littlewood's bound, with a constant depending only on ε, on the part of each excursion lying below its horizon. In particular, for every fixed $\lambda > \lambda_*$,

$$\text{RH} \iff |M(t)| \le C_\varepsilon t^{1/2+\varepsilon} \quad \text{for} \quad x_0 < t \le \min\{x_1, \lambda x_0\} \text{ on every excursion}, \tag{1.16}$$

where $\lambda_* = e^{\pi/\gamma_1} = 1.2489\ldots$. The threshold λ_* is where the two counts above meet, so it is the sharp threshold of the interval-counting comparison used here (Proposition 9.33). For M itself the excursions return quickly: $x_1 \le x_0^{1+o(1)}$ (Corollary 9.8).

Exact exponents (Corollaries 8.11 and 8.13, Theorem 8.16).

$$\sum_{n\le X} |M(n)| = X^{1+\Theta+o(1)}, \qquad \sum_{n\le X} M(n)^2 = X^{1+2\Theta+o(1)}, \tag{1.17}$$

$$\mathcal{E}(X) = X^{1+\Theta+o(1)}, \qquad \int_0^X |M_{\mathrm{s}}| = X^{1+\Theta/2+o(1)}, \tag{1.18}$$

where $\mathcal{E}$ is the chord energy of the skeleton and M_{s} the skeleton; and the series $\sum_I D_I(s)$ of $1/\zeta$ grouped by completed excursions has abscissa of absolute block convergence Θ. In particular the skeleton area satisfies $\int_0^X |M_{\mathrm{s}}| \ll X^{5/4+\varepsilon}$ for all $\varepsilon > 0$ if and only if RH holds.

The localized window (Section 11). For a zero u of M and $u < t \le \lambda u < 2u$,

$$M(t) = -\sum_{d\le \lambda u/2} \mu(d)\big(\lfloor t/d\rfloor - \lfloor u/d\rfloor\big), \tag{1.19}$$

so the RH-equivalent window depends only on μ up to χu, for every $\chi > \chi_* = \lambda_*/2$. We decompose $M(t)$ into scale channels $M(t/k) - M(u/k)$, prove a channel-descent inequality with an explicit loss, and split M into a smooth and a large-prime part which, in the range of the split considered, both have size $(1 + o(1))t/\log^2 t$, with opposite signs.

Correlations (Section 12). At every zero v of M, $\sum_{n<v} M(n)^2 = -\sum_{h<v} hC_h^{[v]}$. Averaged Chowla makes the lags $h \le \sqrt{v}$ negligible at the scale v^2, and Pintz's exponent makes the lags $h \le v^\alpha$, $\alpha < \Theta$, carry a vanishing proportion of $\sum_{n<v} M(n)^2$.

Orientation and selection (Section 10). Reflecting whole excursions leaves M^2 unchanged, so the profile M^2 determines M only up to 2^J orientations; divisor equations reduce these to at most $2N$, while no set of $o(N)$ divisor rows can control a fixed-shift correlation.

Stochastic models (Section 13). Under a uniform tail hypothesis on a normalized excursion statistic, each level of lacunarity of the counted family removes one logarithm from the extreme-value scale.

1.11. **Organization.** Sections 3–5 set up the excursion geometry. Sections 6–8 relate it to ζ and give the area bounds, the global criteria, the exact exponents and the block series of $1/\zeta$. Section 9 contains the counting lemmas, the Kaczorowski–Pintz input, the horizon localization, the Catalan reference laws and the wide peaks. Section 10 treats the reflection fibre, Section 11 the localized window and the large-prime decomposition, Section 12 the correlations, and Section 13 stochastic models. A one-frequency model, numerical data and the conclusions follow.

Conventions. $\rho = \beta + i\gamma$ runs over nontrivial zeros; $\Theta \in [\frac{1}{2}, 1]$ is defined in (1.3); $\gamma_1 = 14.134725\ldots$; RH means $\Theta = 1/2$. $M(x) = M(\lfloor x\rfloor)$ for real x and $M(x) = 0$ for $x < 1$. Statements taken from the literature carry the reference in their heading; heuristics are marked as such.

2. Notation and variables

Global symbols keep the meanings below. Symbols marked "local" are declared again where used and never leave that statement or subsection. Constants c, C are positive and may change from line to line.

Symbol	Meaning	Scope
$\rho = \beta + i\gamma$	Nontrivial zero of ζ; β is used only here	Global
Θ	$\sup\{\Re\rho\}$	Global
γ_1	Lowest positive ordinate, $14.134725\ldots$	Global
$\lambda_*,\ \chi_*$	e^{π/γ_1}, $\lambda_*/2$	Global
λ	Horizon factor, $\lambda > 1$ (Liouville's $\lambda(n)$ only in Section 1.6)	Global
ψ	Function with $\psi(x) \to \infty$ in the superlinear horizon $x\psi(x)$	Section 9
$M,\ M_0$	Mertens function; $M_0 = M + 2$ (normalization of [22])	Global
$I_\nu = (x_{0,\nu}, x_{1,\nu}]$	Completed excursion; also $(u_\nu, v_\nu]$	Global
σ_ν	Orientation ± 1 of I_ν	Global
$\ell_\nu,\ H_\nu,\ \tau_\nu$	Length, height, first apex	Global
$a_\nu,\ b_\nu$	$\tau_\nu - x_{0,\nu}$, $x_{1,\nu} - \tau_\nu$	Global
$\mathcal{Z} = [p, q]$	Zero run	Section 4
$u(x),\ L_x$	Last zero $\le x$; terminal length $\lfloor x\rfloor - u(x)$	Global
$M_{\mathrm{s}},\ M_{\mathrm{sk}}$	Regularized / unregularized skeleton	Global
$\mathcal{A}_+,\ \mathcal{A}_-,\ \mathcal{A},\ \mathcal{A}_{\mathrm{abs}}$	Positive, signed negative, net, absolute area	Global
$\widetilde{A}$	$\int_0^x M$	Section 6
ch_ν	Chord of $\mathcal{A}$ over I_ν	Theorem 8.9
$\mathfrak{q}_\nu = q_\nu$	Number of active steps $\#\{n \in I_\nu : \mu(n) \neq 0\}$; q in a general block	Global
$Q_\nu,\ \mathfrak{b}_\nu$	$\sum_{n\in I_\nu} M(n)^2$, $\sum_{n\in I_\nu} \lvert M(n)\rvert$	Global
$Q_{\mathrm{term}},\ \mathfrak{b}_{\mathrm{term}}$	Same for the terminal open prefix	Global
$B_1,\ B_2,\ \mathcal{E}$	$\sum_{n\le X} \lvert M\rvert$, $\sum_{n\le X} M^2$, chord energy	Global
$C_h^{(\nu)},\ C_h^{[v]},\ C_h^{(u,X]}$	Lag-h correlation in I_ν, in $(0, v]$, in $(u, X]$	Global
$C_g^{\langle h\rangle}$	Lag-g correlation in a truncated prefix of length h	Local
S_t	Oriented partial sum inside a declared block	Local
$\mathcal{F},\ E_{\mathcal{F}}(X)$	Family of excursions and its counting function	Section 9
$\mathfrak{h},\ \mathcal{F}_{\mathfrak{h}}$	Horizon and crossing family	Section 9
A	Power exponent in $x_1 > x_0^A$	Section 9
η	Loss in the Kaczorowski–Pintz exponent $\Theta - \eta$	Global
κ	Coefficient of a crossing count $(\kappa + o(1))\log X$	Theorem 9.12
$\mathcal{S},\ L^{\#},\ r_I$	Squarefree clock, compressed length, support time	Catalan subsection
L	Length of a declared block (also support length $L^{\#}$ in Catalan lemmas)	Local
$p_{\mathrm{Cat}},\ p_{\mathrm{Cat}}^{\max},\ p_{\mathrm{open}}$	Reference-law tail weights	Catalan subsection
$J_N,\ E_\nu$	Number of completed excursions up to a zero cutoff N; active support of I_ν	Arithmetic selection
$\mathbf{z},\ \boldsymbol{\sigma}$	0/1 reflection vector; ± 1 orientation vector	Arithmetic selection
$\mathfrak{D}_N,\ G_N$	Divisor-defect functional and its orientation Gram matrix	Arithmetic selection
$m_\nu,\ \phi_\nu,\ \mathscr{C}^{\mathrm{div}}$	Support centre, positive oriented increments, central divisor matrix	Arithmetic selection
$\mathbf{D}_N,\ \mathsf{A}_h,\ \mathfrak{S}_h$	Divisor matrix; fixed-shift form; pair density	Arithmetic selection
$\theta_{\mathrm{w}},\ \theta'$	Short-interval exponents > 0.55	Wide peaks
$\Lambda,\ h_u,\ \omega_u$	Horizon factor, block length, Lipschitz modulus	Wide peaks

Symbol	Meaning	Scope
G	$(H-h_u)_+$	Proposition 9.43
μ_u	$\mu\mathbf{1}_{[1,u]}$	Section 11
$\Delta_k,\ \mathcal{R}_K$	Channel increment, deep remainder	Section 11
ϑ	Normalizing exponent in $\lvert M(y)\rvert/y^{\vartheta}$	Section 11
$c_y(t,u),\ K_y$	Kernel coefficients (argument y of $M(y)$)	Section 11
$\Upsilon,\ \mathcal{C}$	Multiplicity-weighted and coherent coefficients	Section 11
$\mathbf{a}=(a_d),\ D_{\mathbf{a}}$	Relaxed prefix vector and its divisor sum	Section 11
P	Prefix cutoff in layer statements	Section 11
$P^+(n)$	Largest prime factor	Section 11
$U,\ W,\ M_U^{\mathrm{sm}},\ M_U^{\mathrm{lp}},\ \widetilde{m}$	Large-prime level, t/U^2, smooth and large-prime parts, $\sum_{a\le y}\mu(a)/a$	Section 11
$\Omega(n),\ \omega(n)$	Prime factors with / without multiplicity	Global
$D_I(s)$	Dirichlet block of a completed excursion	Section 8
$s=\varsigma+i\tau$	Complex variable	Global
$e(x),\ \xi$	$e^{2\pi ix}$; frequency	Section 12
K	Lag cutoff in averaged-Chowla statements	Section 12
$R,\ F_R,\ V_R$	Fejér scale and sums	Section 12
ι	Exponent gap in $L\ge u^{3/5+\iota}$	Section 12
$(\mathfrak{X},\mathcal{G},\Pr),\ Z_k$	Probability space and excursion statistic	Section 13
$\mathcal{N}(X)$	Deterministic count bound	Section 13
$\kappa_0,\ \varpi$	Tail constant and exponent	Section 13

3. Excursions and the regularized skeleton

Let μ be the Möbius function and

$$M(x)=\sum_{n\le x}\mu(n),\qquad x\ge 0, \tag{3.1}$$

piecewise constant and right-continuous, with jumps $\mu(n)\in\{-1,0,1\}$. Throughout the paper, for real $x\ge 0$ we use the convention

$$M(x)=M(\lfloor x\rfloor), \tag{3.2}$$

and we set $M(x)=0$ for $x<1$. Thus expressions such as $M(t/k)$ always mean $M(\lfloor t/k\rfloor)$. Unless stated otherwise, every summation index is an integer; a real upper limit in a sum is interpreted through its floor. The segmentation of M by zero runs and the record of its extrema are standard computational objects; Hurst [18] computed $M(x)$ for all $x\le 10^{16}$. The area and quadratic functionals of lattice excursions are classical [48, 37, 44, 21, 7, 2]; here they are applied to the deterministic Möbius walk.

Definition 3.1 (Excursions and zero runs)**.** An *excursion* is an interval

$$I_\nu=(x_{0,\nu},x_{1,\nu}] \tag{3.3}$$

with integer endpoints, $M(x_{0,\nu})=M(x_{1,\nu})=0$, $M(x_{0,\nu}+1)=\sigma_\nu\in\{\pm1\}$, and $\sigma_\nu M(n)\ge 1$ for every integer $x_{0,\nu}<n<x_{1,\nu}$. Thus $x_{1,\nu}$ is the first return to zero after the departure from $x_{0,\nu}$. Consecutive excursions may be separated by a *zero run* $[x_{1,\nu},x_{0,\nu+1}]$, a maximal set of consecutive integers on which $M=0$; it may consist of a single point. The first excursion is $(0,2]$. Since M assumes positive and negative values at arbitrarily large arguments ($M(x)=\Omega_\pm(\sqrt{x})$, Remark 7.6) and its integer jumps belong to $\{-1,0,1\}$, between two integers where M has opposite signs it must vanish at an integer. Hence M has arbitrarily large zeros and every excursion is finite. We also write $u_\nu=x_{0,\nu}$ and $v_\nu=x_{1,\nu}$.

Put

$$\ell_\nu=x_{1,\nu}-x_{0,\nu},\qquad H_\nu=\max_{x_{0,\nu}<n<x_{1,\nu}}\lvert M(n)\rvert, \tag{3.4}$$

$$\tau_\nu = \min\{n \in I_\nu : |M(n)| = H_\nu\}, \qquad a_\nu = \tau_\nu - x_{0,\nu}, \qquad b_\nu = x_{1,\nu} - \tau_\nu. \tag{3.5}$$

Every excursion has $H_\nu \geq 1$ and $\ell_\nu \geq 2$.

Definition 3.2 (Completed excursions and open prefixes)**.** An excursion in the sense of Definition 3.1 is always *completed*: both $M(x_{0,\nu}) = 0$ and $M(x_{1,\nu}) = 0$ are part of the definition. A segment $(u, x]$ with $M(u) = 0$ and $M(n) \neq 0$ for $u < n \leq x$, with no return condition at x, is an *open prefix*; the terminal block of Definition 3.6 is one. Any estimate whose proof uses the backward bound $|M(u+r)| \leq x_{1,\nu} - u - r$ holds for completed excursions only and is never applied to an open prefix.

Lemma 3.3 (Height and step count)**.** *For every excursion,*

$$H_\nu \leq \min(a_\nu, b_\nu) \leq \frac{\ell_\nu}{2}. \tag{3.6}$$

Proof. The steps of M are 0 or ± 1, so at least H_ν steps are needed to reach height H_ν from zero and at least H_ν more to return to zero. □

Definition 3.4 (Crossing and touching zero runs)**.** The zero run $[x_{1,\nu}, x_{0,\nu+1}]$ between consecutive excursions is a *crossing run* if $\sigma_{\nu+1} = -\sigma_\nu$ and a *touching run* if $\sigma_{\nu+1} = \sigma_\nu$.

Definition 3.5 (Regularized skeleton)**.** The *skeleton* M_{s} is the continuous, piecewise linear function defined on every excursion by

$$M_{\mathrm{s}}(t) = \begin{cases} \sigma_\nu \sqrt{H_\nu} \dfrac{t - x_{0,\nu}}{a_\nu}, & x_{0,\nu} \leq t \leq \tau_\nu, \\ \sigma_\nu \sqrt{H_\nu} \dfrac{x_{1,\nu} - t}{b_\nu}, & \tau_\nu \leq t \leq x_{1,\nu}, \end{cases} \tag{3.7}$$

and by $M_{\mathrm{s}} \equiv 0$ on zero runs. Here "regularized" refers only to the normalization of the apex from H_ν to $H_\nu^{1/2}$; no analytic smoothing operation is intended. The *unregularized skeleton* M_{sk} is defined in the same way with peak $\sigma_\nu H_\nu$.

Definition 3.6 (Terminal block of a truncated computation)**.** Let $N \geq 2$ and $u = \max\{n \leq N : M(n) = 0\}$. If $u = N$ there is no terminal block. Otherwise $M(n) \neq 0$ for $u < n \leq N$, and since M has steps $0, \pm 1$ all these values have the sign of $M(N)$. Put

$$\sigma_{\mathrm{term}} = \operatorname{sgn} M(N), \qquad H_{\mathrm{term}} = \max_{u<n\leq N} |M(n)|, \qquad \tau_{\mathrm{term}} = \min\{n \in (u, N] : |M(n)| = H_{\mathrm{term}}\}. \tag{3.8}$$

On $[u, N]$ the regularized skeleton is the segment from $(u, 0)$ to $(\tau_{\mathrm{term}}, \sigma_{\mathrm{term}}\sqrt{H_{\mathrm{term}}})$ followed, if $\tau_{\mathrm{term}} < N$, by the segment ending at $(N, \sigma_{\mathrm{term}}\sqrt{|M(N)|})$. Its signed area is

$$\Delta\mathcal{A}_{\mathrm{term}}(u, N] = \frac{\sigma_{\mathrm{term}}}{2}\Big[\sqrt{H_{\mathrm{term}}}(N - u) + \sqrt{|M(N)|}(N - \tau_{\mathrm{term}})\Big]. \tag{3.9}$$

The terminal skeleton is provisional: after the next return to zero the completed excursion acquires a different descending side. Independently of it, for real x with last zero $u(x) \leq \lfloor x \rfloor$ define

$$\mathfrak{b}_{\mathrm{term}}(x) := \sum_{u(x)<n\leq\lfloor x\rfloor} |M(n)|, \qquad Q_{\mathrm{term}}(x) := \sum_{u(x)<n\leq\lfloor x\rfloor} M(n)^2, \qquad L_x := \lfloor x \rfloor - u(x). \tag{3.10}$$

These depend only on M.

Definition 3.7 (The three curves)**.** With $M_{\mathrm{s}}^{\pm} = \max(\pm M_{\mathrm{s}}, 0)$,

$$\mathcal{A}_+(x) = \int_0^x M_{\mathrm{s}}^+(t)\, dt, \qquad \mathcal{A}_-(x) = -\int_0^x M_{\mathrm{s}}^-(t)\, dt, \qquad \mathcal{A}(x) = \int_0^x M_{\mathrm{s}}(t)\, dt. \tag{3.11}$$

$\mathcal{A}_+$ is nondecreasing, $\mathcal{A}_-$ is nonincreasing, and both are constant on zero runs and on excursions of the opposite sign. The curve $\mathcal{A}_-$ records negative area with its sign; it is not the absolute value of the negative area.

Remark 3.8 (The family of g-skeletons). For any $g : \mathbb{N}_0 \to [0,\infty)$ with $g(0) = 0$ and $g(h) > 0$ for $h \geq 1$, one may replace the apex height $\sqrt{H}$ by $g(H)$. The signed area increment becomes $\frac{1}{2}\sigma_\nu g(H_\nu)\ell_\nu$. The correlation identities of Section 5 concern the Möbius increments and do not depend on g. The choice $g(H) = \sqrt{H}$ is a normalization.

4. Exact formulas

Lemma 4.1 (Increment per excursion)**.** *For every completed excursion,*

$$\mathcal{A}(x_{1,\nu}) - \mathcal{A}(x_{0,\nu}) = \sigma_\nu \frac{\sqrt{H_\nu}\,\ell_\nu}{2}, \qquad \sum_{n\in I_\nu} M_{\mathrm{s}}(n) = \sigma_\nu \sqrt{H_\nu}\frac{\ell_\nu}{2}. \tag{4.1}$$

Proof. The two pieces are triangles of height $\sqrt{H_\nu}$ and bases a_ν, b_ν. For the discrete sum, the ascending values at $j = 1,\dots,a_\nu$ sum to $\sqrt{H_\nu}\sum_{j\leq a_\nu} j/a_\nu = \sqrt{H_\nu}(a_\nu+1)/2$; the descending values at $\tau_\nu + r$, $r = 1,\dots,b_\nu$, sum to $\sqrt{H_\nu}\sum_{r\leq b_\nu}(b_\nu - r)/b_\nu = \sqrt{H_\nu}(b_\nu - 1)/2$. □

Proposition 4.2 (Explicit form of the three curves)**.** *Let $x \in I_\nu$. Then*

$$\mathcal{A}(x) = \begin{cases} \mathcal{A}(x_{0,\nu}) + \sigma_\nu\sqrt{H_\nu}\dfrac{(x - x_{0,\nu})^2}{2a_\nu}, & x \leq \tau_\nu, \\[2ex] \mathcal{A}(x_{0,\nu}) + \sigma_\nu\sqrt{H_\nu}\left(\dfrac{a_\nu}{2} + (x-\tau_\nu) - \dfrac{(x-\tau_\nu)^2}{2b_\nu}\right), & x \geq \tau_\nu. \end{cases} \tag{4.2}$$

At excursion endpoints,

$$\mathcal{A}(x_{1,k}) = \sum_{\nu\leq k}\sigma_\nu\frac{\sqrt{H_\nu}\ell_\nu}{2}, \qquad \mathcal{A}_+(x_{1,k}) = \sum_{\substack{\nu\leq k\\ \sigma_\nu=+1}}\frac{\sqrt{H_\nu}\ell_\nu}{2}, \qquad \mathcal{A}_-(x_{1,k}) = -\sum_{\substack{\nu\leq k\\ \sigma_\nu=-1}}\frac{\sqrt{H_\nu}\ell_\nu}{2}. \tag{4.3}$$

The curves are piecewise quadratic and C^1.

Proof. Integrate (3.7); the endpoint formulas follow from Lemma 4.1. □

A point y is a *strict inflection point* of a function if there exists $\delta > 0$ such that the function is strictly convex on one side of y and strictly concave on the other, within distance δ. The second derivative need not exist at y.

Lemma 4.3 (Critical geometry of $\mathcal{A}$)**.** *Let $\mathcal{Z} = [p,q]$ be the zero run between an excursion of sign σ_- ending at p and the next excursion of sign σ_+ starting at q.*

(i) *Every apex τ_ν is a strict inflection point.*
(ii) *If $\sigma_+ = -\sigma_-$, then $\mathcal{Z}$ is an extremal plateau; it is a strict local extremum exactly when $p = q$.*
(iii) *If $\sigma_+ = \sigma_-$, then $\mathcal{A}$ has no local extremum on $\mathcal{Z}$. Its concavity has sign $-\sigma_-$ immediately to the left and $+\sigma_+$ immediately to the right. If $p = q$ the point is a strict inflection; if $p < q$, no point of $\mathcal{Z}$ is a strict inflection, and the change of concavity is carried by the whole plateau.*
(iv) *Strict local extrema occur exactly at isolated crossing zeros, and plateaus of extrema exactly at crossing runs of positive length; every point of such a plateau is a non-strict local extremum.*

Proof. On the descending side of the left excursion, $\operatorname{sgn}\mathcal{A}' = \sigma_-$ and $\operatorname{sgn}\mathcal{A}'' = -\sigma_-$. On the ascending side of the right excursion, $\operatorname{sgn}\mathcal{A}' = \sigma_+$ and $\operatorname{sgn}\mathcal{A}'' = \sigma_+$. Inside an excursion $\mathcal{A}' = M_{\mathrm{s}} \neq 0$, so no extremum occurs there. On $\mathcal{Z}$, $\mathcal{A}' = 0$. The assertions follow, and at every apex the sign of $\mathcal{A}''$ reverses. □

5. Internal correlations of an excursion

For an integer $L \geq 1$ and a real block $x_1, \ldots, x_L$ (local notation: here x_j are the entries of the block, not excursion endpoints) put

$$S_t = \sum_{j=1}^{t} x_j, \qquad C_h = \sum_{j=1}^{L-h} x_j x_{j+h}. \tag{5.1}$$

Lemma 5.1 (Triangular identity)**.** *For every real finite sequence,*

$$\sum_{t=1}^{L-1} S_t^2 + \sum_{h=1}^{L-1} hC_h = S_L \sum_{t=1}^{L-1} S_t. \tag{5.2}$$

Proof. Expanding the partial sums,

$$\sum_{t=1}^{L} S_t^2 = \sum_{i,j=1}^{L} x_i x_j (L + 1 - \max(i,j)). \tag{5.3}$$

Write $\max(i,j) = \frac{i+j+|i-j|}{2}$. Since $\frac{1}{2}\sum_{i,j}(i+j)x_i x_j = S_L \sum_i i x_i$ and each unordered pair at lag $h = |i-j| \geq 1$ occurs twice,

$$\sum_{t=1}^{L} S_t^2 = (L+1)S_L^2 - S_L \sum_{i=1}^{L} i x_i - \frac{1}{2}\sum_{i,j=1}^{L} x_i x_j |i-j| = (L+1)S_L^2 - S_L \sum_{i=1}^{L} i x_i - \sum_{h=1}^{L-1} hC_h. \tag{5.4}$$

Moreover $\sum_i i x_i = LS_L - \sum_{t=1}^{L-1} S_t$. Hence $\sum_{t=1}^{L} S_t^2 = S_L^2 + S_L \sum_{t<L} S_t - \sum_h hC_h$; subtract S_L^2. □

For a bridge, $S_L = 0$, so the preceding identity gives

$$\sum_{t=1}^{L-1} S_t^2 = -\sum_{h=1}^{L-1} hC_h. \tag{5.5}$$

Remark 5.2 (Hermitian form). For complex x_j and $C_h = \sum_{j=1}^{L-h} x_j \overline{x_{j+h}}$, the same computation applies. The ordered pairs (i,j) and (j,i) at lag h contribute $x_i\overline{x_j} + x_j\overline{x_i}$, so each lag contributes $2\Re C_h$ to $\sum_{i,j} x_i\overline{x_j}|i-j|$, and the cross term becomes $\Re(\overline{S_L}\sum_i i x_i)$. Hence

$$\sum_{t=1}^{L-1} |S_t|^2 + \sum_{h=1}^{L-1} h\Re C_h = \Re\left(\overline{S_L}\sum_{t=1}^{L-1} S_t\right). \tag{5.6}$$

For a completed excursion orient the increments by

$$x_j = \sigma_\nu \mu(x_{0,\nu} + j), \qquad S_t = |M(x_{0,\nu} + t)|. \tag{5.7}$$

Then $S_t > 0$ for $1 \leq t < \ell_\nu$ and $S_{\ell_\nu} = 0$. Put

$$Q_\nu = \sum_{t=1}^{\ell_\nu - 1} S_t^2, \qquad C_h^{(\nu)} = \sum_{j=1}^{\ell_\nu - h} x_j x_{j+h}, \qquad \text{so that} \qquad Q_\nu = -\sum_{h=1}^{\ell_\nu - 1} hC_h^{(\nu)}. \tag{5.8}$$

Lemma 5.3 (Zeroth moment and extreme lag)**.** *With $q_\nu = \sum_j x_j^2$,*

$$\sum_{h=1}^{\ell_\nu - 1} C_h^{(\nu)} = -\frac{q_\nu}{2}, \qquad C_{\ell_\nu - 1}^{(\nu)} = -1, \tag{5.9}$$

and q_ν is even and at least 2.

Proof. $0 = (\sum_j x_j)^2 = q_\nu + 2\sum_h C_h^{(\nu)}$. The first oriented step is $+1$; immediately before the first return the path is at height 1, so the last step is -1. The bridge condition forces equally many positive and negative active steps. □

Proposition 5.4 (Quadratic decomposition by excursions)**.** *For real $x \geq 1$ let $u(x)$ and L_x be as in Definition 3.6, and for $1 \leq j \leq L_x$ put $y_j = \mu(u(x)+j)$ and $C_h^{\mathrm{term}} = \sum_{j=1}^{L_x-h} y_j y_{j+h}$. Then*

$$\sum_{n\leq x} M(n)^2 = \sum_{\nu: x_{1,\nu}\leq u(x)} Q_\nu + Q_{\mathrm{term}}(x), \tag{5.10}$$

$$Q_{\mathrm{term}}(x) = M(\lfloor x\rfloor)^2 + M(\lfloor x\rfloor)\sum_{t=1}^{L_x-1} M(u(x)+t) - \sum_{h=1}^{L_x-1} hC_h^{\mathrm{term}}, \tag{5.11}$$

with $Q_{\mathrm{term}}(x) = 0$ if $L_x = 0$. At a zero x of M,

$$\sum_{n\leq x} M(n)^2 = \sum_{x_{1,\nu}\leq x} Q_\nu = -\sum_{x_{1,\nu}\leq x}\sum_h hC_h^{(\nu)}. \tag{5.12}$$

Proof. Every integer up to $u(x)$ lies either in a zero run or in a unique excursion interior. Apply Lemma 5.1 to the terminal block and add its final term $S_{L_x}^2$. □

Lemma 5.5 (Elementary bound for an open terminal block)**.** *Let $u \in \mathbb{N}_0$ satisfy $M(u) = 0$ and let $L \geq 0$. Then $|M(u+r)| \leq r$ for $1 \leq r \leq L$, and*

$$\sum_{r=1}^{L} M(u+r)^2 \leq \frac{L(L+1)(2L+1)}{6}. \tag{5.13}$$

In particular $Q_{\mathrm{term}}(x) \ll L_x^3$. The bridge bound (5.21)*, whose leading coefficient is $1/12$ rather than $1/3$, applies only to completed excursions.*

Proof. $M(u+r) = \sum_{j\leq r}\mu(u+j)$ with $|\mu| \leq 1$; square and sum. □

Thus, in this decomposition, the quadratic sum is expressed through lag correlations internal to excursions only: no correlation between increments of different completed excursions appears on the right-hand side. (The global autocorrelations of μ do contain such pairs; their weighted total is recovered in Proposition 12.2.)

Let

$$\mathfrak{b}_\nu = \sum_{t=1}^{\ell_\nu-1} S_t = \sum_{n\in I_\nu} |M(n)|. \tag{5.14}$$

Cauchy–Schwarz gives

$$\mathfrak{b}_\nu^2 \leq (\ell_\nu - 1)Q_\nu, \tag{5.15}$$

with equality exactly for a flat excursion of height 1. To separate support from sign geometry, let $P_h^{(\nu)} = \#\{j : x_j x_{j+h} \neq 0\}$ and let $D_h^{(\nu)}$ count the opposite-sign pairs among them. Then

$$C_h^{(\nu)} = P_h^{(\nu)} - 2D_h^{(\nu)}, \qquad Q_\nu = \sum_{h=1}^{\ell_\nu-1} h\big(2D_h^{(\nu)} - P_h^{(\nu)}\big). \tag{5.16}$$

Proposition 5.6 (Sharp bounds at fixed length and height)**.** *Let $L \geq 2$ be an integer and let $(S_t)_{0\leq t\leq L}$ be integers with $S_0 = S_L = 0$, $S_t \geq 1$ for $0 < t < L$, $S_t - S_{t-1} \in \{-1, 0, 1\}$, and $H = \max S_t$. Then $L \geq 2H$ and*

$$H^2 + L - 2H \leq \mathfrak{b} \leq H(L-H), \tag{5.17}$$

$$\frac{2H^3+H}{3} + L - 2H \leq Q \leq \frac{2H^3+H}{3} + (L-2H)H^2. \tag{5.18}$$

All four bounds are attained. In particular

$$Q \geq \frac{2H^3+H}{3}, \qquad Q \geq L-1. \tag{5.19}$$

Proof. Before and after the first visit to H the path visits each level $1, \dots, H-1$. Together with one visit to H, these $2H-1$ designated interior times contribute $2(1+\cdots+(H-1))+H = H^2$ to $\mathfrak{b}$ and $2\sum_{j<H} j^2 + H^2 = (2H^3+H)/3$ to Q. The remaining $L-2H$ interior times lie at heights between 1 and H, which gives the four inequalities. The lower bounds are attained by inserting the $L-2H$ remaining steps as horizontal steps at height 1 into the tent path of height H; the upper bounds by inserting all of them as horizontal steps at height H. Both constructions keep $S_t \geq 1$ on $0 < t < L$ and have increments in $\{-1, 0, 1\}$. □

Corollary 5.7 (Sharp fixed-length bounds)**.** *For $L \geq 2$,*

$$\mathfrak{b} \leq \left\lfloor \frac{L^2}{4} \right\rfloor, \qquad Q \leq \begin{cases} L(L^2+2)/12, & L \text{ even}, \\ L(L^2-1)/12, & L \text{ odd}, \end{cases} \tag{5.20}$$

and in particular

$$Q \leq \frac{L^3+2L}{12}. \tag{5.21}$$

Both bounds are attained by the tent path, with one horizontal step at maximal height when L is odd.

Proof. Use $S_t \leq \min(t, L-t)$ and sum this bound and its square. □

6. Where ζ enters

The net area curve $\mathcal{A}$ depends nonlinearly on M. The dependence enters through $\sqrt{H_\nu}$ and through the selection of $x_{0,\nu}, \tau_\nu, x_{1,\nu}$; no closed expression of its Mellin transform in terms of ζ is known. The nodes $x_{0,\nu}, x_{1,\nu}$ are zeros of M. If RH holds and all zeros are simple, there is a sequence $T_k \to \infty$ such that, for non-integer x,

$$M(x) = \lim_{k\to\infty} \sum_{|\gamma|<T_k} \frac{x^\rho}{\rho\,\zeta'(\rho)} - 2 + \sum_{n\geq 1} \frac{(-1)^{n-1}(2\pi/x)^{2n}}{(2n)!\,n\,\zeta(2n+1)} \tag{6.1}$$

[53, §14.27]. For $\widetilde{A}(x) := \int_0^x M(t)\,dt$ one has $\widetilde{A}(1) = 0$ (the value of M at the single point 1 does not affect the integral). Integration by parts therefore gives, for $\Re s > 1$,

$$\int_1^\infty \widetilde{A}(x) x^{-s-2}\,dx = \frac{1}{s(s+1)\zeta(s)}. \tag{6.2}$$

The residues of $x^{s+1}/(s(s+1)\zeta(s))$ are

$$\sum_\rho \frac{x^{\rho+1}}{\rho(\rho+1)\zeta'(\rho)} - 2x + 12 + (\text{residues at the trivial zeros}), \tag{6.3}$$

where $x/\zeta(0) = -2x$ is the residue at $s=0$ and $1/((-1)\zeta(-1)) = 12$ the residue at $s=-1$. Under RH and simplicity of the zeros, the contour argument of [53, §14.27], with the extra decay $(s+1)^{-1}$, turns (6.3) into an identity for $\widetilde{A}(x)$ with the ρ-sum taken as $\lim_k \sum_{|\gamma|<T_k}$; absolute convergence of the ρ-sum needs moment bounds for $1/\zeta'(\rho)$ of the type assumed in [36]. Formula (6.3) is not used in any proof. The area of M_{sk} differs from $\widetilde{A}$ by explicit local terms.

Proposition 6.1 (Dipole identity)**.** *For each excursion let its* legs *be the integer sets $J = (u_\nu, \tau_\nu] \cap \mathbb{Z}$ and $(\tau_\nu, v_\nu] \cap \mathbb{Z}$, with midpoints $c_J = (u_\nu + \tau_\nu)/2$ and $(\tau_\nu + v_\nu)/2$. Then for every zero v_k,*

$$\int_0^{v_k} M_{\mathrm{sk}}(t)\,dt = \widetilde{A}(v_k) + \sum_{J\subset(0,v_k]} \sum_{m\in J} \mu(m)\,(m - c_J). \tag{6.4}$$

Proof. Since M is constant on each $[n, n+1)$, $\int_a^b M(t)\,dt = \sum_{n=a}^{b-1} M(n)$ for integers $a < b$. On an ascent leg, $\int_u^\tau M_{\mathrm{sk}} = \sigma_\nu H a/2$ and $\int_u^\tau M = \sum_{m=u+1}^{\tau} \mu(m)(\tau - m)$, while $\sigma_\nu H = \sum_{m=u+1}^{\tau} \mu(m)$; the difference is $\sum_m \mu(m)(a/2 - \tau + m) = \sum_m \mu(m)(m - c_J)$. On a descent leg, $\int_\tau^v M_{\mathrm{sk}} = \sigma_\nu H b/2$ and $\int_\tau^v M = b\,\sigma_\nu H + \sum_{m=\tau+1}^{v} \mu(m)(v-m)$ with $\sum_{m=\tau+1}^{v} \mu(m) = -\sigma_\nu H$; the difference is $\sum_m \mu(m)(m - (v - b/2))$, and $v - b/2 = (\tau + v)/2 = c_J$. On zero runs both integrands vanish. Sum over legs. □

7. Bounds for the area curves

Lemma 7.1 (Transfer)**.** *Let $x \geq 1$ and let $x^* \geq x$ be a zero of M (in a computation truncated at $N \geq x$, take $x^* = N$ and the terminal skeleton of Definition 3.6). If $|M(n)| \leq C_0\, n^{1/2}$ for all $1 \leq n \leq x^*$ (so necessarily $C_0 \geq 1$), then*

$$|M_{\mathrm{s}}(t)| \leq C_0^{1/2} t^{1/4} \quad (0 \leq t \leq x), \qquad \mathcal{A}_+(x) + |\mathcal{A}_-(x)| \leq \tfrac{4}{5}\, C_0^{1/2} x^{5/4}. \tag{7.1}$$

Proof. For $t \geq \tau_\nu$, $|M_{\mathrm{s}}(t)| \leq \sqrt{H_\nu} \leq C_0^{1/2} \tau_\nu^{1/4} \leq C_0^{1/2} t^{1/4}$. For $u_\nu \leq t \leq \tau_\nu$ the chord from $(u_\nu, 0)$ to $(\tau_\nu, \sqrt{H_\nu})$ lies below the chord of $C_0^{1/2} t^{1/4}$ over $[u_\nu, \tau_\nu]$, whose endpoints $C_0^{1/2} u_\nu^{1/4} \geq 0$ and $C_0^{1/2} \tau_\nu^{1/4} \geq \sqrt{H_\nu}$ (as $\tau_\nu \leq x^*$) are at least as high; that chord lies below $C_0^{1/2} t^{1/4}$ by concavity. Integrate. □

Lemma 7.2 (A concave majorant)**.** *For $y > e$ let $\phi(y) = (\log y)^{3/5} (\log\log y)^{-1/5}$ and, for $\kappa > 0$, $F(y) = y^{1/2} e^{-\kappa\phi(y)}$. There is $y_0 = y_0(\kappa)$ such that F is increasing and concave on $[y_0, \infty)$.*

Proof. With $y = e^s$, $s > 1$, and $g(s) = s^{3/5} (\log s)^{-1/5}$, one has $F = \exp(s/2 - \kappa g(s))$. Put $h(s) = \tfrac{1}{2} - \kappa g'(s)$. Then

$$y F'(y) = F\, h(s), \qquad y^2 F''(y) = F\big(h(s)^2 - h(s) - \kappa g''(s)\big). \tag{7.2}$$

Since $g'(s) \to 0$ and $g''(s) \to 0$, $h(s) \to \tfrac{1}{2}$ and $h^2 - h - \kappa g'' \to -\tfrac{1}{4}$. Hence $F' > 0$ and $F'' < 0$ for large y. □

Corollary 7.3. *From the Vinogradov–Korobov bound* [55]*: there is an absolute constant $c > 0$ with*

$$M(x) \ll x \exp\big(-c\,\phi(x)\big) \qquad (x > e), \tag{7.3}$$

one obtains

$$\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \ll x^{3/2} \exp\big(-\tfrac{c}{2}\,\phi(x)\big). \tag{7.4}$$

Proof. Let K be such that $|M(y)| \leq K y\, e^{-c\phi(y)}$ for $y \geq y_1$, and let F be as in Lemma 7.2 with $\kappa = c/2$, so that $\sqrt{|M(y)|} \leq K^{1/2} F(y)$ for $y \geq y_1$. Only finitely many excursions start below $Y_0 := \max(y_0, y_1)$; they are finite, so they contribute $O(1)$ to the area. Consider an excursion with $u_\nu \geq Y_0$. Its apex satisfies $\sqrt{H_\nu} = \sqrt{|M(\tau_\nu)|} \leq K^{1/2} F(\tau_\nu)$; the bound is taken at the apex τ_ν, which lies in the excursion, and not at its future endpoint v_ν. For $t \geq \tau_\nu$ monotonicity of F gives $|M_{\mathrm{s}}(t)| \leq \sqrt{H_\nu} \leq K^{1/2} F(t)$. For $u_\nu \leq t \leq \tau_\nu$, $|M_{\mathrm{s}}(t)|$ is the chord from $(u_\nu, 0)$ to $(\tau_\nu, \sqrt{H_\nu})$. This chord lies below the chord from $(u_\nu, K^{1/2} F(u_\nu))$ to $(\tau_\nu, K^{1/2} F(\tau_\nu))$, which has both endpoints at least as high, and the latter lies below $K^{1/2} F$ on $[u_\nu, \tau_\nu] \subset [y_0, \infty)$ by concavity. Hence $|M_{\mathrm{s}}(t)| \leq K^{1/2} F(t)$ for all $t \geq Y_0$ outside the finitely many early excursions, and since F is increasing,

$$\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \leq O(1) + K^{1/2} \int_{Y_0}^{x} F(t)\,dt \leq O(1) + K^{1/2} x F(x) = O(1) + K^{1/2} x^{3/2} e^{-\frac{c}{2}\phi(x)}. \tag{7.5}$$

□

Corollary 7.4. *If RH holds, then*

$$\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \ll_\varepsilon x^{5/4+\varepsilon}. \tag{7.6}$$

Consequently, if $\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \gg x^{5/4+\delta}$ along an unbounded sequence for some fixed $\delta > 0$, then RH is false. By Corollary 8.13 the converse also holds: (7.6) *is equivalent to RH.*

Proof. It suffices to treat $0 < \varepsilon < \frac{3}{2}$, so that $t \mapsto t^{1/4+\varepsilon/2}$ is concave. Under RH, $|M(y)| \le C_\varepsilon y^{1/2+\varepsilon}$, so $\sqrt{H_\nu} \le C_\varepsilon^{1/2} \tau_\nu^{1/4+\varepsilon/2}$. The chord argument of Lemma 7.1, with the concave majorant $C_\varepsilon^{1/2} t^{1/4+\varepsilon/2}$ in place of $C_0^{1/2} t^{1/4}$, gives $|M_{\mathrm{s}}(t)| \le C_\varepsilon^{1/2} t^{1/4+\varepsilon/2}$ and, after integration, $\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \ll_\varepsilon x^{5/4+\varepsilon/2}$. □

Remark 7.5 (Endpoint-indexed bounds). From $\ell_\nu \ge 2H_\nu$ and $\sqrt{H_\nu}\ell_\nu \ll v_\nu^{5/4+\varepsilon}$ one deduces $H_\nu \ll v_\nu^{5/6+O(\varepsilon)}$, indexed by the future endpoint v_ν. For a general family of disjoint intervals this gives no bound at interior points. For M itself Corollary 9.8 gives $v_\nu \le u_\nu^{1+\varepsilon}$ for all late excursions, so the bound becomes $|M(t)| \ll t^{5/6+O(\varepsilon)}$; the sharp statement is the exact area exponent of Corollary 8.13.

Remark 7.6 (Oscillation of the heights). The Mertens conjecture is false [38, 23]; the current bounds are due to Hurst [18], who proves $\limsup M(x)/\sqrt{x} > 1.826054$ and $\liminf M(x)/\sqrt{x} < -1.837625$. The excursion containing a point n with $M(n) > 1.826054\sqrt{n}$ has $u_\nu < n$, so $H_\nu > 1.826054\, u_\nu^{1/2}$ for infinitely many positive excursions, and similarly with 1.837625 for negative ones. A lower bound indexed by v_ν does not follow, since v_ν may be much larger than n.

8. Global criteria equivalent to RH

The Mellin identity and the global L^1/L^2 equivalences below are consequences of Littlewood's criterion and partial summation; the excursion decompositions and pruning lemmas rewrite them in terms of excursions. For $\Re s > 1$,

$$\int_1^\infty M(x) x^{-s-1}\, dx = \frac{1}{s\,\zeta(s)}. \tag{8.1}$$

If $\int_1^\infty |M(x)| x^{-\varsigma-1} dx < \infty$ for every $\varsigma > 1/2$, the left side continues analytically to $\Re s > 1/2$, so ζ has no zeros there.

Proposition 8.1 (L^1 criterion). *RH holds if and only if, for every $\varepsilon > 0$,*

$$\sum_{v_\nu \le x} \mathfrak{b}_\nu + \mathfrak{b}_{\mathrm{term}}(x) = \sum_{n \le x} |M(n)| \ll x^{3/2+\varepsilon}. \tag{8.2}$$

Proof. RH gives $M(x) \ll_\varepsilon x^{1/2+\varepsilon}$, hence the bound. Conversely, fix $\varsigma_0 > 1/2$ and $0 < \varepsilon < \varsigma_0 - 1/2$. Partial summation with $\sum_{n \le x} |M(n)| \ll x^{3/2+\varepsilon}$ gives $\sum_n |M(n)| n^{-\varsigma_0-1} < \infty$; since M is constant on $[n, n+1)$, $\int_1^\infty |M(x)| x^{-\Re s-1} dx \le \int_1^\infty |M(x)| x^{-\varsigma_0-1} dx < \infty$ uniformly for $\Re s \ge \varsigma_0$. Hence $F(s) := \int_1^\infty M(x) x^{-s-1} dx$ converges absolutely and uniformly there and is holomorphic on $\Re s > \varsigma_0$. By (8.1), $F = 1/(s\zeta(s))$ on $\Re s > 1$; by uniqueness of analytic continuation the meromorphic function $1/(s\zeta(s))$ coincides with F on $\Re s > \varsigma_0$, so it has no poles there, i.e. ζ has no zeros there. As $\varsigma_0 > 1/2$ is arbitrary, RH follows. □

Lemma 8.2 (Pruning). $\mathfrak{b}_\nu \le H_\nu(\ell_\nu - 1) \le \ell_\nu^2/2$, *hence*

$$\sum_{v_\nu \le x,\ \ell_\nu \le L_0} \mathfrak{b}_\nu \le \frac{x L_0}{2}. \tag{8.3}$$

Corollary 8.3 (Reduction to long excursions). *For fixed $\delta \ge 0$ and $L_0 = x^{1/2-\delta}$, RH holds if and only if*

$$\sum_{\substack{v_\nu \le x \\ \ell_\nu > L_0}} \mathfrak{b}_\nu + \mathbf{1}_{\{L_x > L_0\}} \mathfrak{b}_{\mathrm{term}}(x) \ll x^{3/2+\varepsilon} \quad \text{for every } \varepsilon > 0. \tag{8.4}$$

At most x/L_0 long excursions are involved.

Proof. If RH holds, the left side is at most $\sum_{n\le x}|M(n)| \ll x^{3/2+\varepsilon}$. Conversely, the completed excursions with $\ell_\nu \le L_0$ contribute at most $xL_0/2 \le x^{3/2}$ by Lemma 8.2, and if $L_x \le L_0$ the terminal block contributes $\mathfrak{b}_{\mathrm{term}}(x) \le \sum_{r\le L_x} r = L_x(L_x+1)/2 \le L_0^2 \le x$ when $L_0 \ge 1$, and nothing when $L_0 < 1$, since then $L_x = 0$. Hence $\sum_{n\le x}|M(n)| \ll x^{3/2+\varepsilon}$, and Proposition 8.1 applies. □

Remark 8.4 (The $\sqrt{x}$ frontier). The cut may depend on ε: with $L_0 = x^{1/2+\varepsilon/2}$ the discarded completed excursions contribute at most $xL_0/2 = O(x^{3/2+\varepsilon/2})$, while a short terminal block contributes $O(L_0^2) = O(x^{1+\varepsilon})$. Both are admissible in the $x^{3/2+\varepsilon}$ budget. The frontier of pruning is therefore $x^{1/2+o(1)}$; what requires new information is a cut x^θ with a fixed $\theta > 1/2$. For $\ell_\nu \asymp x$ the trivial bound gives $\mathfrak{b}_\nu \ll x^2$ against a budget $x^{3/2+\varepsilon}$, and there are only $O(1)$ such excursions.

Heuristic 8.5 (Renewal tail)**.** If excursion lengths behave like those of a recurrent random walk,

$$\#\{v_\nu \le x :\ \ell_\nu > L\} \asymp \sqrt{x}\, L^{-1/2}, \tag{8.5}$$

and with $\mathfrak{b}_\nu \asymp \ell_\nu^{3/2}$ the dyadic range $[L, 2L]$ contributes $\asymp \sqrt{x}\, L$ to (8.2), so the terminal scales $L \asymp x$ dominate the L^1 sum.

Theorem 8.6 (Correlation criterion)**.** *RH holds if and only if, for every $\varepsilon > 0$,*

$$\sum_{n\le x} M(n)^2 = \sum_{v_\nu \le x} Q_\nu + Q_{\mathrm{term}}(x) = -\sum_{v_\nu \le x}\sum_h h C_h^{(\nu)} + Q_{\mathrm{term}}(x) \ll x^{2+\varepsilon}. \tag{8.6}$$

Proof. Under RH, $|M(n)| \ll_\delta n^{1/2+\delta}$ for every $\delta > 0$; with $\delta = \varepsilon/2$, $M(n)^2 \ll_\varepsilon n^{1+\varepsilon}$ and the sum is $\ll x^{2+\varepsilon}$. Conversely, by Cauchy–Schwarz,

$$\sum_{n\le x}|M(n)| \le x^{1/2}\Big(\sum_{n\le x} M(n)^2\Big)^{1/2} \ll x^{3/2+\varepsilon/2}; \tag{8.7}$$

apply Proposition 8.1. □

Corollary 8.7 (Local sufficient conditions)**.** *Suppose that for every $\varepsilon > 0$ there is C_ε such that*

$$Q_\nu \le C_\varepsilon\, \ell_\nu^{2+\varepsilon} \quad \textit{for every completed excursion with } \ell_\nu > v_\nu^{1/2}. \tag{8.8}$$

Then RH holds.

Proof. Let $\varepsilon_0 \in (0,1)$, put $\varepsilon = \varepsilon_0/4$, and let C_ε be as in the hypothesis. Split the completed excursions at $\ell_\nu = x^{1/2}$. If $\ell_\nu \le x^{1/2}$, then (5.21) gives $Q_\nu \le \ell_\nu x/12 + \ell_\nu/6$, and disjointness bounds the total by $x^2/12 + x/6$. If $\ell_\nu > x^{1/2}$ then $\ell_\nu > v_\nu^{1/2}$ since $v_\nu \le x$, and the hypothesis bounds the total by $C_\varepsilon \sum \ell_\nu^{2+\varepsilon} \le C_\varepsilon(\sum \ell_\nu)^{2+\varepsilon} \le C_\varepsilon x^{2+\varepsilon}$. The terminal block is not a completed excursion, and (5.21) does not apply to it. By Lemma 5.5, $Q_{\mathrm{term}}(x) \le L_x(L_x+1)(2L_x+1)/6$, which is $O(x^{3/2})$ if $L_x \le x^{1/2}$. If $L_x > x^{1/2}$, then $M(\lfloor x\rfloor) \ne 0$. Let $I_\nu = (u(x), v_\nu]$ be the unique completed excursion whose open prefix contains $\lfloor x\rfloor$; since all summands are nonnegative, $Q_{\mathrm{term}}(x) \le Q_\nu$. Fix $A = 1+\varepsilon$. By Corollary 9.8, $v_\nu \le u(x)^A \le x^A$ once x is large, so $\ell_\nu \le x^A$. If $\ell_\nu > v_\nu^{1/2}$ the hypothesis gives $Q_\nu \le C_\varepsilon x^{A(2+\varepsilon)}$; otherwise (5.21) gives $Q_\nu \ll \ell_\nu^3 \le v_\nu^{3/2} \le x^{3A/2}$. In all cases $Q_{\mathrm{term}}(x) \ll_\varepsilon x^{(1+\varepsilon)(2+\varepsilon)} \le x^{2+4\varepsilon} = x^{2+\varepsilon_0}$. This step uses the arithmetic input of Corollary 9.8: for a general family of disjoint intervals, the future return of the terminal block is not controlled by x, and a bound for completed excursions would not control $Q_{\mathrm{term}}(x)$. Altogether $\sum_{n\le x} M(n)^2 \ll_{\varepsilon_0} x^{2+\varepsilon_0}$ for every $\varepsilon_0 \in (0,1)$, and Theorem 8.6 gives RH. □

Heuristic 8.8 (Ballistic excursions)**.** In a model where, after a zero of M, the next values of μ behave locally at random with squarefree density $6/\pi^2$, the pattern "k non-zero steps of one sign followed by k of the other" follows a given zero with probability $\ge c_0^{-k}$. If excursions with $H_\nu = k$ and $\ell_\nu \asymp k$ occur for unbounded k, then by (5.19) $Q_\nu/\ell_\nu^{2+\varepsilon} \gg k^{1-\varepsilon} \to \infty$, so the unpruned version

of (8.8) fails for every $\varepsilon \in (0,1)$. If, as the probability c_0^{-k} suggests, such excursions first occur at positions v_ν growing at least exponentially in k, then $\ell_\nu \ll v_\nu^{1/2}$ and they do not affect the pruned condition (8.8); this is why the pruning in Corollary 8.7 is needed.

Theorem 8.9 (Energy criterion)**.** *With u the last zero $\le x$, $H_{\mathrm{run}}(x) = \max_{u<n\le x}|M(n)|$ and $H_{\mathrm{run}}(x) = 0$ when $u = \lfloor x \rfloor$, define*

$$\mathcal{E}(x) := \frac{1}{4}\Big(\sum_{v_\nu \le x} H_\nu \ell_\nu + (x-u)H_{\mathrm{run}}(x)\Big). \tag{8.9}$$

For a completed excursion let ch_ν be the chord joining $(u_\nu, \mathcal{A}(u_\nu))$ to $(v_\nu, \mathcal{A}(v_\nu))$. By (4.1), $\mathrm{ch}'_\nu = \sigma_\nu\sqrt{H_\nu}/2$, so $\frac{1}{4}H_\nu\ell_\nu = \int_{I_\nu}(\mathrm{ch}'_\nu)^2$ is the Dirichlet energy of the affine interpolant ch_ν joining the endpoint values of $\mathcal{A}$ on I_ν; it is not the Dirichlet energy $\int_{I_\nu} M_{\mathrm{s}}^2$ of $\mathcal{A}$ itself. Then

$$\sum_{n\le x}|M(n)| \le 4\mathcal{E}(x) \le x\max_{n\le x}|M(n)|, \tag{8.10}$$

and RH holds if and only if $\mathcal{E}(x) \ll x^{3/2+\varepsilon}$ for every $\varepsilon > 0$.

Proof. $\mathfrak{b}_\nu \le H_\nu\ell_\nu$, $\mathfrak{b}_{\mathrm{term}}(x) \le (x-u)H_{\mathrm{run}}(x)$ and $\sum \ell_\nu + (x-u) \le x$; apply Proposition 8.1. □

8.1. **Exact logarithmic exponents from Pintz's mean and late-maximal-order theorem.** The criteria above are upper-bound equivalences. Theorems 2.1 and 2.2 of Pintz [41, Theorems 2.1–2.2, equations (2.14)–(2.15)] determine, in logarithmic scale, the mean modulus and late maximal order of $|M|$ in terms of the rightmost zeros of ζ. In [41], $D_M(x) = x^{-1}\int_0^x |M|$ (equivalently $\int_1^x$), $S_{M,\delta}(x) = \max_{x^{1-\delta}\le u\le x}|M(u)|$ for each fixed $\delta > 0$, and $Z(x) = \max_{\gamma>0} x^\beta/\gamma$; Theorem 2.2 there, equation (2.15), states $\log D_M(x) \sim \log S_{M,\delta}(x) \sim \log Z(x)$ as $x \to \infty$ for every fixed $\delta > 0$, and (6.1) there gives $\log Z(x) \sim \Theta \log x$ (both cases $\Theta < 1$ and $\Theta = 1$). Since $\Theta > 0$, this is the statement below. For $\Theta < 1$ the logarithmic-scale lower bound for D_M already follows from Pintz's inequality $D_M(Y) \ge Y^{\beta_0}/(6|\gamma_0|^3)$ for $Y > e^{|\gamma_0|+4}$, valid for every zero $\beta_0 + i\gamma_0$ [39], and the upper bound from the classical estimate $M(x) \ll_\varepsilon x^{\Theta+\varepsilon}$; the case $\Theta = 1$ is the new content of [41]. We record the consequence for the excursion quantities used in this paper.

Theorem 8.10 (Pintz, logarithmic mean and late maximal order [41, Theorems 2.1 and 2.2])**.** *Let $\Theta = \sup\{\Re\rho : \zeta(\rho) = 0,\ 0 < \Re\rho < 1\}$. Then*

$$D_M(X) := \frac{1}{X}\int_1^X |M(t)|\,dt = X^{\Theta+o(1)}. \tag{8.11}$$

Moreover, for every fixed $\delta > 0$,

$$S_{M,\delta}(X) := \max_{X^{1-\delta}\le t\le X}|M(t)| = X^{\Theta+o_\delta(1)}. \tag{8.12}$$

Here the equalities are in logarithmic scale, equivalently

$$\frac{\log D_M(X)}{\log X} \to \Theta, \qquad \frac{\log S_{M,\delta}(X)}{\log X} \to \Theta \quad \text{for each fixed } \delta > 0. \tag{8.13}$$

Corollary 8.11 (Exact exponents of the absolute mass, the quadratic mass and the chord energy of the Möbius sum)**.** *Put*

$$B_1(X) := \sum_{n\le X}|M(n)|, \qquad B_2(X) := \sum_{n\le X} M(n)^2. \tag{8.14}$$

Then

$$B_1(X) = X^{1+\Theta+o(1)}, \qquad B_2(X) = X^{1+2\Theta+o(1)}, \qquad \mathcal{E}(X) = X^{1+\Theta+o(1)}. \tag{8.15}$$

Consequently

$$\Theta = \lim_{X\to\infty}\left(\frac{\log B_1(X)}{\log X} - 1\right) = \frac{1}{2}\lim_{X\to\infty}\left(\frac{\log B_2(X)}{\log X} - 1\right) = \lim_{X\to\infty}\left(\frac{\log \mathcal{E}(X)}{\log X} - 1\right). \tag{8.16}$$

In particular RH is equivalent to any one of the three logarithmic exponents being respectively $3/2$*,* 2*, and* $3/2$*.*

Proof. Let $N = \lfloor X \rfloor$. Since $M(t) = M(n)$ on $[n, n+1)$,

$$\int_1^X |M(t)|\,dt = \sum_{n=1}^{N-1} |M(n)| + (X-N)|M(N)|, \tag{8.17}$$

so

$$B_1(X) = \int_1^X |M(t)|\,dt + O(|M(N)| + 1). \tag{8.18}$$

The boundary term is $O(X)$, whereas Theorem 8.10 gives $\int_1^X |M(t)|\,dt = X^{1+\Theta+o(1)}$ and $\Theta \geq 1/2$; hence $B_1(X) = X^{1+\Theta+o(1)}$. We first note that

$$\max_{n\leq X} |M(n)| = X^{\Theta+o(1)}. \tag{8.19}$$

The lower bound follows from $S_{M,\delta}$ with any fixed δ. For the upper bound, if $\Theta < 1$, fix once and for all $\delta \in (1-\Theta, 1)$, independent of X, and split the range $n \leq X$ at $X^{1-\delta}$. On $n < X^{1-\delta}$ the trivial bound gives $|M(n)| \leq X^{1-\delta} = o(X^{\Theta})$; on $X^{1-\delta} \leq n \leq X$, Theorem 8.10 with this fixed δ gives $|M(n)| \leq S_{M,\delta}(X) = X^{\Theta+o(1)}$. When $\Theta = 1$ the trivial bound $|M(n)| \leq n$ suffices.

Cauchy–Schwarz and (8.19) now give

$$\frac{B_1(X)^2}{X} \leq B_2(X) \leq X \max_{n\leq X} |M(n)|^2, \tag{8.20}$$

so both sides have exponent $1 + 2\Theta$. Finally (8.10) squeezes $4\mathcal{E}(X)$ between $B_1(X)$ and

$$X \max_{n\leq X} |M(n)|, \tag{8.21}$$

both of exponent $1 + \Theta$. □

Corollary 8.12 (Length localization of the area and quadratic mass)**.** *Fix* $0 \leq \alpha < \Theta$*. Among the completed excursions ending at most* X*, the total contribution of those with* $\ell_\nu \leq X^\alpha$ *is negligible:*

$$\sum_{\substack{v_\nu \leq X \\ \ell_\nu \leq X^\alpha}} \mathfrak{b}_\nu + \mathbf{1}_{\{L_X \leq X^\alpha\}} \mathfrak{b}_{\mathrm{term}}(X) = o(B_1(X)), \tag{8.22}$$

and

$$\sum_{\substack{v_\nu \leq X \\ \ell_\nu \leq X^\alpha}} Q_\nu + \mathbf{1}_{\{L_X \leq X^\alpha\}} Q_{\mathrm{term}}(X) = o(B_2(X)). \tag{8.23}$$

Thus, on the logarithmic scale forced by the zeta zeros, essentially all L^1 *and* L^2 *mass is carried by excursion blocks longer than* X^α *for every* $\alpha < \Theta$*.*

Proof. Lemma 8.2 bounds the first complete-excursion sum by $O(X^{1+\alpha})$, while the short terminal block contributes $O(X^{2\alpha})$. Both are $o(X^{1+\Theta+o(1)})$ when $\alpha < \Theta$. For the quadratic mass use (5.21) and disjointness:

$$\sum_{\ell_\nu \leq X^\alpha} Q_\nu \ll X^{1+2\alpha}, \qquad Q_{\mathrm{term}}(X) \ll X^{3\alpha} \quad (L_X \leq X^\alpha). \tag{8.24}$$

Because $\alpha < \Theta \leq 1$, both exponents are strictly below $1 + 2\Theta$. □

Corollary 8.13 (Exact exponent of the skeleton area)**.** *The proof uses the return localization (RL) of Corollary 9.8, which holds unconditionally. Put* $\mathcal{A}_{\mathrm{abs}}(x) := \mathcal{A}_+(x) + |\mathcal{A}_-(x)| = \int_0^x |M_{\mathrm{s}}(t)|\,dt$. *Then*

$$\mathcal{A}_{\mathrm{abs}}(x) = x^{1+\Theta/2+o(1)} \qquad (x \to \infty). \tag{8.25}$$

In particular RH $\iff$ $\mathcal{A}_{\mathrm{abs}}(x) \ll_\varepsilon x^{5/4+\varepsilon}$ *for every* $\varepsilon > 0$.

Proof. Upper bound. Fix $0 < \varepsilon < 2-\Theta$. By (8.19) there is C_ε with $H_\nu = |M(\tau_\nu)| \le C_\varepsilon \tau_\nu^{\Theta+\varepsilon}$. Since $0 < (\Theta+\varepsilon)/2 < 1$, the function $C_\varepsilon^{1/2} t^{(\Theta+\varepsilon)/2}$ is concave and increasing, and the chord argument of Lemma 7.1 gives $|M_{\mathrm{s}}(t)| \le C_\varepsilon^{1/2} t^{(\Theta+\varepsilon)/2}$, hence $\mathcal{A}_{\mathrm{abs}}(x) \ll_\varepsilon x^{1+(\Theta+\varepsilon)/2}$. As ε is arbitrary, $\limsup_x \log \mathcal{A}_{\mathrm{abs}}(x)/\log x \le 1 + \Theta/2$.

Lower bound. Let $\delta > 0$ and fix $\varepsilon \in (0,1)$ so small that

$$e(\varepsilon) := \frac{1 + \Theta - \varepsilon - \frac{1}{2}(1+\varepsilon)(\Theta+\varepsilon)}{1+\varepsilon} \ge 1 + \frac{\Theta}{2} - \frac{\delta}{2}; \tag{8.26}$$

this is possible because $e(\varepsilon) \to 1 + \Theta/2$ as $\varepsilon \downarrow 0$. For $X \ge 2$ let $v^*(X)$ be the least zero of M with $v^*(X) \ge \lfloor X \rfloor$. By (RL) (Corollary 9.8 with $A = 1+\varepsilon$) there is $X_1(\varepsilon)$ such that $v^*(X) \le X^{1+\varepsilon}$ for every $X \ge X_1(\varepsilon)$: if $M(\lfloor X \rfloor) = 0$ then $v^*(X) = \lfloor X \rfloor$; otherwise $v^*(X)$ ends the excursion containing $\lfloor X \rfloor$, whose start u satisfies $u < X$, and $v^*(X) \le u^{1+\varepsilon}$ once $u \ge U_{1+\varepsilon}$, which holds for all large X because the excursions starting below $U_{1+\varepsilon}$ are finitely many and finite. Every $n \le X$ with $M(n) \ne 0$ lies in a completed excursion ending at most $v^*(X)$. With $\mathfrak{b}_\nu \le H_\nu \ell_\nu$, $m^* := \max_{n \le v^*(X)} |M(n)|$ and Lemma 4.1,

$$B_1(X) \le \sum_{v_\nu \le v^*(X)} H_\nu \ell_\nu \le (m^*)^{1/2} \sum_{v_\nu \le v^*(X)} \sqrt{H_\nu}\, \ell_\nu = 2(m^*)^{1/2} \mathcal{A}_{\mathrm{abs}}(v^*(X)). \tag{8.27}$$

By Corollary 8.11 and (8.19) there is $X_2(\varepsilon)$ such that for all $X \ge X_2(\varepsilon)$, $B_1(X) \ge X^{1+\Theta-\varepsilon}$ and $m^* \le v^*(X)^{\Theta+\varepsilon} \le X^{(1+\varepsilon)(\Theta+\varepsilon)}$. Hence, for $X \ge \max(X_1, X_2)$, using monotonicity of $\mathcal{A}_{\mathrm{abs}}$,

$$\mathcal{A}_{\mathrm{abs}}(X^{1+\varepsilon}) \ge \mathcal{A}_{\mathrm{abs}}(v^*(X)) \ge \tfrac{1}{2} X^{(1+\varepsilon)e(\varepsilon)}. \tag{8.28}$$

Given $y \ge \max(X_1, X_2)^{1+\varepsilon}$, put $X = y^{1/(1+\varepsilon)}$; then $\mathcal{A}_{\mathrm{abs}}(y) \ge \frac{1}{2} y^{e(\varepsilon)} \ge \frac{1}{2} y^{1+\Theta/2-\delta/2} \ge y^{1+\Theta/2-\delta}$ for all large y. Since $\delta > 0$ is arbitrary, $\liminf_y \log \mathcal{A}_{\mathrm{abs}}(y)/\log y \ge 1 + \Theta/2$. □

Remark 8.14 (Inputs). The new part is the inequality $B_1(X) \le 2(m^*)^{1/2} \mathcal{A}_{\mathrm{abs}}(v^*(X))$ and its use. The exponent inputs are Corollary 8.11 (from [41], or from [39] and the classical upper bound) and the return localization (RL) of Corollary 9.8; for an arbitrary family of disjoint intervals the terminal block of $B_1(X)$ is not controlled by the area up to X. For the lim sup form of the lower bound the classical $B_1(X) = \Omega(X^{1+\Theta-\varepsilon})$ (Mellin transform and Proposition 8.1) suffices instead of [41].

8.2. **Excursion blocks of $1/\zeta$.**

Proposition 8.15 (Excursion blocks of $1/\zeta$)**.** *For a completed excursion* $I = (x_0, x_1]$ *put* $D_I(s) = \sum_{x_0 < n \le x_1} \mu(n) n^{-s}$. *Then*

$$D_I(s) = \sum_{n=x_0+1}^{x_1-1} M(n)\big(n^{-s} - (n+1)^{-s}\big). \tag{8.29}$$

For real $\varsigma > 0$, $D_I(\varsigma)$ *has the sign of the excursion and*

$$|D_I(\varsigma)| \ge \varsigma \mathfrak{b}_I x_1^{-\varsigma-1} \ge \varsigma H_I^2 x_1^{-\varsigma-1}. \tag{8.30}$$

For $\Re s > 1$,

$$\frac{1}{\zeta(s)} = \sum_I D_I(s), \tag{8.31}$$

zero runs contributing nothing.

Proof. For $a_n = \mu(n)$, $A(n) = M(n)$ and $f(n) = n^{-s}$, finite summation by parts gives

$$\sum_{n=x_0+1}^{x_1} a_n f(n) = A(x_1)f(x_1) - A(x_0)f(x_0) + \sum_{n=x_0+1}^{x_1-1} A(n)(f(n) - f(n+1)). \tag{8.32}$$

Since $M(x_0) = M(x_1) = 0$, this is (8.29). For real $\varsigma > 0$, all interior $M(n)$ have the excursion sign, and

$$n^{-\varsigma} - (n+1)^{-\varsigma} = \varsigma \int_n^{n+1} x^{-\varsigma-1}\, dx \geq \varsigma x_1^{-\varsigma-1}. \tag{8.33}$$

The bound $\mathfrak{b}_I \geq H_I^2$ is (5.17). If M vanishes at two consecutive integers the intervening $\mu(n)$ is zero, so every n with $\mu(n) \neq 0$ lies in a completed excursion. For $\Re s > 1$ the ordinary Dirichlet series is absolutely convergent, so summing the finite disjoint blocks gives (8.31). □

Theorem 8.16 (Abscissa of absolute excursion-block convergence)**.** *Let $D_I(s)$ be the completed-excursion blocks of Proposition 8.15. The series*

$$\sum_I D_I(s) \tag{8.34}$$

has abscissa of absolute convergence, with respect to the excursion blocks, exactly

$$\boxed{\varsigma_{a,\mathrm{exc}} = \Theta}. \tag{8.35}$$

Equivalently, the infimum of the real numbers ς_0 for which the series converges absolutely throughout the half-plane $\Re s > \varsigma_0$ is Θ. More precisely, it converges absolutely and locally uniformly for $\Re s > \Theta$ and there represents $1/\zeta(s)$, while for every real $\varsigma < \Theta$ with $\varsigma \neq 0$ one has

$$\sum_I |D_I(\varsigma)| = \infty. \tag{8.36}$$

No assertion is made on the boundary line $\Re s = \Theta$. Consequently,

$$\mathrm{RH} \iff \sum_I D_I(s) \text{ converges absolutely for every } \Re s > \tfrac{1}{2}. \tag{8.37}$$

This is block absolute convergence; the ordinary termwise Dirichlet series $\sum_n \mu(n)n^{-s}$ has abscissa of absolute convergence 1 *and, by $M(x) \ll_\varepsilon x^{\Theta+\varepsilon}$, abscissa of (conditional) convergence Θ. Grouping by completed excursions converts this conditional convergence into absolute convergence of the blocks.*

Remark 8.17. The isolated point $s = 0$ does not enter the definition of the abscissa. For completeness, every completed excursion satisfies $D_I(0) = \sum_{n\in I} \mu(n) = 0$.

Proof. Let $s = \varsigma + i\tau$. Since

$$n^{-s} - (n+1)^{-s} = s \int_n^{n+1} x^{-s-1}\, dx, \tag{8.38}$$

with the real positive branch of $\log x$, one has, for $\varsigma \geq -1$,

$$|n^{-s} - (n+1)^{-s}| \leq |s| n^{-\varsigma-1}. \tag{8.39}$$

Hence

$$\sum_I |D_I(s)| \leq |s| \sum_{n\geq 1} |M(n)| n^{-\varsigma-1}. \tag{8.40}$$

Let $K_0 \Subset \{\Re s > \Theta\}$ be compact and set $\varsigma_0 = \min_{s\in K_0} \Re s > \Theta$. Choose $0 < \varepsilon_0 < \varsigma_0 - \Theta$. From $B_1(X) = X^{1+\Theta+o(1)}$ we have $B_1(X) \ll_{\varepsilon_0} X^{1+\Theta+\varepsilon_0}$ for large X, and partial summation gives

$$\sum_{n\geq 1} |M(n)| n^{-\varsigma_0-1} < \infty. \tag{8.41}$$

Since $\sup_{s\in K_0}|s|<\infty$, the Weierstrass criterion gives normal convergence of the block series on K_0. Thus it is holomorphic in $\Re s>\Theta$. There are no zeros of ζ in that half-plane by the definition of Θ, and $1/\zeta$ extends holomorphically to $s=1$ with value 0; hence $1/\zeta$ is holomorphic throughout the connected half-plane. The identity already holds for $\Re s>1$, so the identity theorem extends it to all $\Re s>\Theta$.

For real $\varsigma>0$, order the completed excursions by their right endpoints. If V is a zero of M, the fixed sign of each excursion and finite Abel summation give

$$\sum_{I:\,x_1(I)\le V}|D_I(\varsigma)|=\sum_{n<V}|M(n)|\big(n^{-\varsigma}-(n+1)^{-\varsigma}\big). \tag{8.42}$$

Since M has arbitrarily large zeros (Definition 3.1, using $M(x)=\Omega_\pm(\sqrt{x})$), letting $V\to\infty$ through zeros and using monotone convergence gives

$$\sum_{I}|D_I(\varsigma)|=\sum_{n\ge1}|M(n)|\big(n^{-\varsigma}-(n+1)^{-\varsigma}\big), \tag{8.43}$$

even when both sides are infinite. If the right side were finite for some $0<\varsigma<\Theta$, then for $n\le X$

$$n^{-\varsigma}-(n+1)^{-\varsigma}=\varsigma\int_n^{n+1}t^{-\varsigma-1}\,dt\ge\varsigma(X+1)^{-\varsigma-1}, \tag{8.44}$$

so

$$B_1(X)\le\frac{(X+1)^{\varsigma+1}}{\varsigma}\sum_{n\ge1}|M(n)|\big(n^{-\varsigma}-(n+1)^{-\varsigma}\big)=O(X^{1+\varsigma}), \tag{8.45}$$

contradicting $B_1(X)=\Omega(X^{1+\Theta-\varepsilon})$ for every $\varepsilon>0$. For real $\varsigma<0$ the same Abel identity holds (the factor $n^{-\varsigma}-(n+1)^{-\varsigma}$ is negative, so each block again has constant sign), and $|n^{-\varsigma}-(n+1)^{-\varsigma}|=|\varsigma|\int_n^{n+1}x^{-\varsigma-1}dx\ge|\varsigma|\min\{1,(V+1)^{-\varsigma-1}\}$ for $n<V$. Hence $\sum_{I:\,x_1(I)\le V}|D_I(\varsigma)|\ge|\varsigma|\min\{1,(V+1)^{-\varsigma-1}\}B_1(V-1)$, which tends to infinity along zeros V: for $\varsigma\le-1$ because $B_1\to\infty$, and for $-1<\varsigma<0$ because $V^{-\varsigma-1}B_1(V-1)\ge V^{\Theta-\varsigma-\varepsilon}$ for large V. Therefore the block abscissa is exactly Θ. □

Remark 8.18 (Status of the result). The analytic input is classical: $\sum_n|M(n)|n^{-\varsigma-1}$ converges for $\varsigma>\Theta$ and diverges for $\varsigma<\Theta$; the case $\varsigma=\Theta$ is left open (upper bound $M(x)\ll x^{\Theta+\varepsilon}$; lower bound by the Mellin argument of Proposition 8.1). The new point is the excursion grouping: the fixed sign of each completed excursion turns Abel summation into the positive identity (8.43). We did not find this block formulation in the literature (absence of a match is weak evidence).

9. Lacunary excursion families and horizon-localized criteria

9.1. **Geometric counts.** The lemmas of this subsection use only that excursions are disjoint integer intervals ordered by their starting points, the next one starting at or after the end of the previous one. They therefore hold verbatim for every sequence of disjoint integer intervals, deterministic or random, and in particular pathwise for any stochastic model of excursions (Section 13). For a family $\mathcal{F}$ of excursions and $X\ge2$ put

$$E_{\mathcal{F}}(X):=\#\{\nu\in\mathcal{F}:x_{0,\nu}\le X\}. \tag{9.1}$$

For a function $\mathfrak{h}:[0,\infty)\to[0,\infty]$, the family of excursions *crossing the horizon* $\mathfrak{h}$ is

$$\mathcal{F}_{\mathfrak{h}}:=\{\nu:x_{1,\nu}>\mathfrak{h}(x_{0,\nu})\}. \tag{9.2}$$

For $A>1$ the excursion I_ν is *A-exceptional* if $x_{1,\nu}>x_{0,\nu}^A$, i.e. $\nu\in\mathcal{F}_{x^A}$, and *A-ordinary* otherwise; for $A=2$ we say *square-exceptional* and *square-ordinary*. For $\lambda>1$ it is *λ-long* if $x_{1,\nu}>\lambda x_{0,\nu}$. We write $E_A=E_{\mathcal{F}_{x^A}}$, $E_\square=E_2$ and $E_\lambda=E_{\mathcal{F}_{\lambda x}}$; the context separates E_A (A an exponent) from E_λ (λ a factor).

Lemma 9.1 (Power lacunarity)**.** *For every $A > 1$ and $X \geq 2$,*

$$E_A(X) \leq 2 + \frac{1}{\log A} \log \frac{\log X}{\log 2}. \tag{9.3}$$

In particular

$$E_{\square}(X) \leq 2 + \log_2 \frac{\log X}{\log 2}, \qquad E_{\square}(X) \leq 4 \log\log X \quad (X \geq e^e). \tag{9.4}$$

Proof. No excursion starts at 1, since $M(1) = 1$. List the A-exceptional starts ≥ 2 as $2 \leq a_1 < a_2 < \cdots$. The next excursion starts at or after the end of the preceding one, so $a_{j+1} \geq x_{1,j} > a_j^A$ and $\log a_j \geq A^{j-1} \log 2$. If $a_j \leq X$ then $j \leq 1 + \log(\log X/\log 2)/\log A$; add 1 for the excursion $(0, 2]$. For the second bound in (9.4), $\log_2(\log X/\log 2) = (\log\log X - \log\log 2)/\log 2 \leq 1.443 \log\log X + 0.529$. □

The next lemma is the partition of the square horizon $[x_0, x_0^2] = [x_0, cx_0^2] \cup [cx_0^2, x_0^2]$: the upper part is accessible only to a doubly lacunary family, for every fixed $c > 0$.

Lemma 9.2 (Fractional square horizon)**.** *For $0 < c \leq 1$ and $X \geq 2$,*

$$E_{\mathcal{F}_{cx^2}}(X) = \#\{\nu : x_{0,\nu} \leq X,\ x_{1,\nu} > cx_{0,\nu}^2\} \leq \frac{2}{c} + 2 + \log_2 \frac{\log X}{\log 2}. \tag{9.5}$$

Consequently, the range $cx_0^2 < t \leq x_0^2$ of the excursions starting at most X is visited by at most $O_c(\log\log X)$ excursions.

Proof. At most $\lceil 2/c \rceil \leq 2/c + 1$ members start below $2/c$. List the others as $a_1 < a_2 < \cdots$ and put $b_j = ca_j \geq 2$. Then $a_{j+1} \geq x_{1,j} > ca_j^2$, so

$$b_{j+1} = ca_{j+1} > c^2 a_j^2 = b_j^2, \tag{9.6}$$

and $\log b_j \geq 2^{j-1} \log 2$. Since $b_j \leq cX \leq X$, the number of such j is at most $1 + \log_2(\log X/\log 2)$. An excursion containing a point $t > cx_0^2$ has $x_1 \geq t > cx_0^2$. □

Lemma 9.3 (Superlinear horizons)**.** *Let $\psi : [0, \infty) \to (0, \infty)$ satisfy $\psi(x) \to \infty$. Then*

$$E_{\mathcal{F}_{x\psi(x)}}(X) = \#\{\nu : x_{0,\nu} \leq X,\ x_{1,\nu} > x_{0,\nu}\psi(x_{0,\nu})\} = o(\log X). \tag{9.7}$$

Proof. Given $K > 0$ choose x_K with $\psi(x) \geq e^K$ for $x \geq x_K$. At most $x_K + 1$ members start below x_K; we may take $x_K \geq 1$. For consecutive members $a_j < a_{j+1}$ starting at or above x_K, $a_{j+1} \geq x_{1,j} > a_j\psi(a_j) \geq e^K a_j$, so $\log a_j$ increases by more than K at each step and there are at most $1 + \log X/K$ of them. Hence $\limsup E(X)/\log X \leq 1/K$ for every K. □

Lemma 9.4 (Linear lacunarity)**.** *For every $\lambda > 1$ and $X \geq 2$,*

$$E_\lambda(X) = \#\{\nu : x_{0,\nu} \leq X,\ x_{1,\nu} > \lambda x_{0,\nu}\} \leq 2 + \frac{\log(X/2)}{\log \lambda}. \tag{9.8}$$

Proof. For starts $a_j \geq 2$, $a_{j+1} \geq x_{1,j} > \lambda a_j$, hence $a_j > 2\lambda^{j-1}$ for $j \geq 2$; add 1 for $(0, 2]$. □

All four bounds are sharp up to constants for general families of disjoint intervals, as geometric and doubly geometric chains of intervals show.

9.2. **Arithmetic input: Kaczorowski–Pintz.** We separate the published statement, its translation from $M_0 = M + 2$ to M, and the consequences for excursions.

Theorem 9.5 (Kaczorowski–Pintz [22])**.** *Let $M_0(x) = M(x) - 1/\zeta(0) = M(x) + 2$, the function M_a of* [22] *at $a = 0$. Let $V(f, Y)$ be defined as in* [22]*, through the largest number of points of $[1, Y]$ at which f is nonzero with alternating signs; the number of sign changes of f on $[1, Y]$ is at least $V(f, Y) - 1$, which does not affect the asymptotic statements below. Then:*

(i) [22, Corollary 4, p. 178, (2.13)–(2.15)]

$$\liminf_{Y\to\infty} \frac{V(M_0,Y)}{\log Y} \geq \frac{\gamma_1}{\pi}, \tag{9.9}$$

with combined oscillation of size $x^{\Theta-\eta}$ for every fixed $\eta>0$: there exist ordered points $t_1<\cdots<t_N\leq Y$, with $N\geq(\gamma_1/\pi-o(1))\log Y$, whose signs alternate and which satisfy $|M_0(t_j)|\geq t_j^{\Theta-\eta}$;

(ii) *The local statement is the specialization to M_0 of Theorem 1, (2.2)–(2.3), via Corollary 1 and Corollary 4 of* [22]. *Theorem 1 states literally that, for every $\varepsilon>0$ and all sufficiently large Y, every interval $[Y^{1-\varepsilon},Y]$ contains at least one sign change of M_0, and that the sign changes in (2.2) are combined with oscillation of size $x^{\Theta-\varepsilon}$. By the definition of "combined oscillation" immediately following (1.1), the alternating witness points satisfy the corresponding magnitude bound. Consequently there are two points of that interval, of opposite signs, at which $|M_0(t)|\geq t^{\Theta-\varepsilon}$. In particular, for arbitrary fixed $\delta>0$ and $\eta>0$, taking $\varepsilon=\min\{\delta,\eta\}$ gives a threshold $Y_0(\delta,\eta)$ such that $[Y^{1-\delta},Y]$ contains points t_+,t_- with $M_0(t_+)\geq t_+^{\Theta-\eta}$ and $M_0(t_-)\leq -t_-^{\Theta-\eta}$.*

Lemma 9.6 (From M_0 to M)**.** *If $|M_0(t)|>2$, then $\operatorname{sgn}M(t)=\operatorname{sgn}M_0(t)$ and $|M(t)|\geq|M_0(t)|-2$. In particular, for every fixed $0<\eta<\Theta$, at points with $|M_0(t)|\geq t^{\Theta-\eta}$ and t large, $\operatorname{sgn}M(t)=\operatorname{sgn}M_0(t)$ and $|M(t)|\geq\frac{1}{2}t^{\Theta-\eta}$.*

Proof. $M_0>2$ gives $M>0$; $M_0<-2$ gives $M<-4$. The second assertion uses $t^{\Theta-\eta}\to\infty$ for $\eta<\Theta$. □

Corollary 9.7 (Distinct large excursions)**.** *If $\Theta>1/2$ and $0<\eta<\Theta-1/2$, then*

$$N_\eta(Y)\geq\left(\frac{\gamma_1}{\pi}-o(1)\right)\log Y, \tag{9.10}$$

where $N_\eta(Y)$ counts distinct excursions starting at most Y that contain a point $t\leq Y$ with $|M(t)|\geq\frac{1}{2}t^{\Theta-\eta}$.

Proof. Take the alternating points of Theorem 9.5(i) in $[1,Y]$, discarding the boundedly many below the threshold of Lemma 9.6. By that lemma, consecutive points $t<t'$ have $M(t)$ and $M(t')$ of opposite signs. Since M is integer valued with steps $0,\pm1$, it vanishes at some integer in (t,t'), so t and t' lie in different excursions. Each point with $M(t)\neq0$ belongs to exactly one excursion, and excursions are ordered, so the excursions containing the ordered points are pairwise distinct. Their starts are below the points, hence at most Y, and $|M(t)|\geq\frac{1}{2}t^{\Theta-\eta}$ at each point. □

Corollary 9.8 (Multiplicative localization of the returns of the Möbius sum)**.** (Return localization, (RL).) *For every fixed $A>1$ there exists U_A such that every excursion starting at $u\geq U_A$ ends at $v\leq u^A$. Equivalently, M has only finitely many A-exceptional excursions; in particular only finitely many square-exceptional ones.*

Proof. Put $\delta=1-1/A$ and $Y=u^A$, so that $[Y^{1-\delta},Y]=[u,u^A]$. Fix once and for all $\eta=\Theta/2$. After this choice the resulting threshold depends only on A; denote it by U_A. Enlarge U_A if necessary so that $t^{\Theta-\eta}>2$ throughout $[u,u^A]$ whenever $u\geq U_A$. Theorem 9.5(ii) and Lemma 9.6 then give $t_+,t_-\in[u,u^A]$ with $M(t_+)>0>M(t_-)$ and $|M_0(t_\pm)|>2$. Neither point can lie in $[u,u+1)$: on that interval $M=M(u)=0$ and hence $M_0=2$. Therefore $u<\lfloor t_\pm\rfloor$. Since M changes by $0,\pm1$ at each integer, it vanishes at some integer n with $\min\{\lfloor t_+\rfloor,\lfloor t_-\rfloor\}<n<\max\{\lfloor t_+\rfloor,\lfloor t_-\rfloor\}$, and $u<n<u^A$. If $u=x_{0,\nu}$, the first return satisfies $x_{1,\nu}\leq n<u^A$. □

Remark 9.9. A bare sign change of $M_0=M+2$ would not force a zero of M. The decisive point is that the local sign change is combined with oscillation of size $t^{\Theta-\eta}\to\infty$, so that Lemma 9.6 applies to both witnesses.

Remark 9.10 (Where (RL) is used). Property (RL) rests on the local statement (2.2)–(2.3) of Theorem 1 of [22], inherited by Corollary 1 and applied to M_0 in Corollary 4: every sufficiently late interval $[Y^{1-\varepsilon}, Y]$ contains a sign change combined with oscillation of size $x^{\Theta-\varepsilon}$. A density of sign changes alone would not bound individual return times. It is used exactly in: the terminal-block step of Corollary 8.7; the lower bound of Corollary 8.13; the converse direction of Theorem 12.3; Corollary 9.30; Proposition 9.32; Remark 7.5; and the statements that power and fractional-square horizons are inactive for M (Remark 9.11). All other results are independent of (RL); in particular the horizon theorem uses only Theorem 9.5(i).

Remark 9.11 (Geometric counts and arithmetic localization). For the excursions of M itself Corollary 9.8 is stronger than Lemma 9.1: each A-exceptional family is finite. Consequently, for M itself the power horizons x_0^A and the fractional-square horizons cx_0^2 are inactive for all late excursions, and the corresponding RH-equivalences (Corollary 9.13, Corollary 9.14(i)) follow already from Corollary 9.8; the horizons with genuine content for M are $x_0\psi(x_0)$ and λx_0. The geometric lemmas remain separate results because they are elementary, independent of ζ, and valid for every ordered family of disjoint intervals. This is exactly what the localization theorem below and the extreme-value theorems of Section 13 use: the former needs only a count below $(\gamma_1/\pi)\log X$, the latter needs a count valid pathwise in a model where no analogue of Kaczorowski–Pintz is available.

9.3. **Horizon localization.**

Theorem 9.12 (Horizon-localized RH criterion)**.** *Let $\mathfrak{h} : [0,\infty) \to [0,\infty]$ be a function, let $\mathcal{F}$ be a family of excursions, and suppose that for some $0 \le \kappa < \gamma_1/\pi$*

$$E_{\mathcal{F}\cup\mathcal{F}_{\mathfrak{h}}}(X) \le (\kappa + o(1))\log X \qquad (X \to \infty). \tag{9.11}$$

Then RH is equivalent to the following statement: for every $\varepsilon > 0$ there is C_ε such that

$$|M(t)| \le C_\varepsilon t^{1/2+\varepsilon} \qquad \text{for all } t \in I_\nu,\ \nu \notin \mathcal{F},\ t \le \mathfrak{h}(x_{0,\nu}). \tag{9.12}$$

No condition is imposed on the excursions in $\mathcal{F}$, nor on the part of any excursion lying beyond its horizon $\mathfrak{h}(x_0)$.

Proof. RH implies (9.12) by Littlewood's criterion. Conversely, suppose $\Theta > 1/2$ and fix $\eta, \varepsilon > 0$ with $1/2 + \varepsilon < \Theta - \eta$. By (9.10) there are at least $(\gamma_1/\pi - o(1))\log X$ distinct excursions starting at most X, each containing a witness t with $|M(t)| \ge \frac{1}{2}t^{\Theta-\eta}$. Consider such an excursion ν with witness t. If $\nu \notin \mathcal{F}$ and $t \le \mathfrak{h}(x_{0,\nu})$, the witness lies in the range of (9.12). Otherwise $\nu \in \mathcal{F}$, or $t > \mathfrak{h}(x_{0,\nu})$, in which case $x_{1,\nu} > t > \mathfrak{h}(x_{0,\nu})$ (as $M(t) \ne 0$) and $\nu \in \mathcal{F}_{\mathfrak{h}}$. By (9.11) at least

$$\Big(\frac{\gamma_1}{\pi} - \kappa - o(1)\Big)\log X \longrightarrow \infty \tag{9.13}$$

witnesses lie in the range of (9.12). They belong to distinct excursions, so they are unbounded, and at them $t^{\Theta-\eta}/t^{1/2+\varepsilon} \to \infty$, contradicting (9.12). □

The case $\mathfrak{h}(x) = \infty$ excludes a sparse family $\mathcal{F}$ of whole excursions; the case $\mathcal{F} = \emptyset$ truncates every excursion at its horizon.

Corollary 9.13 (Square horizon)**.** *For every fixed $A > 1$, RH is equivalent to* (9.12) *restricted to A-ordinary excursions. In particular*

$$\boxed{\text{RH} \iff |M(t)| \ll_\varepsilon t^{1/2+\varepsilon} \text{ for all } t \text{ in excursions with } x_1 \le x_0^2.} \tag{9.14}$$

A separate application of Theorem 9.12, with no excursions discarded and horizon $\mathfrak{h}(x) = x^2$, gives the second RH-equivalent statement

$$|M(t)| \ll_\varepsilon t^{1/2+\varepsilon} \quad (x_{0,\nu} < t \le \min\{x_{1,\nu}, x_{0,\nu}^2\}) \tag{9.15}$$

over all excursions. The two restricted statements are therefore two distinct consequences of the horizon theorem, not an elementary equivalence between the restrictions themselves.

Proof. Take $\mathcal{F} = \mathcal{F}_{x^A}$, $\mathfrak{h} = \infty$, respectively $\mathcal{F} = \emptyset$, $\mathfrak{h}(x) = x^2$, in Theorem 9.12; Lemma 9.1 gives $\kappa = 0$. □

The count $E_\square(X) = O(\log\log X)$ is thus what makes the square horizon admissible. The square horizon can be lowered.

Corollary 9.14 (Descending horizons)**.** *RH is equivalent to Littlewood's bound* $|M(t)| \ll_\varepsilon t^{1/2+\varepsilon}$ *on the ranges*

$$x_{0,\nu} < t \le \min\{x_{1,\nu}, \mathfrak{h}(x_{0,\nu})\} \tag{9.16}$$

over all excursions, for each of the following horizons:

(i) $\mathfrak{h}(x) = cx^2$*, for any fixed* $c > 0$*;*
(ii) $\mathfrak{h}(x) = x\psi(x)$*, for any* $\psi : [0,\infty) \to (0,\infty)$ *with* $\psi(x) \to \infty$*;*
(iii) $\mathfrak{h}(x) = \lambda x$*, for any fixed* $\lambda > \lambda_* := e^{\pi/\gamma_1} = 1.2489\ldots$

Here c*,* ψ *and* λ *do not depend on the excursion. The same holds with "truncate at* $\mathfrak{h}(x_0)$*" replaced by "discard the excursions with* $x_1 > \mathfrak{h}(x_0)$*". In particular it is enough to control the excursions with* $x_1 \le \frac{5}{4}x_0$*.*

Proof. Apply Theorem 9.12 with $\mathcal{F} = \emptyset$ (respectively $\mathcal{F} = \mathcal{F}_\mathfrak{h}$, $\mathfrak{h} = \infty$). For (i), $c \ge 1$ reduces to Lemma 9.1 and $0 < c < 1$ to Lemma 9.2; $\kappa = 0$. For (ii), Lemma 9.3 gives $\kappa = 0$. For (iii), Lemma 9.4 gives $\kappa = 1/\log\lambda$, and $1/\log\lambda < \gamma_1/\pi$ exactly when $\lambda > \lambda_*$. Finally $5/4 > \lambda_*$. □

At the level of the restricted statements, (i) and (ii) imply (iii) eventually, because their horizons are at least λx_0 for every fixed admissible $\lambda > \lambda_*$ once x_0 is large. The converse does not follow from inclusion of ranges; rather, each of (i)–(iii) is separately RH-equivalent by Theorem 9.12. The finitely many small excursions affect only the constant C_ε. The horizon has been lowered from x_0^2 to λx_0: the possible failure of Littlewood's bound, if any, is visible within a bounded multiplicative distance of the start of an excursion. Quantitatively:

Corollary 9.15 (Two-sided localization of large values)**.** *Suppose* $\Theta > 1/2$*, and fix* $0 < \eta < \Theta - \frac{1}{2}$ *and* $\lambda > \lambda_*$ *before letting* $X \to \infty$*. Among the excursions starting at* $x_0 \le X$*, at least*

$$\Big(\frac{\gamma_1}{\pi} - \frac{1}{\log\lambda} - o(1)\Big)\log X \tag{9.17}$$

contain a point t_* *with* $|M(t_*)| \ge \frac{1}{2}t_*^{\Theta-\eta}$ *and*

$$x_0 + \tfrac{1}{2}x_0^{\Theta-\eta} \le t_* \le \lambda x_0. \tag{9.18}$$

Proof. Of the excursions of Corollary 9.7, those whose witness exceeds λx_0 are λ-long, hence at most $\log X/\log\lambda + 2$ by Lemma 9.4. For the others, $M(x_0) = 0$ and unit steps give $|M(t_*)| \le t_* - x_0$, so $t_* - x_0 \ge \frac{1}{2}t_*^{\Theta-\eta} \ge \frac{1}{2}x_0^{\Theta-\eta}$. □

Remark 9.16 (Dyadic erosion and the uniform transition). Fix an integer $j \ge 0$ before letting $X \to \infty$. Lemma 9.2 with $c = 2^{-j}$ gives

$$\#\left\{\nu : x_{0,\nu} \le X,\ x_{1,\nu} > \frac{x_{0,\nu}^2}{2^j}\right\} = O_{2^j}(\log\log X), \tag{9.19}$$

so Theorem 9.12 successively permits the horizons

$$x_0^2, \quad \frac{x_0^2}{2}, \quad \frac{x_0^2}{4}, \quad \frac{x_0^2}{8}, \ldots. \tag{9.20}$$

The constant in the fixed-c estimate depends on c; hence one may not set $j = j(x_0) \sim \log_2 x_0$ by that lemma alone. If instead $j(x) = \lfloor \log_2(x/\psi(x)) \rfloor$ with $\psi(x) \to \infty$, then the corresponding

horizon $x^2/2^{j(x)}$ lies between $x\psi(x)$ and $2x\psi(x)$ and Lemma 9.3 supplies the uniform $o(\log X)$ crossing count. Thus the rigorous erosion chain is

$$x_0^2 \to \frac{x_0^2}{2} \to \frac{x_0^2}{4} \to \cdots \to x_0\psi(x_0) \to \lambda x_0, \qquad \lambda > \lambda_*. \tag{9.21}$$

9.4. **Fixed-mask Catalan compression inside fractional square horizons.** The fractional-square localization has a useful interaction with the exact combinatorics of a completed excursion. The relevant Catalan time is not physical time: zero Möbius increments must first be removed. This support-time distinction is essential below.

Definition 9.17 (Support-time path and fixed-mask Catalan law)**.** Let $I = (x_0, x_1]$ be a completed excursion, and list the nonzero increments in I as

$$x_0 < a_1 < a_2 < \cdots < a_L \le x_1, \qquad \mu(a_j) \neq 0. \tag{9.22}$$

Put

$$r_I(t) := \#\{j : a_j \le t\}, \qquad x_0 < t \le x_1. \tag{9.23}$$

For the excursion I_ν, let $\sigma_\nu \in \{\pm 1\}$ be its orientation and define

$$S_0 = 0, \qquad S_r := \sigma_\nu \sum_{j=1}^{r} \mu(a_j) \qquad (1 \le r \le L). \tag{9.24}$$

Then $L = 2n$ is even and

$$S_0 = S_L = 0, \qquad S_r > 0 \quad (1 \le r < L), \qquad S_r - S_{r-1} \in \{-1, +1\}. \tag{9.25}$$

Thus $(S_r)_{0\le r\le L}$ is a primitive Dyck path. The *fixed-mask primitive Catalan law* keeps the physical support $\{a_1, \dots, a_L\}$ fixed and assigns to those support positions a uniformly chosen primitive Dyck sign sequence of length L. We denote its probability by $\mathbb{P}_{\mathrm{pCat},I}$.

For the observed excursion one has, exactly,

$$|M(t)| = S_{r_I(t)}, \qquad r_I(t) \le t - x_0 \le t, \tag{9.26}$$

with $r_I(t) < t$ whenever $x_0 \ge 1$, i.e. for every excursion other than $(0, 2]$. The first equality uses the orientation of the excursion and the fact that M is constant across the zero increments.

We first record the global Catalan estimate, since it is useful in its own right and clarifies why a different estimate is needed at the square horizon.

Lemma 9.18 (Catalan maximum tail, classical)**.** *Let $L = 2n$ and let $H_{\max}$ be the maximum of a uniformly chosen Dyck path of length $2n$. For $1 \le h \le n$,*

$$\mathbb{P}_{\mathrm{Cat},L}\{H_{\max} \ge h\} \le (n+1)\frac{\binom{2n}{n+h}}{\binom{2n}{n}} \le (n+1)\exp\left(-\frac{h^2}{n+h}\right) \le \left(\frac{L}{2}+1\right)e^{-h^2/L}. \tag{9.27}$$

For a uniformly chosen primitive Dyck path of length $L = 2n$,

$$\mathbb{P}_{\mathrm{pCat},L}\{H_{\max} \ge h\} \le \frac{L}{2}\exp\left(-\frac{(h-1)^2}{L-2}\right), \qquad 2 \le h \le n. \tag{9.28}$$

Proof. There are $\mathrm{Cat}_n = (n+1)^{-1}\binom{2n}{n}$ Dyck paths of length $2n$ and $\binom{2n}{n}$ unrestricted bridges. Reflecting a bridge after its first visit to level h gives a bijection between bridges that reach h and unrestricted walks ending at $2h$, hence their number is $\binom{2n}{n+h}$. Therefore

$$\mathbb{P}_{\mathrm{Cat},2n}\{H_{\max} \ge h\} \le (n+1)\frac{\binom{2n}{n+h}}{\binom{2n}{n}}. \tag{9.29}$$

Moreover

$$\frac{\binom{2n}{n+h}}{\binom{2n}{n}} = \prod_{j=1}^{h} \frac{n-j+1}{n+j}, \tag{9.30}$$

and $\log(1-x) \le -x$ gives

$$\log \frac{\binom{2n}{n+h}}{\binom{2n}{n}} \le -\sum_{j=1}^{h} \frac{2j-1}{n+j} \le -\frac{h^2}{n+h}. \tag{9.31}$$

Since $n+h \le 2n = L$, (9.27) follows. A primitive path of length $2n$ has the form $+1$, followed by a Dyck path of length $2n-2$, followed by -1; its maximum is one plus the maximum of the interior Dyck path. Applying the preceding bound with $n-1$ and $h-1$ proves (9.28). □

Remark 9.19 (Why the maximum tail is not the square-horizon estimate). The exponent in (9.28) is of order H^2/L. For a general family of disjoint intervals retained by the square horizon one may have $L \asymp x_0^2$ (for M itself such excursions are finitely many by Corollary 9.8), while the relevant witness lies much earlier, at physical time $t \asymp x_0$. Then a violation of size $t^{1/2+\varepsilon}$ need not make H^2/L large. Thus the global maximum tail, although correct, is not the right Catalan quantity for combining with the fractional-square horizon. The correct quantity is the pointwise height at the support time $r_I(t)$.

Lemma 9.20 (Pointwise primitive-Catalan tail)**.** *Let $L = 2n \ge 4$ and let (S_r) be a uniformly chosen primitive Dyck path of length L. For $1 \le r \le L-1$ and real $h \ge 1$,*

$$\mathbb{P}_{\mathrm{pCat},L}\{S_r \ge h\} \le 8n^{3/2} \exp\left(-\frac{h^2}{2r}\right) \le 3L^{3/2} \exp\left(-\frac{h^2}{2r}\right). \tag{9.32}$$

By time reversal the same estimate holds with r replaced by $L-r$; hence

$$\mathbb{P}_{\mathrm{pCat},L}\{S_r \ge h\} \le 3L^{3/2} \exp\left[-\frac{h^2}{2\min\{r, L-r\}}\right]. \tag{9.33}$$

If $h > \min\{r, L-r\}$ the probability is 0, so the bound is only relevant for $h \le \min\{r, L-r\}$.

Proof. Ignore both positivity and the endpoint constraint. Among all 2^L sign words, the proportion whose first r signs have sum at least h is at most $\exp(-h^2/(2r))$ by the elementary Chernoff bound for a Rademacher sum. Therefore the number of primitive Dyck paths satisfying $S_r \ge h$ is at most

$$2^L \exp\left(-\frac{h^2}{2r}\right). \tag{9.34}$$

The total number of primitive Dyck paths of length $2n$ is Cat_{n-1}. The standard central-binomial lower bound $\binom{2m}{m} \ge 4^m/(2\sqrt{m})$ gives

$$\mathrm{Cat}_{n-1} = \frac{1}{n}\binom{2n-2}{n-1} \ge \frac{4^{n-1}}{2n\sqrt{n-1}}. \tag{9.35}$$

Since $2^L = 4^n$, division gives the first inequality in (9.32); the second follows from $8n^{3/2} < 3(2n)^{3/2}$. Reversing the path and changing the signs of the reversed increments preserves the primitive Dyck class and carries height S_r to height S_{L-r}, which proves (9.33). □

For an observed excursion I define its pointwise Catalan reference-tail weight by

$$p_{\mathrm{Cat}}(I,t) := \mathbb{P}_{\mathrm{pCat},I}\left\{S_{r_I(t)} \ge |M(t)|\right\}. \tag{9.36}$$

This is a reference-tail probability under the fixed-mask Catalan law; it is not a p-value for a stochastic model of the arithmetic Möbius path.

Corollary 9.21 (Fractional-square Catalan rarity)**.** *Fix $c > 0$ and $\varepsilon > 0$. Let $I = (x_0, x_1]$ be a completed excursion with $x_1 \le cx_0^2$, and keep its zero mask fixed. For all sufficiently large x_0,*

$$\mathbb{P}_{\mathrm{pCat},I}\left\{\exists\, t \in I \cap \mathbb{Z} :\ S_{r_I(t)} \ge \lceil t^{1/2+\varepsilon} \rceil\right\} \le 3c^{5/2} x_0^5 \exp\left(-\frac{1}{2} x_0^{2\varepsilon}\right). \tag{9.37}$$

In particular the right-hand side is smaller than x_0^{-A} for every fixed $A > 0$ once x_0 is large enough.

Proof. If $L = L^{\#}(I) = 2$ then $S_r \le 1$ and the event is empty for $x_0 \ge 1$; assume $L \ge 4$. At a support position $t = a_r$ one has $r \le t - x_0 < t$. Hence Lemma 9.20 with $h = \lceil t^{1/2+\varepsilon} \rceil$ gives

$$\mathbb{P}_{\mathrm{pCat},I}\{S_r \ge t^{1/2+\varepsilon}\} \le 3L^{3/2} \exp\left(-\frac{1}{2} t^{2\varepsilon}\right) \le 3L^{3/2} \exp\left(-\frac{1}{2} x_0^{2\varepsilon}\right). \tag{9.38}$$

There are at most L support positions. Since $L \le x_1 - x_0 \le x_1 \le cx_0^2$, one has $3L \cdot L^{3/2} = 3L^{5/2} \le 3(cx_0^2)^{5/2} = 3c^{5/2} x_0^5$, and the union bound gives (9.37). Between support positions the path is constant, so no additional physical times have to be counted. □

The preceding statement is purely combinatorial: it measures how far a large value lies in the tail of a reference law, not how the Möbius constraints select a path. Combined with the localization theorem it gives the following quantitative form.

Corollary 9.22 (Catalan cost of a Kaczorowski–Pintz witness under $\Theta > 1/2$)**.** *Suppose $\Theta > 1/2$, fix $0 < \eta < \Theta - \frac{1}{2}$, put $\delta = \Theta - \eta - \frac{1}{2}$, and fix $\lambda > \lambda_*$. Then there are infinitely many excursions $I = (x_0, x_1]$ with $x_1 \le \lambda x_0$ containing a point t with*

$$-\log p_{\mathrm{Cat}}(I, t) \ \ge\ \frac{1}{8} x_0^{2\delta} - O(\log x_0). \tag{9.39}$$

The same holds with $x_1 \le \lambda x_0$ replaced by $x_1 \le cx_0^2$ for any fixed $c > 0$.

Proof. By Corollary 9.7 and Lemma 9.4, at least $(\gamma_1/\pi - 1/\log\lambda - o(1)) \log X$ excursions starting at most X have $x_1 \le \lambda x_0$ and contain t with $h = |M(t)| \ge \frac{1}{2} t^{1/2+\delta}$; for cx_0^2 use Lemma 9.2 instead. With $r = r_I(t) < t$ and $L \le x_1 - x_0$, Lemma 9.20 gives $p_{\mathrm{Cat}}(I, t) \le 3L^{3/2} \exp(-h^2/(2r)) \le 3L^{3/2} \exp(-t^{2\delta}/8)$, and $t > x_0$, $L \le \lambda x_0$ (respectively $L \le cx_0^2$). □

Remark 9.23 (Selector bounds are one-sided). Lemma 9.20 gives only the upper-tail estimate

$$p_{\mathrm{Cat}}(I, t) \le 3L^{3/2} \exp\left(-\frac{|M(t)|^2}{2r_I(t)}\right), \tag{9.40}$$

and hence the one-sided implication

$$-\log p_{\mathrm{Cat}}(I, t) \ge \frac{|M(t)|^2}{2r_I(t)} - \log(3L^{3/2}). \tag{9.41}$$

No reverse inequality, and therefore no equivalence "up to polynomial factors", is proved here. Consequently a selector bound $-\log p_{\mathrm{Cat}}(I, t) \le \Phi(x_0)$ with $\Phi \ge 0$, $\Phi(x) = x^{o(1)}$ on excursions with $x_1 \le \lambda x_0$ (so that $\Phi(x) + O(\log x) \le x^{2\varepsilon}$ eventually, for every $\varepsilon > 0$) implies $|M(t)| \le t^{1/2} x_0^{o(1)}$ there, so RH follows from Corollary 9.14; this is a sufficient condition, not an independent equivalent formulation. A polynomial lower bound $p_{\mathrm{Cat}} \ge x_0^{-A}$ would similarly give $|M(t)| \ll \sqrt{t \log t}$ on the retained ranges. To turn the Catalan weight into a two-sided measure of deviation one would need a matching lower-tail estimate of the form $-\log p_{\mathrm{Cat}}(I, t) \ll |M(t)|^2/r_I(t) + \log L$, which is not proved here.

Corollary 9.24 (Catalan cost along a Littlewood-violating sequence)**.** *Let $t_j \to \infty$, $A_j \to \infty$ and $\varepsilon_0 > 0$ with $|M(t_j)| \ge A_j t_j^{1/2+\varepsilon_0}$, and let $I_j = (x_{0,j}, x_{1,j}]$ be the excursion containing t_j. If $x_{1,j} \le x_{0,j}^B$ for a fixed B, then*

$$-\log p_{\mathrm{Cat}}(I_j, t_j) \;\ge\; \frac{A_j^2}{2}\, t_j^{2\varepsilon_0} - O_B(\log x_{0,j}) \;\ge\; \frac{A_j^2}{2}\, x_{0,j}^{2\varepsilon_0} - O_B(\log x_{0,j}). \tag{9.42}$$

Proof. With $r = r_{I_j}(t_j) < t_j$ and $h = |M(t_j)|$, $h^2/(2r) \ge A_j^2 t_j^{1+2\varepsilon_0}/(2t_j)$; apply Lemma 9.20 with $L \le x_{1,j} \le x_{0,j}^B$. □

9.4.1. *Selection energies: completed excursions and open prefixes.* The support-time clock of Definition 9.17 is the squarefree-compressed clock

$$\mathcal{S}(x) = \#\{n \le x : \mu(n) \ne 0\}, \qquad L^{\#}(I) = \mathcal{S}(x_1) - \mathcal{S}(x_0) \le x_1 - x_0, \tag{9.43}$$

and $r_I(t) = \mathcal{S}(t) - \mathcal{S}(x_0)$. Two tail quantities are natural: one for the height of a completed excursion, and one for an open prefix, which does not require knowledge of the future return.

Definition 9.25 (Selection energies)**.** For a completed excursion I with height $H(I)$ and compressed length $L^{\#} = L^{\#}(I)$, put

$$p_{\mathrm{Cat}}^{\max}(I) := \mathbb{P}_{\mathrm{pCat},L^{\#}}\{H_{\max} \ge H(I)\}, \qquad \mathcal{I}_{\mathrm{Cat}}(I) := -\log p_{\mathrm{Cat}}^{\max}(I). \tag{9.44}$$

For $t \ge 1$ with $M(t) \ne 0$ let $a(t)$ be the last zero of M before t and $r = \mathcal{S}(t) - \mathcal{S}(a(t))$. After orientation, the compressed segment from $a(t)$ to t is a positive meander of length r: a ± 1 path $(S_k)_{k \le r}$ with $S_0 = 0$ and $S_k > 0$ for $1 \le k \le r$. Let $\mathcal{M}_r$ be the set of such paths. Since $S_r \equiv r \pmod 2$, an endpoint event with incompatible parity has probability zero. Define

$$p_{\mathrm{open}}(t) := \mathbb{P}_{\mathcal{M}_r}\{S_r \ge |M(t)|\}, \tag{9.45}$$

the uniform measure on $\mathcal{M}_r$.

Lemma 9.26 (Height–entropy inequalities)**.** (i) *If $L^{\#} = 2$, the compressed path is $0, 1, 0$ and $H(I) = 1$. For every completed excursion with $L^{\#} \ge 4$,*

$$\mathcal{I}_{\mathrm{Cat}}(I) \;\ge\; \frac{(H(I)-1)^2}{L^{\#}-2} - \log\frac{L^{\#}}{2}. \tag{9.46}$$

(ii) *$\#\mathcal{M}_r = \binom{r-1}{\lfloor (r-1)/2 \rfloor} \ge 2^r/(4\sqrt{r})$, and for $h \ge 1$*

$$\mathbb{P}_{\mathcal{M}_r}\{S_r \ge h\} \le 4\sqrt{r}\, e^{-h^2/(2r)}, \qquad -\log p_{\mathrm{open}}(t) \;\ge\; \frac{M(t)^2}{2r} - \log(4\sqrt{r}). \tag{9.47}$$

Proof. (i) is (9.28). (ii) The count of positive meanders is classical (ballot theorem; the number ending at s is $\frac{s}{r}\binom{r}{(r+s)/2}$). From $\binom{2m}{m} \ge 4^m/(2\sqrt{m})$ and $\binom{2m+1}{m} = \frac{1}{2}\binom{2m+2}{m+1}$ one gets $\#\mathcal{M}_r \ge 2^r/(4\sqrt{r})$. Among all 2^r words the proportion with $S_r \ge h$ is at most $e^{-h^2/(2r)}$ by the Chernoff bound; divide. □

Proposition 9.27 (Selector criteria)**.** (i) Completed excursions. *Fix $\lambda > \lambda_*$. If there is A such that $p_{\mathrm{Cat}}^{\max}(I) \ge x_0^{-A}$ for every sufficiently large completed excursion with $x_1 \le \lambda x_0$, then $H(I) \ll_{A,\lambda} \sqrt{x_0 \log x_0}$ on that family, and RH holds.*

(ii) Open prefixes. *If there is A such that $p_{\mathrm{open}}(t) \ge t^{-A}$ for all large t with $M(t) \ne 0$, then $|M(t)| \ll_A \sqrt{t \log t}$, and RH holds.*

Proof. (i) If $L^{\#} = 2$ then $H(I) = 1$. Otherwise $L^{\#} \ge 4$ and, by Lemma 9.26(i), $(H-1)^2 \le (L^{\#}-2)(A \log x_0 + \log L^{\#})$, and $L^{\#} \le (\lambda - 1)x_0$. Littlewood's bound therefore holds on every excursion with $x_1 \le \lambda x_0$, and Corollary 9.14(iii), in the form that discards the λ-long excursions, gives RH. (ii) By Lemma 9.26(ii), $M(t)^2 \le 2r(A \log t + \log(4\sqrt{r}))$ with $r \le t$. □

Remark 9.28 (The threshold λ_*). The function $\cos(\gamma_1 \log x)$ changes sign γ_1/π times per unit of $\log x$, i.e. once in every multiplicative interval of ratio $e^{\pi/\gamma_1} = \lambda_*$; the constant in (9.9) is this density. On the geometric side, a chain of λ-long intervals realizes the bound of Lemma 9.4. The method of Theorem 9.12 compares these two counts, and λ_* is the exact balance point, hence the sharp threshold of this counting mechanism: for $\lambda \le \lambda_*$ the λ-long family can, as far as counting alone is concerned, absorb every Kaczorowski–Pintz excursion. Going below λ_* requires information on where inside an excursion the large values occur, not only how many excursions carry them.

Proposition 9.29 (Logarithmic-length budget)**.** *Let $\mathcal{F}$ be a family of excursions and let $u_1 < \cdots < u_N$ be the starts of its members with $2 \le u_j \le X$, with ends v_j. The initial excursion $(0, 2]$ is explicitly excluded, since $\log(v/u)$ is not defined at $u = 0$. Then*

$$\sum_{j=1}^{N-1} \log \frac{v_j}{u_j} \le \log \frac{X}{2}. \tag{9.48}$$

If moreover $v_N \le u_N^A$ for some $A \ge 1$, then $\sum_{j=1}^{N} \log(v_j/u_j) \le A \log X$.

Proof. Disjointness gives $u_{j+1} \ge v_j$, so $\log(v_j/u_j) \le \log u_{j+1} - \log u_j$; the sum telescopes to at most $\log u_N - \log u_1 \le \log(X/2)$. The last term is at most $(A-1)\log u_N \le (A-1)\log X$. □

Lemmas 9.1–9.4 are the special cases in which every term is bounded below. The budget form shows what the horizon theorem actually measures.

Corollary 9.30 (Mean logarithmic length of Kaczorowski–Pintz excursions)**.** *Suppose $\Theta > 1/2$, fix $0 < \eta < \Theta - \frac{1}{2}$, and let $\mathcal{K}(X)$ be any set of distinct excursions starting at most X, each containing a point t with $|M(t)| \ge \frac{1}{2} t^{\Theta-\eta}$, such that $\#\mathcal{K}(X) \ge (\gamma_1/\pi - o(1)) \log X$ (such sets exist by Corollary 9.7). Then*

$$\frac{1}{\#\mathcal{K}(X)} \sum_{\nu \in \mathcal{K}(X)} \log \frac{v_\nu}{u_\nu} \;\le\; \frac{\pi}{\gamma_1} + o(1) = \log \lambda_* + o(1). \tag{9.49}$$

Proof. Discard the initial excursion $(0, 2]$, whose start is 0 and whose contribution to the count is $O(1)$. Fix $A > 1$ before taking $X \to \infty$. By Corollary 9.8, all but finitely many excursions satisfy $v \le u^A$. These exceptions form a fixed global finite set once A is fixed, so the sum of their individual finite values $\log(v/u)$ is $O_A(1)$. Proposition 9.29 gives a total of at most $A \log X + O_A(1)$. Since $\#\mathcal{K}(X) \ge (\gamma_1/\pi - o(1)) \log X$, it follows for every fixed $A > 1$ that

$$\limsup_{X \to \infty} \frac{1}{\#\mathcal{K}(X)} \sum_{\nu \in \mathcal{K}(X)} \log \frac{v_\nu}{u_\nu} \le \frac{A\pi}{\gamma_1}. \tag{9.50}$$

Only after this limit is taken do we let $A \downarrow 1$, obtaining π/γ_1. □

Remark 9.31 (What lowering λ_* requires). Corollary 9.14(iii) is the statement that the excursions carrying Kaczorowski–Pintz witnesses cannot all have $\log(v/u) > \log \lambda$ with $\lambda > \lambda_*$; Corollary 9.30 is its averaged form. Consequently, any proof that such excursions have mean logarithmic length at least $\pi/\gamma_1 + \delta$ for some fixed $\delta > 0$ would contradict Corollary 9.30 and prove RH, and a surplus $\delta > \log(\lambda_*/\lambda)$ among the excursions whose witness lies beyond λx_0 would extend Corollary 9.14(iii) to that $\lambda < \lambda_*$. Kaczorowski–Pintz, as stated, gives neither: it supplies at least one return in each band $[Y^{1-\delta}, Y]$ but no upper separation between consecutive witnesses. Nor do short-interval estimates for μ supply it: $|M(x+h) - M(x)| = o(h)$ is compatible with M staying positive throughout $(x, x+h]$, whereas a return requires a decrement of size at least $M(x)$ in a prescribed direction. Energy bounds do not help either, since a flat excursion of height 1 has $\mathfrak{b}_I, Q_I \asymp \ell_I$.

9.5. **A deductive audit below the counting threshold.** The preceding counting argument identifies the exact information that would be sufficient to push the multiplicative horizon below λ_*. We make that requirement quantitative and record a matching no-go statement showing that it does not follow from the geometric inputs alone.

Put

$$d_1 := \frac{\gamma_1}{\pi}, \qquad a_* := \frac{1}{d_1} = \frac{\pi}{\gamma_1}, \qquad \lambda_* = e^{a_*}. \tag{9.51}$$

Assume $\Theta > 1/2$, fix $0 < \eta < \Theta - \frac{1}{2}$, and let $\mathcal{K}(X)$ be a Kaczorowski–Pintz witness family as in Corollary 9.30. For each $\nu \in \mathcal{K}(X)$ choose one witness t_ν in the corresponding excursion $(u_\nu, v_\nu]$.

Fix $1 < \lambda < \lambda_*$ and define the bad subfamily

$$\mathcal{B}_\lambda(X) := \{\nu \in \mathcal{K}(X) : t_\nu > \lambda u_\nu\}. \tag{9.52}$$

Proposition 9.32 (Positional surplus criterion)**.** *Suppose there is a fixed $\delta > \log(\lambda_*/\lambda)$ such that*

$$\frac{1}{\#\mathcal{B}_\lambda(X)} \sum_{\nu \in \mathcal{B}_\lambda(X)} \left(\log \frac{v_\nu}{u_\nu} - \log \lambda \right) \geq \delta - o(1) \tag{9.53}$$

whenever $\mathcal{B}_\lambda(X)$ is nonempty. Then

$$\#\mathcal{B}_\lambda(X) \leq \left(\frac{1}{\log \lambda + \delta} + o(1) \right) \log X < (d_1 - o(1)) \log X. \tag{9.54}$$

Consequently $\gg \log X$ witnesses satisfy $t_\nu \leq \lambda u_\nu$, and the horizon argument gives the same contradiction under $\Theta > 1/2$ as in Theorem 9.12.

Proof. Fix first an exponent $A > 1$. Return localization and Proposition 9.29 give

$$\sum_{\nu \in \mathcal{K}(X)} \log \frac{v_\nu}{u_\nu} \leq A \log X + O_A(1). \tag{9.55}$$

After taking $X \to \infty$ we may let $A \downarrow 1$, exactly as in the proof of Corollary 9.30; hence the total logarithmic budget is $(1 + o(1)) \log X$. By (9.53), every bad member consumes on average at least $\log \lambda + \delta - o(1)$ of this budget. Therefore

$$(\log \lambda + \delta - o(1)) \#\mathcal{B}_\lambda(X) \leq (1 + o(1)) \log X. \tag{9.56}$$

Since $\log \lambda + \delta > \log \lambda_* = a_* = 1/d_1$, the resulting coefficient is strictly smaller than d_1. Subtracting the bad excursions from $\#\mathcal{K}(X) \geq (d_1 - o(1)) \log X$ leaves a positive proportion, on the logarithmic scale, of witnesses covered by the shorter horizon. □

The surplus has the exact decomposition

$$\log \frac{v_\nu}{u_\nu} - \log \lambda = \log \frac{t_\nu}{\lambda u_\nu} + \log \frac{v_\nu}{t_\nu}. \tag{9.57}$$

Thus the existence of a few long excursions is irrelevant: the required surplus must hold uniformly on average over the bad witness subfamily.

Unit step size alone does not provide such a fixed surplus. If a witness beyond the horizon has height $H \asymp t^\alpha$ with $\alpha < 1$, and even if one knows $M(\lambda u) = o(H)$ in the relevant orientation, then the step bound gives only

$$t - \lambda u \geq H - o(H), \qquad v - t \geq H, \tag{9.58}$$

and consequently

$$\frac{v}{u} \geq \lambda + (2 + o(1)) \frac{H}{u} = \lambda + o(1). \tag{9.59}$$

The gain is additive and sublinear, whereas Proposition 9.32 requires a fixed logarithmic gain.

Proposition 9.33 (Geometric inputs do not force positional surplus)**.** *Disjointness of the excursions, increments in $\{-1,0,1\}$, a localized return, a witness count $(d_1 - o(1))\log X$, and a mean logarithmic length bounded by $a_* + o(1)$ do not imply that a positive proportion of witnesses lies before λu for any fixed $\lambda < \lambda_*$. This remains true even if the witness height is $\asymp t^\alpha$ for a fixed $1/2 < \alpha < 1$ and every excursion is eventually shorter than u^A for each fixed $A > 1$.*

Proof. Choose

$$1 < \lambda < c < r < \lambda_*, \qquad \frac{1}{2} < \alpha < 1, \tag{9.60}$$

and, starting from a sufficiently large integer u_1, define recursively

$$u_{j+1} = \lceil r u_j \rceil, \qquad v_j = u_{j+1}, \qquad t_j = \lfloor c u_j \rfloor. \tag{9.61}$$

For all sufficiently large j, one has $u_j < t_j < v_j$. Since both gaps $t_j - u_j$ and $v_j - t_j$ are linear in u_j, whereas $H_j := \lfloor t_j^\alpha/2 \rfloor = o(u_j)$, one can construct on $(u_j, v_j]$ a primitive bridge with increments in $\{-1,0,1\}$ that rises to height H_j by time t_j, stays strictly away from zero while inserting horizontal steps, and returns to zero at v_j.

Moreover,

$$\#\{j : u_j \le X\} = \frac{\log X}{\log r} + O(1) > d_1 \log X + O(1), \tag{9.62}$$

while

$$\frac{1}{J}\sum_{j \le J} \log \frac{v_j}{u_j} \longrightarrow \log r < a_*, \qquad t_j > \lambda u_j. \tag{9.63}$$

Finally $v_j \asymp r u_j$, so $v_j \le u_j^A$ for every fixed $A > 1$ and all sufficiently large j. This is not a construction of the Möbius sum; it is a countermodel to any deduction using only the listed geometric and counting inputs. □

First-arrival record blocks and an exact positional cost. The preceding no-go statement isolates the missing arithmetic information but still treats the position of a witness only through its total excursion length. There is a second exact decomposition, internal to a first arrival, that keeps track of the successive record levels.

Definition 9.34 (First-arrival record decomposition)**.** Let $I = (u, v]$ be a completed excursion of orientation $\sigma \in \{-1, 1\}$, and let $t_H = u + h < v$ be the first point at which

$$S_r := \sigma M(u + r), \qquad S_0 = 0, \tag{9.64}$$

reaches the integer height $H \ge 1$. Thus $S_h = H$ and $S_r < H$ for $0 \le r < h$. Put

$$\tau_j := \min\{r : S_r = j\} \quad (0 \le j \le H), \qquad L_j := \tau_{j+1} - \tau_j, \tag{9.65}$$

and, for $0 \le r \le L_j$,

$$T_{j,r} := S_{\tau_j + r} - j, \qquad d_{j,r} := -T_{j,r}. \tag{9.66}$$

Then $T_{j,0} = 0$, $T_{j,L_j} = 1$, and $T_{j,r} \le 0$ for $0 < r < L_j$. Define the record-return cost

$$\mathcal{V}(u, H) := \sum_{j=0}^{H-1} \sum_{r < L_j} d_{j,r}(d_{j,r} + 1). \tag{9.67}$$

Proposition 9.35 (Record-return decomposition and depth rigidity)**.** *With the notation above, if $R_r := \max_{0 \le s \le r} S_s$, then*

$$\mathcal{V}(u, H) = \sum_{r=1}^{h-1} (R_r - S_r)(R_r - S_r + 1). \tag{9.68}$$

Moreover, writing $m_j := L_j - 1$, the path $T_{j,0}, \dots, T_{j,m_j}$ is a completed nonpositive bridge at level j, and the last step $m_j \to L_j$ creates the next record. If

$$A_j := \sum_{r=1}^{m_j-1} d_{j,r}, \qquad Q_j := \sum_{r=1}^{m_j-1} d_{j,r}^2, \tag{9.69}$$

then

$$\mathcal{V} = \sum_{j=0}^{H-1} (Q_j + A_j). \tag{9.70}$$

If $B_j := \max_{0\le r\le m_j} d_{j,r}$, then

$$\mathcal{V}_j := Q_j + A_j \ge \frac{B_j(B_j+1)(2B_j+1)}{3}, \qquad \sum_{j<H} B_j^3 \le \frac{3}{2}\mathcal{V}. \tag{9.71}$$

Equality in the first inequality is attained by the triangular descent to B_j and return to 0, with arbitrary additional waits at depth 0 but no positive-depth waits or extra oscillations.

Proof. Lemma 5.1, applied to the increment block from τ_j to τ_{j+1}, gives

$$\sum_{r<L_j} T_{j,r}^2 + \sum_{g<L_j} g C_g^{[j]} = \sum_{r<L_j} T_{j,r}. \tag{9.72}$$

Hence $-\sum_{g<L_j} g C_g^{[j]} = \sum_{r<L_j} d_{j,r}(d_{j,r}+1)$, which is (9.67). On $\tau_j \le r < \tau_{j+1}$ one has $R_r = j$, which gives (9.68). Since the first arrival at $j+1$ must end with a $+1$ step from j, one has $T_{j,L_j-1} = 0$; this proves the bridge decomposition and $\mathcal{V}_j = Q_j + A_j$. A bridge reaching depth B_j and returning to zero by unit steps must visit $1, 2, \dots, B_j - 1, B_j, B_j - 1, \dots, 1$. Therefore

$$\mathcal{V}_j \ge B_j(B_j+1) + 2\sum_{d=1}^{B_j-1} d(d+1) = \frac{B_j(B_j+1)(2B_j+1)}{3}. \tag{9.73}$$

The last assertion and the cubic sum follow immediately. □

For the positional identity below, the zero runs must be truncated at the first-arrival endpoint. For $u < n \le u + h$ with $\mu(n) \ne 0$, let

$$g_h(n) := \max\{g \ge 0 : n + g \le u + h,\ \mu(n+1) = \cdots = \mu(n+g) = 0\}, \tag{9.74}$$

and let g_u be the initial zero-run length after u, truncated in the same way. Put

$$\Gamma_h := \sum_{u<n\le u+h} \mu(n) g_h(n). \tag{9.75}$$

For $1 \le r \le h$ write $x_r := S_r - S_{r-1} = \sigma\mu(u+r)$ and $d_r := R_r - S_r$, and partition the indices into

$$\begin{aligned} \mathcal{A} &:= \{r : x_r = 1,\ d_r = 0\}, \quad \mathcal{B} := \{r : x_r = 1,\ d_r \ge 1\}, \\ \mathcal{C} &:= \{r : x_r = -1\}, \qquad\qquad \mathcal{D} := \{r : x_r = 0,\ d_r = 0\}, \\ \mathcal{E} &:= \{r : x_r = 0,\ d_r \ge 1\}. \end{aligned} \tag{9.76}$$

Define the nonnegative excess

$$E'''(u,H) := \sum_{r\in\mathcal{C}\cup\mathcal{E}} \big(d_r(d_r+1) - 2\big) + \sum_{r\in\mathcal{B}} d_r(d_r+1) + 2\sum_{r\in\mathcal{B}} g_h(u+r). \tag{9.77}$$

Proposition 9.36 (Exact first-arrival position identity)**.** *With the preceding notation,*

$$\boxed{h = \mathcal{V}(u,H) + H + \sigma\Gamma_h + g_u - E'''(u,H).} \tag{9.78}$$

In particular $E'''(u,H) \ge 0$.

Proof. Let $D := \#\mathcal{C}$. The nonzero increments have final displacement H, so their number is $H+2D$. Every zero in the prefix belongs either to the initial run after u or to the run following a nonzero increment; because g_h is truncated at $u+h$, the last record step contributes no zero beyond the prefix. Hence

$$h = H + 2D + \sum_{r\in\mathcal{A}\cup\mathcal{B}\cup\mathcal{C}} g_h(u+r) + g_u. \tag{9.79}$$

Also

$$\sigma\Gamma_h = \sum_{r\in\mathcal{A}} g_h(u+r) + \sum_{r\in\mathcal{B}} g_h(u+r) - \sum_{r\in\mathcal{C}} g_h(u+r). \tag{9.80}$$

Finally, grouping the positive-depth zero steps according to the preceding nonzero increment gives exactly

$$\mathcal{V} = 2D + 2\sum_{r\in\mathcal{C}} g_h(u+r) + E'''. \tag{9.81}$$

Combining the last three identities proves (9.78). Nonnegativity follows term by term from $d_r \geq 1$ on $\mathcal{B}\cup\mathcal{C}\cup\mathcal{E}$. □

Lemma 9.37 (Uniform cancellation of the truncated zero-run correlation)**.** *Fix $\iota > 0$. Uniformly for*

$$u^{3/5+\iota} \leq h \leq u, \tag{9.82}$$

one has

$$\Gamma_h = o(h) \qquad (u\to\infty). \tag{9.83}$$

Proof. Fix $0 < A < 1$ and choose $0 < c < A$. Put $P = c\log\log u$, $Q := \prod_{p\leq P} p^2$ and $K := 2\pi(P)+1$. Let $N_P(m)$ be the indicator that $p^2 \mid m$ for at least one prime $p \leq P$, and let $g_P(n)$ be the length of the consecutive $N_P = 1$ run following n. The function g_P is periodic modulo Q. Moreover no K consecutive integers are all covered by these prime squares: the number covered in such a block is at most

$$K\sum_{p\leq P}\frac{1}{p^2} + \pi(P) \leq K\left(\frac{\pi^2}{8} - \frac{3}{4}\right) + \pi(P) < K. \tag{9.84}$$

Thus $0 \leq g_P \leq K$.

The difference between the genuine zero runs and the small-square runs is charged to integers divisible by p^2 with $p > P$. Splitting an excess run into pieces of length K gives, uniformly in the stated range,

$$\sum_{u<n\leq u+h} |g_h(n) - g_P(n)| \ll K\left(\frac{h}{P} + \sqrt{u}\right) + K^2 = o(h). \tag{9.85}$$

Here the K^2 term accounts for truncation at $u+h$.

Write the finite Fourier expansion $g_P(n) = \sum_{a \bmod Q} \widehat{g}_P(a)e(an/Q)$ with normalized coefficients. Parseval and Cauchy–Schwarz give $\sum_a |\widehat{g}_P(a)| \leq K\sqrt{Q}$. Therefore (12.23) yields

$$\left|\sum_{u<n\leq u+h} \mu(n)g_P(n)\right| \ll_{A,\iota} K\sqrt{Q}\,\frac{h}{(\log u)^A}. \tag{9.86}$$

By the prime number theorem, $\sqrt{Q} = \exp(\vartheta(P)) = (\log u)^{c+o(1)}$, while $K = (\log\log u)^{1+o(1)}$. Since $c < A$, this is $o(h)$; together with the preceding comparison it proves (9.83). □

Corollary 9.38 (First-arrival cost below the counting threshold)**.** *Assume $\Theta > 1/2$, fix $0 < \eta < \Theta - 1/2$, put $\alpha := \Theta - \eta$, and fix*

$$\lambda_* < \Lambda < 2. \tag{9.87}$$

For the Kaczorowski–Pintz excursions supplied by Corollary 9.15, set $H_u := \lfloor u^\alpha/2 \rfloor$ and let $t_H = u+h$ be the first arrival at height H_u in the orientation of the excursion. Then $H_u = o(u)$, $t_H \le \Lambda u$, and

$$\frac{t_H}{u} \le 1 + \frac{\mathcal{V}(u, H_u)}{u} + o(1). \tag{9.88}$$

Consequently, if for some fixed $\varepsilon_* > 0$ an unbounded sequence of these prefixes satisfies

$$\mathcal{V}(u, H_u) \le (\lambda_* - 1 - \varepsilon_*)u, \tag{9.89}$$

then

$$t_H \le (\lambda_* - \varepsilon_* + o(1))u. \tag{9.90}$$

Thus, combined with Littlewood's bound on a linear excursion window, such a cost estimate would lower the sufficient localized horizon below λ_*; the cost estimate alone is not a proof of RH.

Proof. Corollary 9.15 supplies an unbounded family of excursions with a point $t_* \le \Lambda u$ and $|M(t_*)| \ge \frac{1}{2}t_*^\alpha \ge \frac{1}{2}u^\alpha$, so the first arrival at H_u exists before t_*. The classical estimate $\sum_{n\le x}\mu(n)^2 = (6/\pi^2)x + O(\sqrt{x})$ implies $g_u = O(\sqrt{u})$: a longer initial zero run would contain no squarefree integer and would contradict the same estimate on that interval.

If $h < u^{3/5+\iota}$ for some fixed sufficiently small $\iota > 0$, then $h/u = o(1)$ directly. Otherwise Lemma 9.37 and (9.78), with $E''' \ge 0$, give $h \le \mathcal{V} + o(u)$. This proves (9.88) in both cases. The last implication is immediate. If Littlewood's bound held on a fixed horizon $\lambda' u$ with $\lambda_* - \varepsilon_* < \lambda' < \lambda_*$, then along this sequence it would give $H_u \ll u^{1/2+\epsilon}$, contradicting $H_u \asymp u^\alpha$ for $1/2 + \epsilon < \alpha$. □

Remark 9.39 (Exact limitation of the record-return cost). The improvement in Corollary 9.38 is the coefficient: the sufficient scale is $\lambda_* - 1$, not $(6/\pi^2)(\lambda_* - 1)$. It does not make the unresolved estimate weaker than early arrival. In the linear regime $h \asymp u$, the same identities give

$$\frac{\mathcal{V}(u, H)}{u} = \frac{h}{u} + \frac{E'''(u, H)}{u} + o(1). \tag{9.91}$$

Hence a bound $\mathcal{V} \le (\lambda_* - 1 - \varepsilon_*)u$ simultaneously forces early arrival and leaves a reserve for the positive retrogression excess E'''. An independent estimate of E''', or a signed estimate controlling $h + E'''$ jointly, is therefore required to advance this route.

There is a second way to remove the deterministic first-arrival drift before asking for arithmetic cancellation. It leads to centered correlations on the record blocks.

Proposition 9.40 (Centered record-block position bound)**.** *For a record block j, put*

$$x_{j,s} := \sigma\mu(u + \tau_j + s) \quad (1 \le s \le L_j), \qquad z_{j,s} := x_{j,s} - \frac{1}{L_j}. \tag{9.92}$$

Then $\sum_{s\le L_j} z_{j,s} = 0$. Let

$$B_{j,r} := \sum_{s=1}^{r} z_{j,s} = T_{j,r} - \frac{r}{L_j}, \qquad C^\circ_{j,g} := \sum_{s=1}^{L_j - g} z_{j,s} z_{j,s+g}. \tag{9.93}$$

The centered bridge energy satisfies

$$\mathcal{Q}^\circ_j := \sum_{r=1}^{L_j-1} B_{j,r}^2 = -\sum_{g=1}^{L_j-1} g C^\circ_{j,g} \ge \frac{(L_j - 1)(2L_j - 1)}{6L_j}. \tag{9.94}$$

Consequently, with $\mathcal{Q}^\circ := \sum_{j<H} \mathcal{Q}^\circ_j$,

$$\mathcal{Q}^\circ \ge \frac{h}{3} - \frac{H}{2} + \frac{1}{6}\sum_{j<H}\frac{1}{L_j}, \qquad h \le 3\mathcal{Q}^\circ + \frac{3}{2}H. \tag{9.95}$$

For $H = o(u)$, any estimate

$$\mathcal{Q}^\circ \le \left(\frac{\lambda_* - 1}{3} - \varepsilon_*\right) u \tag{9.96}$$

therefore places the first arrival at $t_H \le (\lambda_* - 3\varepsilon_* + o(1))u$.

Proof. The centered increments form a completed bridge, so Lemma 5.1 gives the equality in (9.94). Since $T_{j,r} \le 0$ before the next record, $B_{j,r} \le -r/L_j$, and hence

$$\mathcal{Q}_j^\circ \ge \frac{1}{L_j^2} \sum_{r=1}^{L_j - 1} r^2 = \frac{(L_j - 1)(2L_j - 1)}{6L_j}. \tag{9.97}$$

Summing and using $\sum_j L_j = h$ proves (9.95). The final implication follows from $H = o(u)$. □

Remark 9.41 (Scope of the centered route). The centered quantity removes the positive excess E''' from the statement of the target, but it creates a different analytic problem: the record blocks are adaptive and may be much shorter than the interval on which the Kaczorowski–Pintz witness is localized. In the difficult regime $h \asymp u$ and $H \asymp u^\alpha$ with $\alpha > 1/2$, their mean length is $h/H \asymp u^{1-\alpha}$, below the current all-interval Fourier range $u^{3/5+\iota}$ used in Theorem 12.13. Thus the centered identity is an internal route to lowering λ_*, but the present Fourier input does not close it block by block.

9.6. **Wide peaks from cancellation in all short intervals.** The only arithmetic input of the following lemma is the quantitative short-interval theorem of Matomäki–Teräväinen [31, Theorem 1.1]: for every fixed $\theta > 0.55$ and every fixed $\varepsilon > 0$, uniformly for the relevant intervals,

$$\sum_{x < n \le x + H} \mu(n) \ll_{\theta,\varepsilon} \frac{H}{(\log x)^{1/3-\varepsilon}} \qquad (H \ge x^\theta). \tag{9.98}$$

Lemma 9.42 (Mesoscopic Lipschitz bound)**.** *Fix $\Lambda > 1$ and $\theta_{\mathrm{w}} \in (0.55, 1)$. For large u put $h_u = \lceil (\Lambda u)^{\theta_{\mathrm{w}}} \rceil$ and*

$$\omega_u := \max\Big\{ h_u^{-1},\ \max_{u \le a \le \Lambda u} \frac{|M(a + h_u) - M(a)|}{h_u} \Big\}. \tag{9.99}$$

Then $\omega_u \to 0$, and for all integers $u \le a < b \le \Lambda u$,

$$|M(b) - M(a)| \le \omega_u (b - a) + h_u. \tag{9.100}$$

Proof. Fix $\theta' \in (0.55, \theta_{\mathrm{w}})$. For all sufficiently large u, $a + h_u \le 2\Lambda u$ uniformly for $u \le a \le \Lambda u$. By [31, Theorem 1.1], for every fixed $\varepsilon > 0$ there is a constant C such that, uniformly for $u \le b \le 2\Lambda u$,

$$\Big| M\Big(b + \lceil b^{\theta'} \rceil\Big) - M(b) \Big| \le C \frac{b^{\theta'}}{(\log u)^{1/3-\varepsilon}} \tag{9.101}$$

for all sufficiently large u. Theorem 1.1 of [31] is uniform in the starting point x for intervals of length $x^{\theta'}$; here $b \asymp_\Lambda u$, so $\log b = \log u + O_\Lambda(1)$, and replacing $x^{\theta'}$ by $\lceil b^{\theta'} \rceil$ changes the sum by $O(1)$, absorbed in C.

For a given a, define recursively

$$b_0 = a, \qquad b_{j+1} = b_j + \lceil b_j^{\theta'} \rceil, \tag{9.102}$$

and stop at the first J for which $b_{J+1} > a + h_u$. Since $b_j \asymp u$ throughout the construction, each full block has length $\asymp u^{\theta'}$, whence

$$J \ll \frac{h_u}{u^{\theta'}} + 1, \qquad \sum_{j < J} b_j^{\theta'} \ll h_u. \tag{9.103}$$

All starting points b_j lie in $[u, 2\Lambda u]$. Summing the full-block estimates and bounding the final incomplete piece trivially gives, uniformly for $u \le a \le \Lambda u$,

$$\frac{|M(a+h_u)-M(a)|}{h_u} \ll_{\varepsilon,\theta',\theta_{\mathrm{w}},\Lambda} (\log u)^{-1/3+\varepsilon} + u^{\theta'-\theta_{\mathrm{w}}}. \tag{9.104}$$

Consequently

$$\omega_u \ll_{\varepsilon,\theta',\theta_{\mathrm{w}},\Lambda} (\log u)^{-1/3+\varepsilon} + u^{\theta'-\theta_{\mathrm{w}}} + h_u^{-1} = o(1), \tag{9.105}$$

and the same estimate is uniform for $u \in [U, 2U]$ as $U \to \infty$.

For (9.100), write $b - a = qh_u + r$ with $0 \le r < h_u$. Every complete h_u-block starts at a point in $[u, \Lambda u]$ and contributes at most $\omega_u h_u$, while the terminal remainder contributes at most $r < h_u$. Thus $|M(b) - M(a)| \le \omega_u(b-a) + h_u$. □

Proposition 9.43 (Wide peaks)**.** *With $\Lambda, \theta_{\mathrm{w}}, h_u, \omega_u$ as in Lemma 9.42, let $I_\nu = (u, v]$ be an excursion of orientation σ_ν with $v \le \Lambda u$ and u large, let t be an integer with $u < t < v$ and $|M(t)| = H$, and put $G = (H - h_u)_+$ and $m = \lfloor G/\omega_u \rfloor$. Then*

$$t - u \ge \frac{G}{\omega_u}, \qquad v - t \ge \frac{G}{\omega_u}, \tag{9.106}$$

$$\mathfrak{b} \ge H + 2\sum_{j=1}^{m}(G - \omega_u j), \qquad Q \ge H^2 + 2\sum_{j=1}^{m}(G - \omega_u j)^2. \tag{9.107}$$

Consequently, uniformly for such excursions with $H/h_u \to \infty$,

$$v - u \ge (2 + o(1))\frac{H}{\omega_u}, \qquad \mathfrak{b} \ge (1 + o(1))\frac{H^2}{\omega_u}, \qquad Q \ge \Big(\frac{2}{3} + o(1)\Big)\frac{H^3}{\omega_u}, \tag{9.108}$$

so that $(v-u)/H$, $\mathfrak{b}/H^2$ and Q/H^3 tend to infinity, whereas the unit-step bounds of Lemma 3.3 and Proposition 5.6 give only 2, 1 and $\frac{2}{3}$.

Proof. By (9.100) applied to $(u, t]$ and $(t, v]$, $H \le \omega_u(t-u) + h_u$ and $H \le \omega_u(v-t) + h_u$. Thus, since the two distances are integers,

$$t - u,\ v - t \ge \left\lceil \frac{G}{\omega_u} \right\rceil, \tag{9.109}$$

and therefore $m = \lfloor G/\omega_u \rfloor \le t - u, v - t$. For $1 \le j \le m$ the points $t \pm j$ lie in $[u, v]$, and (9.100) gives $\sigma_\nu M(t \pm j) \ge G - \omega_u j \ge 0$. Summing t and the pairs $t \pm j$ gives the bounds for $\mathfrak{b}$ and Q. Finally $2\sum_{j\le m}(G - \omega_u j) = G^2/\omega_u + O(G)$ and $2\sum_{j\le m}(G - \omega_u j)^2 = \frac{2}{3}G^3/\omega_u + O(G^2)$. Under $H/h_u \to \infty$ we have $G = H - h_u = (1 + o(1))H$ and $G \to \infty$. Equation (9.105) gives $\omega_u = o(1)$, hence eventually $\omega_u \le 1$ and the relative errors satisfy $\omega_u/G \le 1/G = o(1)$. This proves the stated uniform asymptotics. □

Corollary 9.44 (Packing of high excursions)**.** *Let $Y = Y(U)$ with $Y/U^{\theta_{\mathrm{w}}} \to \infty$. The number of excursions with $U \le u \le 2U$, $v \le \Lambda u$ and height at least Y is $o(U/Y)$, whereas the unit-step bound gives $O(U/Y)$.*

Proof. They are disjoint and contained in $[U, 2\Lambda U]$. By Proposition 9.43, each has length at least

$$(2 + o(1))\frac{Y}{\max_{U\le u\le 2U}\omega_u}. \tag{9.110}$$

If $N(U, Y)$ denotes their number, disjointness therefore gives

$$N(U,Y)(2 + o(1))\frac{Y}{\max_{U\le u\le 2U}\omega_u} \le (2\Lambda - 1)U. \tag{9.111}$$

By the uniform estimate (9.105),

$$\omega_U^* := \max_{U\le u\le 2U}\omega_u = o(1). \tag{9.112}$$

Therefore

$$N(U,Y) \ll_\Lambda \omega_U^* \frac{U}{Y} = o(U/Y), \tag{9.113}$$

which completes the packing argument. □

Remark 9.45 (Scope). Proposition 9.43 is informative only for heights above $u^{0.55+o(1)}$. If $\Theta > 0.55$ and $0.55 < \theta_{\mathrm{w}} < \Theta - \eta$, it applies to the Kaczorowski–Pintz witness excursions of Corollary 9.15 with $v \le \lambda u$: their peaks are wide in the sense above. It does not lower λ_*: the resulting lower bound for $\log(v/u)$ is of order $H/(\omega_u u)$, which may still tend to 0, whereas Remark 9.31 requires a fixed surplus. The published quantitative estimate of Matomäki–Teräväinen [31, Theorem 1.1] makes the gain explicit. If $0.55 < \theta' < \theta_{\mathrm{w}}$, then the blocking argument in Lemma 9.42 gives, for every fixed $\varepsilon > 0$,

$$\omega_u \ll_{\varepsilon,\theta',\theta_{\mathrm{w}},\Lambda} (\log u)^{-1/3+\varepsilon} + u^{\theta'-\theta_{\mathrm{w}}}. \tag{9.114}$$

Consequently, in the regime $H/h_u \to \infty$ of Proposition 9.43,

$$Q \gg_\varepsilon H^3 (\log u)^{1/3-\varepsilon}. \tag{9.115}$$

10. Arithmetic selection on the Catalan reflection fibre

The fixed-mask Catalan description separates two finite degrees of freedom of a completed excursion of M: its positive support-time shape and its orientation. The Catalan estimates developed above act on the first coordinate. The present section isolates the second coordinate and shows that it is strongly constrained by divisor inversion. The resulting arithmetic selector is nevertheless insufficient, by itself, to observe a fixed-shift Chowla form on the ambient squarefree space. This produces a sharp separation between orientation selection and two-point cancellation.

Throughout this section, let N be an integer with $M(N) = 0$, so that the portion of the path up to N consists only of completed excursions and zero runs. Write

$$I_\nu = (x_{0,\nu}, x_{1,\nu}], \qquad 1 \le \nu \le J_N, \tag{10.1}$$

for the completed excursions up to N, and let

$$E_\nu := \{d \in I_\nu : \mu(d) \neq 0\} \tag{10.2}$$

be the active support of the νth excursion. For a word $w \in \{-1,0,1\}^N$ define

$$M_w(n) := \sum_{d \le n} w(d), \qquad \Pi_N(w) := \big(M_w(n)^2\big)_{1 \le n \le N}. \tag{10.3}$$

We restrict to words having the same zero mask as μ.

Theorem 10.1 (Exact reflection fibre). *Let $w \in \{-1,0,1\}^N$ have the same support as μ. Then*

$$\Pi_N(w) = \Pi_N(\mu) \tag{10.4}$$

if and only if, on each completed excursion I_ν, the word w agrees either with μ or with $-\mu$. Consequently the fibre of the complete quadratic profile has exactly

$$2^{J_N} \tag{10.5}$$

elements.

Proof. Both partial-sum paths start from the same value, $M_w(0) = M(0) = 0$, and $w, \mu \in \{-1,0,1\}^N$ have the same zero mask. Independent reflection of a completed excursion multiplies M by -1 on the interior of that excursion and leaves it zero at both endpoints; hence it preserves $M(n)^2$ at every integer. Conversely, equality of the quadratic profiles gives the same zero set and the same absolute path. On the interior of each completed excursion one has

$$|M_w(n)| = |M(n)| > 0, \tag{10.6}$$

so M_w has no interior zero and therefore has constant sign there. Since M also has constant sign on that excursion, the ratio M_w/M is a constant in $\{\pm 1\}$. Taking first differences successively across that component shows that the same constant relates w and μ on the active support of the excursion. The choices are independent because every completed excursion begins and ends at zero. □

Theorem 10.1 identifies the precise information lost by every statistic determined by the full quadratic profile: after the absolute excursion shapes are fixed, the remaining ambiguity is a Boolean orientation vector.

For $\mathbf{z} = (\mathbf{z}_1, \dots, \mathbf{z}_{J_N}) \in \{0,1\}^{J_N}$, let $w_{\mathbf{z}}$ be obtained from μ by reflecting precisely those excursions for which $\mathbf{z}_\nu = 1$. Fix arbitrary weights $\varrho_n > 0$ and define the divisor-defect functional

$$\mathfrak{D}_N(w) := \varrho_1 |w(1) - 1|^2 + \sum_{n=2}^{N} \varrho_n \left| \sum_{d \mid n} w(d) \right|^2 . \tag{10.7}$$

For every excursion put

$$g_\nu(n) := \sum_{\substack{d \mid n \\ d \in E_\nu}} \mu(d), \qquad 1 \le n \le N, \tag{10.8}$$

and define the $J_N \times J_N$ matrix

$$(G_N)_{\nu\kappa} := \sum_{n=1}^{N} \varrho_n g_\nu(n) g_\kappa(n). \tag{10.9}$$

Theorem 10.2 (Positive Gram selector for excursion orientations)**.** *The matrix G_N is positive semidefinite and*

$$\mathfrak{D}_N(w_{\mathbf{z}}) = 4\, \mathbf{z}^{\mathsf{T}} G_N \mathbf{z} \qquad (\mathbf{z} \in \{0,1\}^{J_N}). \tag{10.10}$$

Moreover,

$$\mathbf{z}^{\mathsf{T}} G_N \mathbf{z} > 0 \qquad \text{for every } \mathbf{z} \neq 0. \tag{10.11}$$

Thus the divisor equations select the arithmetic orientation uniquely inside the complete quadratic-profile fibre.

Proof. For $n \ge 2$, Möbius inversion gives

$$\sum_{d \mid n} \mu(d) = 0. \tag{10.12}$$

Reflecting the excursions indexed by $\mathbf{z}$ therefore gives

$$\sum_{d \mid n} w_{\mathbf{z}}(d) = -2 \sum_{\nu=1}^{J_N} \mathbf{z}_\nu g_\nu(n). \tag{10.13}$$

At $n = 1$ one has explicitly

$$w_{\mathbf{z}}(1) - 1 = -2 \sum_{\nu=1}^{J_N} \mathbf{z}_\nu g_\nu(1), \tag{10.14}$$

with only the excursion containing 1 able to contribute. Substitution into (10.7) yields (10.10), and (10.9) is a Gram representation, so $G_N \succeq 0$.

If a nonzero Boolean vector $\mathbf{z}$ satisfied $\mathbf{z}^{\mathsf{T}} G_N \mathbf{z} = 0$, then $\mathfrak{D}_N(w_{\mathbf{z}}) = 0$. Hence $w_{\mathbf{z}}(1) = 1$ and

$$\sum_{d \mid n} w_{\mathbf{z}}(d) = 0 \qquad (2 \le n \le N). \tag{10.15}$$

Finite Möbius inversion determines the coefficients recursively and forces $w_{\mathbf{z}}(d) = \mu(d)$ for every $d \le N$, contradicting the reflection of a nonempty set of excursions. □

The Gram form is exact but uses all divisor equations. A much smaller family of specially placed divisor equations already compresses the orientation fibre from exponential to polynomial size.

For the νth excursion list its active support as

$$a_{\nu,1} < \cdots < a_{\nu,L_\nu^{\#}}, \qquad L_\nu^{\#} = 2r_\nu, \tag{10.16}$$

and define its support centre

$$m_\nu := a_{\nu,r_\nu}. \tag{10.17}$$

Set

$$\phi_\nu(d) := \sigma_\nu \mu(d) \qquad (d \in E_\nu), \tag{10.18}$$

so that ϕ_ν is the oriented increment sequence of the positive support-time excursion. Define the central divisor matrix

$$\mathscr{C}^{\mathrm{div}}_{\nu\kappa} := \sum_{\substack{d \mid m_\nu \\ d \in E_\kappa}} \phi_\kappa(d). \tag{10.19}$$

Theorem 10.3 (Polynomial compression of the orientation fibre)**.** *The matrix $\mathscr{C}^{\mathrm{div}}$ is lower triangular. Its diagonal entry in row ν is nonzero whenever*

$$x_{0,\nu} \geq \frac{m_\nu}{2}. \tag{10.20}$$

The condition (10.20) *can fail for at most $\lfloor \log_2 N \rfloor$ completed excursions.*

After the affine normalization $w(1) = 1$ is imposed, the solution set of the central divisor equations is an affine subset of the orientation-coordinate space whose linear direction has dimension at most $\lfloor \log_2 N \rfloor + 1$. More concretely, once the exceptional orientation coordinates have been specified, every nonexceptional coordinate not already fixed by the affine normalization is determined successively by a central equation with nonzero diagonal coefficient. Consequently, among the 2^{J_N} Boolean orientations in the quadratic-profile fibre, at most

$$2^{\lfloor \log_2 N \rfloor + 1} \leq 2N \tag{10.21}$$

satisfy the central divisor equations. Furthermore, at most $\lfloor \log_2 N \rfloor + 1$ additional divisor rows suffice to select the Möbius orientation uniquely.

Proof. If $\kappa > \nu$, every $d \in E_\kappa$ exceeds m_ν, so no such d can divide m_ν; hence $\mathscr{C}^{\mathrm{div}}_{\nu\kappa} = 0$. If (10.20) holds, every proper divisor of m_ν is at most $m_\nu/2 \leq x_{0,\nu}$ and therefore lies outside E_ν. Thus only $d = m_\nu$ contributes to the diagonal, giving

$$\mathscr{C}^{\mathrm{div}}_{\nu\nu} = \phi_\nu(m_\nu) \in \{\pm 1\}. \tag{10.22}$$

Since $m_\nu \leq x_{1,\nu}$, failure of (10.20) implies $x_{1,\nu} > 2x_{0,\nu}$. Count the initial excursion separately. If $2 \leq a_1 < \cdots < a_r$ are the positive starts of the remaining failing completed excursions, then disjointness gives

$$N \geq x_{1,r} > 2a_r > 2^2 a_{r-1} > \cdots > 2^r a_1 \geq 2^{r+1}. \tag{10.23}$$

Thus the total number of failures, including the possible initial one, is strictly smaller than $\log_2 N$ when N is not a power of two and at most $\log_2 N - 1$ when it is; in either case it is at most $\lfloor \log_2 N \rfloor$.

Let $\boldsymbol{\sigma} = (\sigma_1, \ldots, \sigma_{J_N})^{\mathsf{T}}$ be the arithmetic orientation vector. For every central row with $m_\nu > 1$,

$$(\mathscr{C}^{\mathrm{div}}\boldsymbol{\sigma})_\nu = \sum_{d \mid m_\nu} \mu(d) = 0. \tag{10.24}$$

The row $n = 1$ is imposed separately as the affine normalization $w(1) = 1$. Let $\mathcal{E}_N$ be the set of indices for which (10.20) fails. Among the central rows with $m_\nu > 1$, every nonexceptional row has a nonzero new diagonal coefficient. Thus their coefficient matrix has rank at least

$$J_N - |\mathcal{E}_N| - 1, \tag{10.25}$$

the possible loss of one row accounting only for the distinguished index $m_\nu = 1$. Hence the associated homogeneous linear system has nullity at most $|\mathcal{E}_N|+1$. Intersecting this linear solution space with the affine normalization $w(1) = 1$ gives an affine solution set whose linear direction has dimension at most $|\mathcal{E}_N| + 1 \le \lfloor \log_2 N \rfloor + 1$.

For the Boolean count we use the triangular structure directly rather than a dimension argument. Choose first the coordinates indexed by $\mathcal{E}_N$ and, if it is not already among them, the single coordinate attached to the row $m_\nu = 1$. Then process every remaining nonexceptional row in increasing order. All earlier coordinates are already known, while the coefficient of the new coordinate is $\mathscr{C}^{\mathrm{div}}_{\nu\nu} \neq 0$, so that coordinate is uniquely determined. The affine equation $w(1) = 1$ can only remove choices, not create them. Therefore there are at most $2^{|\mathcal{E}_N|+1} \le 2^{\lfloor \log_2 N \rfloor + 1} \le 2N$ Boolean solutions. In particular the Boolean estimate is not inferred from real dimension alone.

For the final assertion, for a generic real orientation vector $\boldsymbol{\sigma}' = (\sigma'_1, \dots, \sigma'_{J_N})$ define $w_{\boldsymbol{\sigma}'}(d) = \sigma'_\nu \phi_\nu(d)$ for $d \in E_\nu$ and $w_{\boldsymbol{\sigma}'}(d) = 0$ off the active support. Factor the complete divisor readout as

$$\boldsymbol{\sigma}' \longmapsto w_{\boldsymbol{\sigma}'} \longmapsto \left(\sum_{d \mid n} w_{\boldsymbol{\sigma}'}(d) \right)_{1 \le n \le N}. \tag{10.26}$$

The first map is injective: the supports E_ν are disjoint and every completed excursion contains an active increment, so two distinct orientation vectors give distinct words. The second map is injective on $\mathbb{R}^N$ because its matrix is $(\mathbf{1}_{d|n})_{1 \le n,d \le N}$, which is lower unitriangular. Hence the composition has full column rank J_N on the orientation coordinates. The central rows with $m_\nu > 1$ have linear rank at least $J_N - |\mathcal{E}_N| - 1$, independently of the separate affine normalization $w(1) = 1$. Extending a basis of their row space to a basis of the full readout therefore needs at most $|\mathcal{E}_N|+1 \le \lfloor \log_2 N \rfloor + 1$ additional divisor rows. The full-rank system has at most one orientation solution, and the arithmetic orientation $\boldsymbol{\sigma}$ is such a solution; it is therefore unique. □

The preceding theorem concerns selection inside the highly structured reflection fibre. It does not imply that a fixed-shift correlation can be recovered from a comparably small family of divisor measurements. In fact a linear obstruction holds on the ambient squarefree space.

Let $\mathbf{D}_N$ be the divisor-incidence matrix

$$(\mathbf{D}_N)_{r,d} := \mathbf{1}_{d|r}, \qquad 1 \le r, d \le N, \tag{10.27}$$

and, for $\mathcal{T} \subseteq \{1, \dots, N\}$, let $\mathbf{D}_\mathcal{T}$ denote the restriction to rows in $\mathcal{T}$. Let A_h be the symmetric fixed-shift matrix defined by

$$x^{\mathsf{T}} \mathsf{A}_h x = \sum_{n \le N-h} x_n x_{n+h}. \tag{10.28}$$

Write

$$\mathfrak{S}_h := \prod_{p^2 \mid h} \left(1 - \frac{1}{p^2}\right) \prod_{p^2 \nmid h} \left(1 - \frac{2}{p^2}\right) > 0, \tag{10.29}$$

so that, by the classical squarefree-pair theorem of Mirsky [33],

$$\#\{n \le X : n, n+h \text{ squarefree}\} = \mathfrak{S}_h X + o_h(X) \tag{10.30}$$

for each fixed nonzero h.

Theorem 10.4 (Linear obstruction to divisor-row observability)**.** *Fix $h \ge 1$. Suppose that for some $c_{\mathrm{obs}} \in [0, \infty)$, some $\varepsilon \in [0, 1/2)$, and some row set $\mathcal{T} \subseteq \{1, \dots, N\}$ one has*

$$|x^{\mathsf{T}} \mathsf{A}_h x| \le c_{\mathrm{obs}} \|\mathbf{D}_\mathcal{T} x\|_2^2 + \varepsilon \|x\|_2^2 \tag{10.31}$$

for every real vector x supported on the squarefree integers in $[1, N]$. Then

$$|\mathcal{T}| \ge \frac{\mathfrak{S}_h}{4} N + o_h(N). \tag{10.32}$$

In particular, no $o(N)$ family of divisor rows can uniformly observe the fixed-shift quadratic form with error below $1/2$.

Proof. Let $N/2 < b \le N - h$ be such that both b and $b+h$ are squarefree. Since $b > N/2$, column b of $\mathbf{D}_N$ is nonzero only in row b; the same holds for $b+h$. If neither row belongs to $\mathcal{T}$, then for

$$x = e_b \pm e_{b+h} \tag{10.33}$$

one has $\mathbf{D}_{\mathcal{T}} x = 0$, while

$$|x^{\mathsf{T}} \mathsf{A}_h x| = 1, \qquad \|x\|_2^2 = 2. \tag{10.34}$$

Thus (10.31) would force $1 \le 2\varepsilon$, contrary to $\varepsilon < 1/2$. Hence $\mathcal{T}$ must meet every edge $\{b, b+h\}$ of the graph of squarefree shifted pairs in $(N/2, N]$.

By (10.29), this graph has

$$\frac{\mathfrak{S}_h}{2} N + o_h(N) \tag{10.35}$$

edges. Its maximum degree is at most 2, so every vertex cover has cardinality at least half the number of edges. This gives (10.32). □

Corollary 10.5 (Selection–observability separation)**.** *For cutoffs N with $M(N) = 0$, the complete quadratic profile has an orientation fibre of size 2^{J_N}. The central divisor equations reduce that fibre to at most $2N$ candidates, and $O(\log N)$ further divisor equations select the arithmetic orientation uniquely. In contrast, for every fixed $h \ge 1$, uniform observation of the ambient fixed-shift form $x^{\mathsf{T}} \mathsf{A}_h x$ with error strictly below $1/2$ requires $\Omega_h(N)$ divisor rows.*

Consequently, the unresolved difficulty in fixed-shift Möbius cancellation is not the finite identification of the orientation bits once the absolute excursion geometry is known. It is the collective cancellation of the shift interaction across the arithmetic path. The Catalan selection problem and the Chowla cancellation problem therefore occupy genuinely different complexity scales.

11. Prefix structure of the localized window

11.1. **The linear window and the one-step recursion.** Once the horizon is below $2x_0$, the square-range bilinear structure of Proposition 11.31 is no longer needed: on $(u, 2u]$ the Möbius function is a *linear* expression in the prefix $\mu \mathbf{1}_{[1,u]}$.

Proposition 11.1 (Linear window, classical)**.** *Let $u \ge 1$. For every integer $u < n \le 2u$,*

$$\mu(n) = -\sum_{\substack{d \mid n \\ d \le u}} \mu(d). \tag{11.1}$$

Consequently, if $M(u) = 0$ and $u < t \le 2u$,

$$M(t) = -\sum_{d \le u} \mu(d) \left(\left\lfloor \frac{t}{d} \right\rfloor - \left\lfloor \frac{u}{d} \right\rfloor \right). \tag{11.2}$$

Proof. Since $n \ge 2$, $\sum_{d \mid n} \mu(d) = 0$. Every proper divisor of n is at most $n/2 \le u$, and every divisor $d \le u$ is proper because $n > u$. Summing (11.1) over $u < n \le t$, the number of multiples of d in $(u, t]$ is $\lfloor t/d \rfloor - \lfloor u/d \rfloor$. □

Corollary 11.2 (Localized prefix formula)**.** *Let $M(u) = 0$, let $1 < \lambda < 2$, and let $u < t \le \lambda u$. Then*

$$M(t) = -\sum_{d \le \lambda u/2} \mu(d) \left(\left\lfloor \frac{t}{d} \right\rfloor - \left\lfloor \frac{u}{d} \right\rfloor \right). \tag{11.3}$$

Indeed, if $\lambda u/2 < d \le u$, then $d > u/2$ and the first multiple of d strictly larger than u is $2d > \lambda u \ge t$.

Corollary 11.3 (Prefix-dependency cutoff)**.** *Put*

$$\chi_* := \frac{1}{2}e^{\pi/\gamma_1} = 0.62445\ldots. \tag{11.4}$$

For every fixed χ with $\chi_* < \chi < 1$, RH is equivalent to Littlewood's bound on every excursion starting at u throughout

$$u < t \le \min\{x_1, 2\chi u\}, \tag{11.5}$$

where, throughout that window, $M(t)$ is given by the sum (11.3) using only the prefix $\mu(d)$ with $d \le \chi u$. In particular, the choice $\chi = 5/8$ corresponds to $\lambda = 5/4$ and gives the RH-equivalent window

$$1 \le r = t - u \le \min\{\ell - 1, u/4\} \tag{11.6}$$

using only $d \le 5u/8$. The return point $r = \ell$, when it belongs to the window, is omitted because $M(u + \ell) = 0$.

The number χ_* is an infimum, not an attained endpoint, because the horizon theorem requires $\lambda > \lambda_*$.

Proof. Take $\lambda = 2\chi > \lambda_*$ in Corollary 9.14(iii), then use Corollary 11.2. □

Corollary 11.4 (Algebraic dependency gap)**.** *Let $\chi_* < \chi < 1$. If $u \in \mathbb{N}$ satisfies $M(u) = 0$, then every increment $\mu(n)$ with*

$$u < n \le 2\chi u \tag{11.7}$$

is determined by the prefix $\mu(1), \ldots, \mu(\lfloor \chi u \rfloor)$. Consequently the intermediate block

$$(\chi u, u] \tag{11.8}$$

does not occur as a direct input in the divisor recursion determining the localized future window. As $\chi \downarrow \chi_*$, the omitted block has relative length tending to $1 - \chi_* = 0.3755\ldots$.

Proof. For $u < n \le 2\chi u$, every proper divisor $d \mid n$ satisfies $d \le n/2 \le \chi u$. Hence

$$\mu(n) = -\sum_{\substack{d \mid n \\ d < n}} \mu(d) \tag{11.9}$$

uses only $d \le \chi u$. □

In convolution form, (11.1) says that $(\mu - \mu_u) * 1 = \mu \mathbf{1}_{(u,\infty)}$ on $n \le 2u$, the order-one analogue of (11.63), which for $u < n \le u^2$ reads $\mu(n) = -(\mu_u * \mu_u * 1)(n)$; on $(u, 2u]$ both give $\mu(n)$, the bilinear form being the quadratic continuation of the linear one. Equivalently, (11.2) follows from $\sum_{d \le t} \mu(d)\lfloor t/d \rfloor = 1$, since $\lfloor t/d \rfloor = 1$ for $u < d \le t \le 2u$.

Proposition 11.5 (One-step recursion, classical)**.** *For real $x \ge 1$,*

$$\sum_{k \le x} M\left(\frac{x}{k}\right) = 1. \tag{11.10}$$

Hence, if $M(u) = 0$ and $t > u$,

$$M(t) = -\sum_{2 \le k \le t} \left[M\left(\frac{t}{k}\right) - M\left(\frac{u}{k}\right) \right], \tag{11.11}$$

with $M(y) = 0$ for $y < 1$. Every argument on the right is at most $t/2$; hence on a localized window $u < t \le \lambda u$ all arguments are at most $\lambda u/2$. For $\lambda = 5/4$ this is $5u/8$.

Proof. $\sum_{k \le x} M(x/k) = \sum_{k \le x} \sum_{d \le x/k} \mu(d) = \sum_{m \le x} \sum_{d \mid m} \mu(d) = 1$. Here and below $M(x/k) = M(\lfloor x/k \rfloor)$ by the global real-argument convention. Subtract the identity at u, where $M(u) = 0$, and extend the second sum to $k \le t$ using $M(u/k) = 0$ for $k > u$. □

Proposition 11.6 (Scale cascade in the localized excursion window)**.** *Let $(u, v]$ be an excursion, let $\lambda_* < \lambda < 2$, and let $u < t \le \min\{v, \lambda u\}$. For $k \ge 2$ put*

$$\Delta_k(u,t) := M(t/k) - M(u/k), \tag{11.12}$$

and for an integer $K \ge 2$ put

$$\mathcal{R}_K(u,t) := -\sum_{K<k\le t} \Delta_k(u,t). \tag{11.13}$$

Then, exactly,

$$M(t) = -\sum_{k=2}^{K} \Delta_k(u,t) + \mathcal{R}_K(u,t), \tag{11.14}$$

and $\mathcal{R}_K(u,t)$ uses only values $M(y)$ with

$$y \le \frac{\lambda u}{K+1}. \tag{11.15}$$

Equivalently, all dependence on values above $\lambda u/(K+1)$ is concentrated in the $K-1$ increments $\Delta_2, \dots, \Delta_K$. Moreover

$$\Delta_k(u,t) = \sum_{u/k<n\le t/k} \mu(n), \tag{11.16}$$

so the supporting real interval has length $(t-u)/k$ and contains $\lfloor t/k \rfloor - \lfloor u/k \rfloor$ integers.

Proof. Equation (11.14) is Proposition 11.5 with the terms $2 \le k \le K$ separated. If $k > K$, then $t/k \le \lambda u/(K+1)$ and $u/k \le \lambda u/(K+1)$. The last identity is the definition of the summatory function on the interval $(u/k, t/k]$. □

Corollary 11.7 (First channels)**.** *For $K = 2$,*

$$M(t) = -\big(M(t/2) - M(u/2)\big) + \mathcal{R}_2(u,t), \tag{11.17}$$

where $\mathcal{R}_2$ uses only $y \le \lambda u/3$. With the convenient localized choice $\lambda = 5/4$, the only channel above $5u/12$ is

$$\Delta_2(u, u+r) = \sum_{u/2<n\le u/2+r/2} \mu(n), \qquad 0 < r \le u/4, \tag{11.18}$$

whose supporting interval has length at most $u/8$. In the limiting sense $\lambda \downarrow \lambda_*$, the successive remainder cutoffs are

$$\frac{\lambda_*}{2}u = 0.62445\ldots u, \quad \frac{\lambda_*}{3}u = 0.4162\ldots u, \quad \frac{\lambda_*}{4}u = 0.3122\ldots u, \quad \frac{\lambda_*}{5}u = 0.2497\ldots u. \tag{11.19}$$

Lemma 11.8 (Channel persistence extracted from a high excursion)**.** *Let $S_r = \sigma_\nu M(u+r) > 0$ on $1 \le r \le h$ with $S_0 = 0$ and $|S_r - S_{r-1}| \le 1$. Fix $K \ge 2$ and write, using Proposition 11.6,*

$$S_r = \sum_{k=2}^{K} X_k(r) + Y_K(r), \quad X_k(r) := -\sigma_\nu \Delta_k(u, u+r), \quad Y_K(r) := \sigma_\nu \mathcal{R}_K(u, u+r). \tag{11.20}$$

If $H = \max_{r\le h} S_r$ and τ is a point with $S_\tau = H$, then among the last $\lfloor H/2 \rfloor + 1$ indices

$$\tau - \lfloor H/2 \rfloor, \dots, \tau \tag{11.21}$$

there is one fixed component among $X_2, \dots, X_K, Y_K$ which is at least $H/(2K)$ at no fewer than $\lceil (\lfloor H/2 \rfloor + 1)/K \rceil$ indices.

Proof. Since $S_0 = 0$ and the path has unit steps, $\tau \ge H$ and $S_{\tau-j} \ge H - j \ge H/2$ for $0 \le j \le \lfloor H/2 \rfloor$. At every one of these indices the sum of the K displayed components is at least $H/2$, hence some component is at least $H/(2K)$. Pigeonholing the $\lfloor H/2 \rfloor + 1$ indices among the K components gives at least $\lceil (\lfloor H/2 \rfloor + 1)/K \rceil$ occurrences, proving the claim. □

Remark 11.9 (What the persistence lemma adds). Lemma 11.8 converts a large value attained inside one excursion into a quantified persistence requirement: either one of the finitely many scaled interval sums $\Delta_2, \dots, \Delta_K$ repeatedly carries a positive fraction of the height, or the responsibility lies repeatedly in the deeper remainder, which is supported entirely below $\lambda u/(K+1)$. This support statement is not a smallness estimate for the remainder. This avoids the iteration defect of Remark 11.22: no new point is claimed to be the start-window point of another excursion.

Corollary 11.10 (Channel descent with explicit loss)**.** *Let $u \in \mathbb{N}$ satisfy $M(u) = 0$, let $t > u$, $2 \le K \le u$, and $0 < \vartheta < 1$. Then either*

(i) *there are $2 \le k \le K$ and $y \in \{\lfloor u/k \rfloor, \lfloor t/k \rfloor\}$ with $|M(y)| \ge |M(t)|/(2K)$, hence*

$$\frac{|M(y)|}{y^\vartheta} \ge \frac{k^\vartheta}{2K} \cdot \frac{|M(t)|}{t^\vartheta}, \qquad y \le \left\lfloor \frac{t}{2} \right\rfloor; \tag{11.22}$$

or

(ii) *$|\mathcal{R}_K(u,t)| \ge |M(t)|/K$, where $\mathcal{R}_K$ depends only on $M(y)$ with $y \le \lfloor t/(K+1) \rfloor$.*

The statement holds at every $t > u$, not only in a localized window.

Proof. The proof of Proposition 11.6 applies verbatim to every $t > u$, with $\mathcal{R}_K$ supported on $y \le t/(K+1)$. By (11.14), one of the K terms $\Delta_2, \dots, \Delta_K, \mathcal{R}_K$ has modulus at least $|M(t)|/K$. If it is $\Delta_k = M(\lfloor t/k \rfloor) - M(\lfloor u/k \rfloor)$, one of its two values has modulus at least $|M(t)|/(2K)$. Since $k \le K \le u$, both candidate arguments are positive integers; hence $y \ge 1$. Moreover $y \le \lfloor t/k \rfloor \le \lfloor t/2 \rfloor$, and therefore

$$\frac{|M(y)|}{y^\vartheta} \ge \frac{|M(t)|}{2K} \left(\frac{t}{k}\right)^{-\vartheta} = \frac{k^\vartheta}{2K} \frac{|M(t)|}{t^\vartheta}. \tag{11.23}$$

□

Remark 11.11 (Channels do not iterate to a bound). Corollary 11.10 has no iteration gap: it applies at the new point y relative to the last zero of M before y. Its defect is the constant loss $k^\vartheta/(2K) < 1$ per step. If alternative (i) held at every step, a chain of length $m \asymp \log t$ would give only $|M(t)| \le (2K)^m \, O(1) = t^{O(\log K)}$, weaker than the trivial bound. For fixed K and $0 < \vartheta < 1$ the guaranteed factor can never reach 1, since

$$\frac{k^\vartheta}{2K} \le \frac{K^\vartheta}{2K} = \frac{1}{2K^{1-\vartheta}} < \frac{1}{2}. \tag{11.24}$$

Thus no iteration of the same pigeonhole estimate can remove the loss at an individual step. A closing argument must instead recover the loss collectively from the persistence supplied by Lemma 11.8, or prove cancellation in the deep remainder. Both require information beyond dependency alone.

Proposition 11.12 (Quantified persistence along channel descent)**.** *Fix $K \ge 2$, $0 < \vartheta < 1$ and $\delta > 0$, and put*

$$c_{\vartheta,K} := \log\left(\frac{2K}{2^\vartheta}\right) > 0. \tag{11.25}$$

Suppose $t_0 \to \infty$ through points satisfying

$$Z_\vartheta(t_0) := \frac{|M(t_0)|}{t_0^\vartheta} \ge t_0^\delta. \tag{11.26}$$

Since $|M(t_0)| \le t_0$, any nonempty asymptotic regime of this form has $\delta \le 1 - \vartheta$.

At stage j, let u_j be the last zero of M before t_j. Continue the descent only while $u_j \ge K$ and alternative (i) *of Corollary 11.10 occurs; stop if either the deep-remainder alternative occurs or $u_j < K$. Whenever a descent step is available there are $2 \le k_j \le K$ and $t_{j+1} \le \lfloor t_j/2 \rfloor$ such that*

$$Z_\vartheta(t_{j+1}) \ge \frac{k_j^\vartheta}{2K} Z_\vartheta(t_j). \tag{11.27}$$

Consequently, after m successful steps,

$$Z_\vartheta(t_m) \geq Z_\vartheta(t_0) \prod_{j<m} \frac{k_j^\vartheta}{2K} \geq t_0^\delta e^{-mc_{\vartheta,K}}. \tag{11.28}$$

Set

$$m_* := \left\lfloor \frac{\delta \log t_0}{2c_{\vartheta,K}} \right\rfloor, \qquad \eta_{\vartheta,K,\delta} := \frac{\delta \log 2}{2c_{\vartheta,K}}. \tag{11.29}$$

Then, for all sufficiently large t_0, either a deep-remainder alternative occurs before step m_*, or the descent is valid through step m_* and there is a descendant satisfying

$$t_{m_*} \leq t_0^{1-\eta_{\vartheta,K,\delta}+o(1)}, \qquad Z_\vartheta(t_{m_*}) \geq t_0^{\delta/2}. \tag{11.30}$$

If $\eta_{\vartheta,K,\delta} \geq 1$, the second alternative is impossible for all sufficiently large t_0; hence in that regime the deep-remainder alternative must occur before step m_*.

Proof. Each successful application of Corollary 11.10 gives $t_{j+1} \leq \lfloor t_j/2 \rfloor$ and multiplies the normalized value by at least $2^\vartheta/(2K) = e^{-c_{\vartheta,K}}$, proving (11.28). Hence for every $m \leq m_*$ reached without a deep remainder,

$$Z_\vartheta(t_m) \geq t_0^{\delta/2}, \qquad t_m \leq t_0/2^m. \tag{11.31}$$

It remains to justify that the stopping condition $u_j < K$ cannot occur before m_*. Since M has infinitely many zeros, choose once and for all a zero $z_K \geq K$. If the last zero before t_j were $< K$, then necessarily $t_j < z_K$; otherwise z_K itself would be a preceding zero. Hence

$$Z_\vartheta(t_j) \leq z_K^{1-\vartheta}, \tag{11.32}$$

a constant depending only on K and ϑ. This contradicts $Z_\vartheta(t_j) \geq t_0^{\delta/2}$ for large t_0. Therefore the only possible premature stop is the deep-remainder alternative. If it does not occur, the iteration reaches m_*. Finally $t_{m_*} \leq t_0 2^{-m_*} = t_0^{1-\eta_{\vartheta,K,\delta}+o(1)}$. If $\eta_{\vartheta,K,\delta} > 1$, this upper bound is eventually < 1, impossible for a positive integer descendant. If $\eta_{\vartheta,K,\delta} = 1$, then $t_{m_*} = t_0^{o(1)}$, so the trivial bound $Z_\vartheta(t_{m_*}) \leq t_{m_*}^{1-\vartheta} = t_0^{o(1)}$ contradicts $Z_\vartheta(t_{m_*}) \geq t_0^{\delta/2}$. Thus the deep-remainder alternative must occur earlier whenever $\eta_{\vartheta,K,\delta} \geq 1$. □

Remark 11.13 (What remains after quantified persistence). Proposition 11.12 shows that a polynomial normalized violation cannot disappear immediately under channel descent. The pointwise loss accumulates, and bounding the deep remainder by absolute values loses a factor of the number of channels. A closing argument must therefore exploit the multi-index persistence of Lemma 11.8 collectively, or establish cancellation among the channels entering $\mathcal{R}_K$.

Corollary 11.14 (RH in the localized linear window)**.** *For every excursion starting at u and every $u < t \leq 5u/4$, the signed identities are*

$$M(t) = -\sum_{d \leq 5u/8} \mu(d) \left(\left\lfloor \frac{t}{d} \right\rfloor - \left\lfloor \frac{u}{d} \right\rfloor \right), \tag{11.33}$$

$$M(t) = -\sum_{2 \leq k \leq t} \left[M\left(\frac{t}{k}\right) - M\left(\frac{u}{k}\right) \right]. \tag{11.34}$$

Consequently RH holds if and only if, for every $\varepsilon > 0$,

$$\left| \sum_{d \leq 5u/8} \mu(d) \left(\left\lfloor \frac{t}{d} \right\rfloor - \left\lfloor \frac{u}{d} \right\rfloor \right) \right| = \left| \sum_{2 \leq k \leq t} \left[M\left(\frac{t}{k}\right) - M\left(\frac{u}{k}\right) \right] \right| \ll_\varepsilon t^{1/2+\varepsilon} \tag{11.35}$$

for every excursion starting at u and every

$$u < t \le \min\left\{x_1, \frac{5}{4}u\right\}. \tag{11.36}$$

Equivalently, in intra-excursion coordinates $t = u + r$ it is enough to control

$$1 \le r \le \min\{\ell - 1, u/4\}. \tag{11.37}$$

Proof. Use Corollary 11.3 with $\chi = 5/8$, together with Proposition 11.5. □

Proposition 11.15 (Kernel form of the recursion)**.** *Let $M(u) = 0$ and let t be an integer with $u < t < 2u$. For $1 \le y \le \lfloor t/2 \rfloor$ put*

$$c_y(t,u) = \Big(\Big\lfloor \frac{t}{y} \Big\rfloor - \Big\lfloor \frac{t}{y+1} \Big\rfloor\Big) - \Big(\Big\lfloor \frac{u}{y} \Big\rfloor - \Big\lfloor \frac{u}{y+1} \Big\rfloor\Big). \tag{11.38}$$

Then:

(i) $M(t) = -\sum_{1 \le y \le \lfloor t/2 \rfloor} c_y(t,u) M(y)$;

(ii) $c_y(t,u) = \dfrac{t-u}{y(y+1)} + e_y$ *with $|e_y| < 2$; for $y > \sqrt{t}$ one has $c_y \in \{-1, 0, 1\}$, and $c_y \ne 0$ only if $y = \lfloor t/k \rfloor$ or $y = \lfloor u/k \rfloor$ for some $2 \le k < \sqrt{t}$ (the value $k = 1$ gives $y \in \{t, u\} > \lfloor t/2 \rfloor$);*

(iii) $c_1(t,u) \ge \frac{1}{2}(t - u - 1)$;

(iv) $M(t) = -(t-u)\,\Upsilon(\sqrt{t}) + \mathrm{Err}(t,u)$, *where Υ is the coefficient* (11.77) *and*

$$|\mathrm{Err}(t,u)| \le 2 \sum_{y \le \sqrt{t}} |M(y)| + \sum_{2 \le k < \sqrt{t}} \Big(\Big|M\Big(\frac{t}{k}\Big)\Big| + \Big|M\Big(\frac{u}{k}\Big)\Big|\Big). \tag{11.39}$$

Proof. (i) In (11.11) group the indices $k \ge 2$ by the value $y = \lfloor t/k \rfloor$, respectively $y = \lfloor u/k \rfloor$. The number of $k \ge 1$ with $\lfloor t/k \rfloor = y$ is $\lfloor t/y \rfloor - \lfloor t/(y+1) \rfloor$, and for $y \le \lfloor t/2 \rfloor$ none of them is $k = 1$. The same holds for u, because $k = 1$ gives $y = u > \lfloor t/2 \rfloor$. (ii) For $x \in \{t, u\}$ the bracket $\lfloor x/y \rfloor - \lfloor x/(y+1) \rfloor$ equals $x/(y(y+1))$ minus a difference of two fractional parts, so it differs from $x/(y(y+1))$ by less than 1. For $y > \sqrt{t} \ge \sqrt{u}$ both $t/y - t/(y+1)$ and $u/y - u/(y+1)$ are < 1, so each bracket is 0 or 1, and it is 1 only if $y = \lfloor x/k \rfloor$ with $k \le x/y < \sqrt{t}$. (iii) $c_1 = (t-u) - (\lfloor t/2 \rfloor - \lfloor u/2 \rfloor)$ and $\lfloor t/2 \rfloor - \lfloor u/2 \rfloor \le \lceil (t-u)/2 \rceil$. (iv) Insert (ii) into (i) for $y \le \sqrt{t}$; the smooth part is $(t-u)\sum_{y \le \sqrt{t}} M(y)/(y(y+1)) = (t-u)\Upsilon(\sqrt{t})$. For $y > \sqrt{t}$ use the last assertion of (ii). □

Corollary 11.16 (Loss of the pointwise induction)**.** *Let $0 < \vartheta < 1$ and suppose $|M(y)| \le C y^{\vartheta}$ for all $y \le \lfloor t/2 \rfloor$. Then, for $u < t < 2u$ with $M(u) = 0$,*

$$\big|M(t) + (t-u)\Upsilon(\sqrt{t})\big| \ll_{\vartheta} C\, t^{(1+\vartheta)/2}, \tag{11.40}$$

while the full absolute kernel can, in general, have size comparable to the window length $t - u$ and yields at best $|M(t)| \ll_{\vartheta} C\,t$ from this information alone. For $\vartheta = \frac{1}{2} + \varepsilon$ the pointwise induction therefore loses $t^{1/2-\varepsilon}$, and $t^{1/4-\varepsilon/2}$ after the smooth component $-(t-u)\Upsilon(\sqrt{t})$ is extracted.

Proof. By Proposition 11.15(iv), $|\mathrm{Err}| \le 2C\sum_{y \le \sqrt{t}} y^{\vartheta} + 2Ct^{\vartheta}\sum_{2 \le k < \sqrt{t}} k^{-\vartheta} \ll_{\vartheta} Ct^{(1+\vartheta)/2}$. The full kernel satisfies $\sum_y |c_y| y^{\vartheta} \ge c_1 \ge \frac{1}{2}(t-u-1)$ by (iii), and $\sum_y |c_y| y^{\vartheta} \ll_{\vartheta} t$ by (ii). □

The smooth component is not small for termwise reasons: by (11.79),

$$\Upsilon(\sqrt{t}) = -\sum_{n > \sqrt{t}} \frac{M(n)}{n(n+1)}, \tag{11.41}$$

so its size is governed by the global identity $\sum_{n \ge 1} M(n)/(n(n+1)) = 0$, i.e. by the pole of ζ at $s = 1$ (Remark 11.52), and involves values of M beyond $t/2$. Even under RH (Littlewood's bound

$M(n) \ll_\varepsilon n^{1/2+\varepsilon}$ inserted in the tail) the tail gives only $(t-u)\Upsilon(\sqrt{t}) \ll t^{3/4+\varepsilon/2}$, the same order as the bound for $\mathrm{Err}(t,u)$; the RH scale for $M(t)$ is a statement about their sum.

Remark 11.17 (Localized intra-excursion system). Let $I = (u, x_1]$ be an excursion of length $\ell = x_1 - u$ and orientation σ_ν, and put

$$S_r := \sigma_\nu M(u+r) > 0 \qquad (1 \le r < \ell), \qquad S_0 = S_\ell = 0. \tag{11.42}$$

For a fixed $\lambda \in (\lambda_*, 2)$ set

$$h_\lambda(u) := \min\{\ell - 1, (\lambda - 1)u\}. \tag{11.43}$$

Then the entire RH-relevant initial segment satisfies simultaneously

$$S_r = -\sigma_\nu \sum_{y \le \lambda u/2} c_y(u+r, u) M(y) > 0, \qquad 1 \le r \le h_\lambda(u), \tag{11.44}$$

and $|S_r - S_{r-1}| \le 1$. For $\lambda = 5/4$ this is the segment $r \le \min\{\ell - 1, u/4\}$ and all dependencies lie below $5u/8$.

Definition 11.18 (Relaxed divisor-sum window class)**.** Fix $u \ge 2$ and $1 \le h < u$. For $\mathbf{a} = (a_d)_{d \le u} \in \{-1, 0, 1\}^u$ with $a_1 = 1$ and $\sum_{d \le u} a_d = 0$, put

$$D_{\mathbf{a}}(n) := \sum_{\substack{d \mid n \\ d \le u}} a_d, \qquad u < n \le u + h. \tag{11.45}$$

We call $\mathbf{a}$ window-admissible with orientation $\sigma_\nu \in \{-1, 1\}$ if $D_{\mathbf{a}}(n) \in \{-1, 0, 1\}$ throughout the window and

$$\sigma_\nu \sum_{u < n \le u + r} D_{\mathbf{a}}(n) < 0 \qquad (1 \le r \le h). \tag{11.46}$$

The induced oriented path is $S_r = -\sigma_\nu \sum_{u < n \le u+r} D_{\mathbf{a}}(n) > 0$. For the genuine Möbius prefix, these conditions reproduce the future-window increment constraint and the excursion-sign constraint coming from $\mu * 1 = \delta_1$, but they do not impose the full family of internal divisor identities below u.

Proposition 11.19 (Ballistic counterexample to the relaxed window class)**.** *The relaxed class of Definition 11.18 admits ballistic paths. Namely, take*

$$a_1 = 1, \qquad a_u = -1, \qquad a_d = 0 \quad (1 < d < u). \tag{11.47}$$

Then for every $u < n < 2u$ one has $D_{\mathbf{a}}(n) = 1$, and with $\sigma_\nu = -1$,

$$S_r = r \qquad (1 \le r < u). \tag{11.48}$$

If $m_y := \sum_{d \le y} a_d$, then $m_y = 1$ for every $y < u$. Hence for any $0 < \vartheta < 1$, any fixed $1 < \lambda < 2$, and $h = \lfloor (\lambda - 1)u \rfloor$,

$$\sup_{y \le \lambda u/2} \frac{|m_y|}{y^\vartheta} = 1, \qquad \frac{S_h}{(u+h)^\vartheta} \asymp_\lambda u^{1-\vartheta} \longrightarrow \infty. \tag{11.49}$$

Thus a maximum principle comparing the normalized localized-window height to the normalized history is false on this relaxed class.

Proof. For $u < n < 2u$, the integer u does not divide n. Hence among the two nonzero coordinates of $\mathbf{a}$ only $d = 1$ contributes to $D_{\mathbf{a}}(n)$, so $D_{\mathbf{a}}(n) = 1$. The remaining assertions are immediate. □

Remark 11.20 (What the relaxation loses). The counterexample violates the internal divisor identities of the Möbius prefix. Already for every prime $p < u$ the genuine identity requires $a_1 + a_p = 0$, whereas the ballistic vector above has $a_p = 0$. Consequently any successful localized argument of this type must use some of the relations

$$\sum_{d \mid n} a_d = 0 \qquad (2 \le n \le u), \tag{11.50}$$

not merely the future-window constraints.

Remark 11.21 (The prefix correlation identity). The completed-excursion bridge identity must not be applied to a truncated localized window. For $1 \le h < \ell$, put $x_j = \sigma_\nu \mu(u+j)$, $S_r = \sum_{j \le r} x_j$, and $C_g^{\langle h \rangle} = \sum_{j=1}^{h-g} x_j x_{j+g}$. Lemma 5.1 applied to the prefix of length h gives exactly

$$\sum_{r=1}^{h-1} S_r^2 + \sum_{g=1}^{h-1} g C_g^{\langle h \rangle} = S_h \sum_{r=1}^{h-1} S_r. \tag{11.51}$$

The right-hand side vanishes only for a completed bridge. Thus $Q_\nu = -\sum_g g C_g^{(\nu)}$ cannot be used as an identity internal to a truncated horizon unless the excursion actually returns there. In particular, passing from a large localized value to the energy of the whole excursion mixes the retained window with the part deliberately discarded by the horizon theorem.

Remark 11.22 (The one-step iteration gap). Even a one-step implication sending a large value in the localized window to a large value at some $y \le \lambda u/2$ would not automatically iterate. The point y need not lie in the localized window of the excursion that contains y; it may lie beyond $\lambda u'$ relative to that excursion's start u'. Sparsity of the family of λ-long excursions does not repair a single descent chain. Iteration therefore requires either a principle valid for all t, or a mechanism that places the new point inside an admissible window at every step.

Proposition 11.23 (Termwise absolute-value contraction gives no nontrivial gain)**.** *Let $(u, x_1]$ be an excursion, $1 \le h < \min\{x_1 - u, u\}$, and let $w_1, \dots, w_h \ge 0$, not all zero. Put $K_y = \sum_{r \le h} w_r c_y(u+r, u)$, so that $\sum_r w_r M(u+r) = -\sum_y K_y M(y)$. If, for some $0 < \vartheta < 1$, $\alpha \ge 0$ and $C_\alpha > 0$,*

$$\sum_y |K_y| y^\vartheta \le C_\alpha \sum_{r \le h} w_r (u+r)^\alpha, \tag{11.52}$$

then

$$\sum_{r \le h} w_r r \le (1 + 2C_\alpha) \sum_{r \le h} w_r (u+r)^\alpha, \qquad \sum_{r \le h} w_r |M(u+r)| \le (1 + 2C_\alpha) \sum_{r \le h} w_r (u+r)^\alpha. \tag{11.53}$$

Thus the contraction hypothesis can hold only on windows where the trivial bound $|M(u+r)| \le r$ already gives the conclusion.

Proof. Proposition 11.15(iii) gives $c_1(u+r, u) \ge (r-1)/2 \ge 0$, hence $\frac{1}{2}\sum_r w_r (r-1) \le K_1 \le C_\alpha \sum_r w_r (u+r)^\alpha$. Since $M(u) = 0$ and M has unit steps, $|M(u+r)| \le r$; since $(u+r)^\alpha \ge 1$, $\sum w_r \le \sum w_r (u+r)^\alpha$, and both inequalities follow. □

Remark 11.24 (No strict coercivity from the first coefficient). Writing $t = u + r$ gives the exact formula

$$c_1(u+r, u) = \begin{cases} \lceil r/2 \rceil, & u \text{ even}, \\ \lfloor r/2 \rfloor, & u \text{ odd}. \end{cases} \tag{11.54}$$

In particular $c_1(u+1, u) = 0$ when u is odd. The preceding proposition therefore rules out only this particular termwise absolute-value mechanism; it does not rule out a global signed contraction or cancellation among the coefficients K_y.

Definition 11.25 (Multiplicative layers of the prefix)**.** Let $\Omega(n)$ denote the number of prime factors of n, counted with multiplicity. For $P \ge 2$ and $k \ge 1$, a vector $(a_d)_{d \le P}$ satisfies the first k Möbius layers if $a_1 = 1$ and

$$\sum_{d \mid n} a_d = 0 \qquad (2 \le n \le P,\ \Omega(n) \le k). \tag{11.55}$$

Thus layer 1 imposes the identities at primes, layer 2 adds prime squares and products of two primes, and so on.

Proposition 11.26 (One-layer propagation into the localized future)**.** *Suppose $(a_d)_{d\le P}$ satisfies the first k Möbius layers. Then*

$$a_d = \mu(d) \qquad (d \le P,\ \Omega(d) \le k). \tag{11.56}$$

Moreover, for every integer n with

$$P < n \le 2P, \qquad \Omega(n) \le k+1, \tag{11.57}$$

one has

$$-\sum_{\substack{d\mid n\\ d\le P}} a_d = \mu(n). \tag{11.58}$$

Thus the first k multiplicative layers of the prefix determine the future increments at every integer n in the stated window $P < n \le 2P$ with $\Omega(n) \le k+1$.

Proof. If $d \mid n$ and $d < n$, then $n/d \ge 2$, hence $\Omega(d) \le \Omega(n) - 1$. We induct on $\Omega(n)$. For $\Omega(n) = 0$, $n = 1$ and $a_1 = 1 = \mu(1)$. If $2 \le n \le P$, $\Omega(n) \le k$, and $a_d = \mu(d)$ for every $d \le P$ with $\Omega(d) < \Omega(n)$, then every proper divisor d of n is such a d, and the layer equation gives $a_n = -\sum_{d\mid n,\, d<n} a_d = -\sum_{d\mid n,\, d<n} \mu(d) = \mu(n)$. For the second, every proper divisor of $n \le 2P$ is at most $n/2 \le P$ and has $\Omega(d) \le k$. Hence the displayed sum equals $-\sum_{d\mid n,\ d<n} \mu(d) = \mu(n)$. □

Remark 11.27 (Multiplicative-layer depth as a quantitative invariant). Fix $\chi \in (\chi_*, 1)$ and the localized window $u < n \le 2\chi u$. Let $k = k(u)$ be a depth up to which the internal Möbius divisor identities are imposed below $P = \chi u$. What growth of $k(u)$, if any, suffices to combine those identities with $M(u) = 0$ and the unit-step and excursion-sign constraints so as to force $\max_{u<t\le 2\chi u} |M(t)| \ll_\varepsilon u^{1/2+\varepsilon}$? A bounded depth would give a compressed arithmetic mechanism; imposing all layers reconstructs the Möbius prefix. Proposition 11.26 makes the depth a quantifiable intermediate invariant; the results below treat only fixed k.

Proposition 11.28 (Layer sums in the localized window)**.** *Fix $1 < \lambda < 2$ and an integer $j \ge 1$ before letting $u \to \infty$. Then*

$$\sum_{\substack{u<n\le \lambda u\\ \Omega(n)=j}} \mu(n) = (-1)^j\,(1+o(1))\,\frac{(\lambda-1)\,u\,(\log\log u)^{j-1}}{(j-1)!\,\log u}. \tag{11.59}$$

Proof. If $\Omega(n) = j$ and $\mu(n) \ne 0$, then n is squarefree with exactly j prime factors and $\mu(n) = (-1)^j$. By Landau's theorem for squarefree integers with exactly j distinct prime factors, the number $\pi_j^\flat(x)$ of such $n \le x$ satisfies $\pi_j^\flat(x) \sim f_j(x) := x(\log\log x)^{j-1}/((j-1)!\log x)$. *Citation TODO: verify the exact theorem/page for this squarefree $\omega(n) = j$ form; it must not be conflated with the corresponding $\Omega(n) = j$ count.* For fixed j and fixed $\lambda > 1$,

$$\frac{f_j(\lambda u)}{f_j(u)} = \lambda\,\frac{(\log\log(\lambda u))^{j-1}}{(\log\log u)^{j-1}}\,\frac{\log u}{\log(\lambda u)} \longrightarrow \lambda. \tag{11.60}$$

Hence $f_j(\lambda u) - f_j(u) = (\lambda - 1 + o(1))f_j(u)$, as claimed. □

Corollary 11.29 (Bounded layer depth)**.** *Fix $k \ge 1$, $\chi \in (\chi_*, 1)$, $P = \chi u$, and the window $u < n \le 2\chi u$.*

(i) *The increments fixed by the first k layers (Proposition 11.26), namely those with $\Omega(n) \le k+1$, form a set of $o(u)$ integers of the window.*

(ii) *Their summed contribution over the window is*

$$\sum_{\substack{u<n\le 2\chi u\\ \Omega(n)\le k+1}} \mu(n) = (-1)^{k+1}(1+o(1))\frac{(2\chi-1)\,u\,(\log\log u)^k}{k!\,\log u}, \tag{11.61}$$

so, for every fixed $\varepsilon > 0$ and all sufficiently large u, it exceeds $u^{1-\varepsilon}$ in modulus and has the fixed sign $(-1)^{k+1}$.

(iii) *Let $a_d = \mu(d)$ for $\Omega(d) \le k$ and $a_d = 0$ otherwise. For squarefree n in the window with $\omega(n) = w \ge k+2$,*

$$D_{\mathbf{a}}(n) = \sum_{\substack{d \mid n \\ d \le P}} a_d = \sum_{j=0}^{k} (-1)^j \binom{w}{j} = (-1)^k \binom{w-1}{k}, \tag{11.62}$$

which has modulus at least $k+1 \ge 2$. Thus the zero extension of a k-layer prefix violates the unit-step constraint at every such n, i.e. at all but $o(u)$ squarefree integers of the window.

Proof. (ii) follows from Proposition 11.28 summed over $j \le k+1$, the term $j = k+1$ dominating. For (i), Landau's asymptotic $\#\{n \le x : \Omega(n) = j\} \sim x(\log\log x)^{j-1}/((j-1)!\log x)$ for fixed j (all integers, not only squarefree ones) shows that the window contains $o(u)$ integers with $\Omega(n) \le k+1$. *Citation TODO: verify the exact theorem/page for this $\Omega(n) = j$ form.* For (iii), all divisors of n other than n are $\le n/2 \le P$, and n itself has $\Omega(n) = w > k$, so $D_{\mathbf{a}}(n)$ is the truncated alternating binomial sum, evaluated by induction on k via $\binom{w-1}{k} = \binom{w}{k} - \binom{w-1}{k-1}$. Finally, by Landau's theorem for squarefree integers with exactly j distinct prime factors (same squarefree ω-count citation TODO as above), $\#\{n \le x : \mu(n)^2 = 1,\ \omega(n) \le k+1\} = o(x)$, whereas $\#\{n \le x : \mu(n)^2 = 1\} \sim 6x/\pi^2$; hence all but $o(u)$ squarefree n in the window have $\omega(n) \ge k+2$. □

Remark 11.30 (What fixed layer depth proves). For every fixed k, Corollary 11.29 shows that the first k layers determine only a density-zero subset of the localized window, while the signed sum of the determined increments has modulus $(1+o(1))(2\chi-1)u(\log\log u)^k/(k!\log u)$, larger than $u^{1-\varepsilon}$ for every fixed $\varepsilon > 0$, once u is sufficiently large. Hence, if the full sum on the window is $O(u^{1/2+\varepsilon})$, substantial cancellation must occur between the increments determined by any fixed set of initial layers and the remaining ones. The results proved here do not determine how the required depth grows with u; in particular they do not show that $\asymp \log\log u$ layers (the typical size of Ω) are necessary or sufficient, which would require estimates uniform in $k = k(u)$. Part (iii) shows that the unit-step constraint is far from vacuous on a k-layer relaxation: it forbids the naive completion at almost every integer of the window, so the free coordinates a_d with $\Omega(d) > k$ are strongly constrained.

11.2. **The square range as an exact bilinear form of the prefix.**

Proposition 11.31 (Bilinear square-range truncation identity)**.** *Let $u \ge 1$ and $1 \le t \le u^2$. Then*

$$M(t) = 2M(\min(t,u)) - \sum_{d,j \le u} \mu(d)\mu(j) \left\lfloor \frac{t}{dj} \right\rfloor. \tag{11.63}$$

In particular, if $M(u) = 0$ and $u < t \le u^2$,

$$M(t) = -\mathcal{B}_u(t), \qquad \mathcal{B}_u(t) := \sum_{d,j \le u} \mu(d)\mu(j) \left\lfloor \frac{t}{dj} \right\rfloor, \qquad \mathcal{B}_u(u) = 0. \tag{11.64}$$

Proof. Let $\mu_u = \mu \mathbf{1}_{[1,u]}$. Each factor of $(\mu - \mu_u) * (\mu - \mu_u)$ is supported on integers $\ge u+1$, so the convolution, and its convolution with 1, vanish on $n < (u+1)^2$, in particular on $n \le u^2$. Expanding and using $\mu * 1 = \delta$ gives $\mu(n) = 2\mu_u(n) - (\mu_u * \mu_u * 1)(n)$ for $n \le u^2$; sum over $n \le t$. □

Remark 11.32 (Higher-order determinacy). For every $k \ge 1$ and $n \le u^k$,

$$\mu(n) = \sum_{j=1}^{k} (-1)^{j-1} \binom{k}{j} (\mu_u^{*j} * 1^{*(j-1)})(n). \tag{11.65}$$

Indeed, $(\mu-\mu_u)^{*k}$ is supported on integers at least $(u+1)^k$, so after convolution with $1^{*(k-1)}$ it still vanishes for $n<(u+1)^k$, in particular for $n\le u^k$. Expanding gives

$$0=\sum_{j=0}^{k}(-1)^j\binom{k}{j}\mu^{*(k-j)}*\mu_u^{*j}*1^{*(k-1)}. \tag{11.66}$$

For $j=0$, $\mu^{*k}*1^{*(k-1)}=\mu$. For $j\ge 1$, pair $k-j$ factors of μ with $k-j$ of the factors 1 to obtain $\mu^{*(k-j)}*1^{*(k-j)}=\delta$; the remaining $j-1$ factors of 1 produce the term in (11.65). Rearranging proves the formula. For $k=1$ this is the trivial truncation on $n\le u$; for $k=2$ it recovers the bilinear square-range truncation of Proposition 11.31; and the support argument above extends to the intended range $n\le u^k$. Thus determination by the prefix is not special to u^2. What is special to $t\le u^2$ is the order-two form (11.64), with rank-one coefficients $\mu(d)\mu(j)$ and elementary kernel $\lfloor t/(dj)\rfloor$. This is a truncation identity of Heath-Brown/Vaughan type [15]; summed forms of this kind are used in the fast computation of M [16].

Remark 11.33 (Square-excursion bridges). Let $u=x_{0,\nu}$ and $x_{1,\nu}\le u^2$. Then $\mathcal{B}_u(u)=\mathcal{B}_u(x_{1,\nu})=0$ and $-\sigma_\nu\mathcal{B}_u(t)>0$ for $u<t<x_{1,\nu}$. Can the rank-one coefficients $\mu(d)\mu(j)$, the kernel $\lfloor t/(dj)\rfloor$ and these bridge and sign constraints be combined to prove the bound

$$\max_{u<t\le x_{1,\nu}}|\mathcal{B}_u(t)|\ll_\varepsilon x_{1,\nu}^{1/2+\varepsilon} \tag{11.67}$$

uniformly over square-ordinary excursions? By Corollary 9.13 this uniform bound is equivalent to RH. By Corollary 11.14 it suffices to consider $t\le\min\{x_{1,\nu},\frac{5}{4}u\}$, where $\mathcal{B}_u(t)$ reduces to the localized linear form and uses only $\mu(d)$ with $d\le 5u/8$.

11.3. **Trial-division horizon of an excursion.**

Lemma 11.34 (Square horizon, classical)**.** *Let $u\ge 2$. Trial division using the primes $p\le u$ computes $\mu(n)$ for every $n\le u^2$.*

Proof. Remove the prime factors $p\le u$ of n; if one occurs twice, $\mu(n)=0$. The cofactor $q\le u^2$ has all prime factors $>u$, so it is 1 or a single prime. □

Remark 11.35 (Determinacy is not information). Primes $>u$ are not logically new information, and determination by a finite prefix is not special to u^2: (11.65) propagates a prefix to every power range, and (11.10) determines M from $M(1)$. The ranges $(u,2u]$ and $(u,u^2]$ are special only in the concrete sense that the exact summatory formula is of order one ((11.2)) or two ((11.64)) in the prefix, and that one-stage trial division by primes $\le u$ suffices on $(u,u^2]$.

Definition 11.36 (Smooth and rough parts)**.** For $y>1$ write $n=s_y(n)\,r_y(n)$, where every prime factor of $s_y(n)$ is $\le y$ and every prime factor of $r_y(n)$ is $>y$. Let $P^+(n)$ be the largest prime factor of n, with $P^+(1)=1$.

Lemma 11.37 (Smooth and rough parts, classical)**.** $\mu(n)=\mu(s_y(n))\,\mu(r_y(n))$. *If $n<y^k$ then $\Omega(r_y(n))\le k-1$, and if $n\le y^2$ then $r_y(n)$ is 1 or a prime.*

Proof. The two parts are coprime. If $\Omega(r)\ge 1$ then $y^{\Omega(r)}<r\le n$. □

Remark 11.38 (The horizon u^2 is not the multiplicative threshold of an excursion). For a square-ordinary excursion $\sqrt{v_\nu}\le u_\nu$. Every $n\le v_\nu$ has at most one prime factor $>\sqrt{v_\nu}$, so the threshold that governs the multiplicative structure inside I_ν is $\sqrt{v_\nu}$, below u_ν. The regime $t>u_\nu^2$ of Section 11.4 lies beyond v_ν for a square-ordinary excursion; that section concerns $M(t)$ globally, and Section 11.5 gives the local version inside an excursion.

11.4. **Large-prime decomposition at level U^2.** The identities of this section are decompositions by large prime factors of Buchstab/Legendre type. The asymptotic statements use only the prime number theorem with classical error term and the Vinogradov–Korobov bound [35, 55].

Definition 11.39 (Smooth baseline and large-prime part)**.** For $U \geq 2$ and $t \geq 1$ put

$$M_U^{\mathrm{sm}}(t) := \sum_{\substack{n \leq t \\ P^+(n) \leq U^2}} \mu(n), \qquad M_U^{\mathrm{lp}}(t) := \sum_{\substack{n \leq t \\ P^+(n) > U^2}} \mu(n), \qquad M(t) = M_U^{\mathrm{sm}}(t) + M_U^{\mathrm{lp}}(t). \tag{11.68}$$

Proposition 11.40 (Single-prime copy, classical)**.** *For a prime p and $t < p^2$,*

$$\mathcal{D}_p(t) := \sum_{a \leq t/p} \mu(ap) = -M(t/p). \tag{11.69}$$

Proof. $a \leq t/p < p$, so $p \nmid a$ and $\mu(ap) = -\mu(a)$. □

Proposition 11.41 (Sign content of one prime cone)**.** *Let $\mathcal{S}(y) = \sum_{a \leq y} \mu(a)^2$. Among the numbers ap with $a \leq y < p$, the numbers of values $+1, -1, 0$ of μ are*

$$N_+ = \frac{\mathcal{S}(y) - M(y)}{2}, \qquad N_- = \frac{\mathcal{S}(y) + M(y)}{2}, \qquad N_0 = \lfloor y \rfloor - \mathcal{S}(y), \tag{11.70}$$

with limiting proportions $3/\pi^2,\ 3/\pi^2,\ 1 - 6/\pi^2$.

Proof. $\mu(ap) = -\mu(a)$, and $\#\{a \leq y : \mu(a) = \mp 1\} = (\mathcal{S}(y) \mp M(y))/2$. □

Proposition 11.42 (First-order window, classical)**.** *If $U^2 < t < U^4$, no $n \leq t$ has two prime factors $> U^2$ (with multiplicity), and*

$$M_U^{\mathrm{lp}}(t) = -\sum_{U^2 < p \leq t} M(t/p). \tag{11.71}$$

Proof. Two such factors would give $n > U^4 > t$. Hence $M_U^{\mathrm{lp}}(t) = \sum_{U^2 < p \leq t} \mathcal{D}_p(t)$, and $t < U^4 < p^2$; apply (11.69). □

Proposition 11.43 (Prime-count form)**.** *For $U^2 < t < U^4$ and $W := t/U^2$,*

$$M_U^{\mathrm{lp}}(t) = -\sum_{a \leq W} \mu(a)\big(\pi(t/a) - \pi(U^2)\big), \qquad M(t) = M_U^{\mathrm{sm}}(t) - \sum_{a \leq W} \mu(a)\pi(t/a) + M(W)\pi(U^2). \tag{11.72}$$

Proof. Expand (11.71) as $-\sum_{U^2 < p \leq t} \sum_{a \leq t/p} \mu(a)$. This is a finite sum, so reversing the order is purely combinatorial. For fixed a the prime runs over $U^2 < p \leq t/a$, which is non-empty only if $a < W$. The displayed sum may harmlessly be written with $a \leq W$: if W is an integer, the endpoint $a = W$ contributes $\pi(U^2) - \pi(U^2) = 0$. □

Thus the only input from primes $> U^2$ is the value of π at the points t/a, $a \leq W$, weighted by $\mu(a)$ at the small scale $W < U^2$.

Definition 11.44 (Copy multiplicities)**.** For $U^2 < t < U^4$, $W = t/U^2$ and $1 \leq y \leq \lfloor W \rfloor$ put

$$N_y(t, U) = \#\Big\{p :\ \max(U^2, \tfrac{t}{y+1}) < p \leq \tfrac{t}{y}\Big\}, \qquad \text{so that} \qquad M_U^{\mathrm{lp}}(t) = -\sum_{1 \leq y \leq \lfloor W \rfloor} N_y(t, U) M(y). \tag{11.73}$$

Remark 11.45 (Asymptotics of the multiplicities). For fixed $W > 1$ and $U \to \infty$,

$$N_y \sim \frac{WU^2}{2 \log U} \cdot \frac{1}{y(y+1)} \quad (1 \leq y \leq \lfloor W \rfloor - 1), \qquad N_{\lfloor W \rfloor} \sim \frac{U^2}{2 \log U}\Big(\frac{W}{\lfloor W \rfloor} - 1\Big) \quad (W \notin \mathbb{Z}), \tag{11.74}$$

and $N_W = 0$ if $W \in \mathbb{Z}$. The top multiplicity is truncated because $t/(\lfloor W \rfloor + 1) < U^2$.

Definition 11.46 (Coherent coefficient)**.**

$$\mathcal{C}(W) := \int_1^W \frac{M(w)}{w^2}\,dw = \sum_{a\le W} \mu(a)\Big(\frac{1}{a} - \frac{1}{W}\Big) = \sum_{a\le W} \frac{\mu(a)}{a} - \frac{M(W)}{W}. \tag{11.75}$$

Proposition 11.47 (Fixed W)**.** *For fixed $W > 1$, as $U \to \infty$,*

$$M_U^{\mathrm{lp}}(WU^2) = -\frac{WU^2}{2\log U}\big(\mathcal{C}(W) + o(1)\big). \tag{11.76}$$

Proof. In (11.72) the sum is finite; for $a < W$ the prime number theorem gives $\pi(WU^2/a) - \pi(U^2) \sim (W/a - 1)U^2/(2\log U)$, and $\sum_{a\le W} \mu(a)(W/a - 1) = W\,\mathcal{C}(W)$. □

Lemma 11.48 (The multiplicity-weighted coefficient)**.** *The coefficient*

$$\Upsilon(W) := \sum_{1\le y\le W} \frac{M(y)}{y(y+1)} = \sum_{n\le W} \frac{\mu(n)}{n} - \frac{M(W)}{\lfloor W\rfloor + 1} \tag{11.77}$$

satisfies

$$\Upsilon(W) = \mathcal{C}\big(\lfloor W\rfloor + 1\big), \qquad \mathcal{C}(W) = \Upsilon(W) - M(W)\Big(\frac{1}{W} - \frac{1}{\lfloor W\rfloor + 1}\Big), \tag{11.78}$$

and

$$\mathcal{C}(W) = -\int_W^\infty \frac{M(w)}{w^2}\,dw, \qquad \Upsilon(W) = -\sum_{n>W} \frac{M(n)}{n(n+1)}. \tag{11.79}$$

Proof. M is constant on $[y, y+1)$ and $\int_y^{y+1} w^{-2}dw = 1/(y(y+1))$, which gives (11.78). For $s > 1$, (8.1) gives $\int_1^\infty M(w)w^{-s-1}\,dw = 1/(s\zeta(s)) \to 0$ as $s \downarrow 1$. For $s \in [1, 1+\delta]$ the bound $M(w) \ll we^{-c\sqrt{\log w}}$ dominates the integrand by an integrable multiple of $e^{-c\sqrt{\log w}}/w$, so dominated convergence gives $\int_1^\infty M(w)w^{-2}dw = 0$. □

Remark 11.49 ($\mathcal{C}$ versus Υ)*.* Summing (11.74) against $M(y)$ suggests the coefficient $\Upsilon(W)$, but the truncated top multiplicity replaces it by $\mathcal{C}(W)$; by (11.78) the two agree exactly when $M(\lfloor W\rfloor) = 0$. For $1 < W < 2$ one has $M_U^{\mathrm{lp}} = -(\pi(t) - \pi(U^2)) \sim -(W-1)U^2/(2\log U)$, i.e. $\mathcal{C}(W) = 1 - 1/W$, whereas $\Upsilon(W) = 1/2$.

Corollary 11.50 (Decay of the coefficients)**.** *By* (11.79) *and* (7.3)*, there is $c_1 > 0$ such that, for $W > e$,*

$$\mathcal{C}(W),\ \Upsilon(W) \ll \exp\big(-c_1(\log W)^{3/5}(\log\log W)^{-1/5}\big). \tag{11.80}$$

Proposition 11.51 (Asymptotic size of the smooth and large-prime components)**.** (i) *Suppose $W > 1$ is fixed and $\mathcal{C}(W) \neq 0$. Then*

$$M_U^{\mathrm{lp}}(WU^2) \sim -\frac{WU^2}{2\log U}\mathcal{C}(W), \tag{11.81}$$

$$M_U^{\mathrm{sm}}(WU^2) \sim +\frac{WU^2}{2\log U}\mathcal{C}(W). \tag{11.82}$$

For $1 < W < 2$ one has $\mathcal{C}(W) = 1 - 1/W > 0$, so the hypothesis holds.

(ii) *There is $C > 0$ such that, as $t \to \infty$, uniformly for $U^2 < t < U^4$ with $W = t/U^2 \ge \exp(C(\log\log t)^2)$,*

$$M_U^{\mathrm{lp}}(t) = (1 + o(1))\frac{t}{(\log t)^2}, \qquad M_U^{\mathrm{sm}}(t) = -(1 + o(1))\frac{t}{(\log t)^2}. \tag{11.83}$$

In particular, in both regimes $|M_U^{\mathrm{lp}}(t)|$ and $|M_U^{\mathrm{sm}}(t)|$ exceed $t^{1/2+\varepsilon}$ for every fixed $\varepsilon < 1/2$ and all large t, and only their sum M can be compared with the scale $t^{1/2+\varepsilon}$.

Proof. We use the classical form [35]: for some $c > 0$,

$$\pi(y) = \mathrm{li}(y) + O(ye^{-c\sqrt{\log y}}), \quad M(y) \ll ye^{-c\sqrt{\log y}}, \quad \widetilde{m}(y) := \sum_{a\le y} \frac{\mu(a)}{a} \ll e^{-c\sqrt{\log y}}, \tag{11.84}$$

and the series $\sum_{n\ge1} \mu(n)\log n/n = -1$ converges conditionally. This follows, for example, from the displayed bound on $\widetilde{m}$ and partial summation (and agrees with the limit at $s = 1$ of $-\frac{d}{ds}\zeta(s)^{-1}$). The integral below converges absolutely by the bound on $\widetilde{m}$, and

$$\int_1^\infty \widetilde{m}(w)\frac{dw}{w} = \lim_{Y\to\infty}\Big(\widetilde{m}(Y)\log Y - \sum_{a\le Y}\frac{\mu(a)\log a}{a}\Big) = 1. \tag{11.85}$$

(i) Proposition 11.47, and $M(t) = o(t/\log t)$. For $1 < W < 2$, $M = 1$ on $[1, 2)$.

(ii) Since $t < U^4$, $W < \sqrt{t}$, so $\log(t/a) \ge \frac12 \log t$ for $a \le W$. By (11.72),

$$M_U^{\mathrm{lp}}(t) = -\sum_{a\le W}\mu(a)\,\mathrm{li}(t/a) - \sum_{a\le W}\mu(a)(\pi(t/a) - \mathrm{li}(t/a)) + M(W)\pi(U^2). \tag{11.86}$$

The second term is $\ll t(1+\log W)e^{-c\sqrt{\log t/2}} = o(t/\log^2 t)$. For the third, $\frac14\log t < \log U < \frac12\log t$, so

$$|M(W)|\pi(U^2) \ll We^{-c\sqrt{\log W}}\frac{U^2}{\log U} \ll e^{-c\sqrt{\log W}}\frac{t}{\log t} = o(t/\log^2 t) \tag{11.87}$$

once $c\sqrt{C} > 1$, where c is the constant in (11.84); we fix such a C. The same condition makes the boundary term of the partial summation below $o(t/\log^2 t)$, and the second term of (11.86) is $o(t/\log^2 t)$ for any C. For the first, with $F(w) = w\,\mathrm{li}(t/w)$,

$$\sum_{a\le W}\mu(a)\,\mathrm{li}(t/a) = \widetilde{m}(W)F(W) - \int_1^W \widetilde{m}(w)F'(w)\,dw, \tag{11.88}$$

$$F'(w) = \mathrm{li}(t/w) - \frac{t/w}{\log(t/w)} = \frac{t/w}{\log^2(t/w)}\Big(1 + O\Big(\frac{1}{\log t}\Big)\Big). \tag{11.89}$$

The boundary term is $\ll e^{-c\sqrt{\log W}}t/\log t = o(t/\log^2 t)$. The integral equals

$$\frac{t}{\log^2 t}\left(\int_1^W \widetilde{m}(w)g_t(w)\frac{dw}{w} + O(1/\log t)\right), \qquad g_t(w) = \frac{\log^2 t}{\log^2(t/w)} \in [1, 4]. \tag{11.90}$$

The required uniformity in (U, W) follows by an explicit ε–W argument. Given $\varepsilon_1 > 0$, choose W_0 so that

$$\int_{W_0}^\infty \frac{|\widetilde{m}(w)|}{w}\,dw < \varepsilon_1. \tag{11.91}$$

For all sufficiently large t, the lower bound on W gives $W > W_0$. On $[1, W_0]$,

$$\sup_{1\le w\le W_0}|g_t(w) - 1| \ll \frac{\log W_0}{\log t} \to 0, \tag{11.92}$$

while on $[W_0, W]$ one has $1 \le g_t(w) \le 4$. The omitted interval (W, ∞) is also contained in the chosen tail. Therefore

$$\int_1^W \widetilde{m}(w)g_t(w)\frac{dw}{w} = \int_1^\infty \widetilde{m}(w)\frac{dw}{w} + o(1) = 1 + o(1), \tag{11.93}$$

uniformly over all admissible pairs (U, W). Hence $M_U^{\mathrm{lp}}(t) = (1+o(1))t/\log^2 t$, and $M_U^{\mathrm{sm}} = M - M_U^{\mathrm{lp}}$ with $M(t) = o(t/\log^2 t)$. □

Remark 11.52 (Why $t/\log^2 t$). $\mathcal{C}(W)$ captures only the $1/\log t$ term, whose coefficient tends to zero by Corollary 11.50. The $1/\log^2 t$ term comes from $\zeta'(s)/\zeta(s)^2 \to -1$, that is, from the pole of ζ at $s = 1$, and does not decay in W.

Remark 11.53 (First copy transitions). For $1 < W < 2$ every copy has length 1 and $M_U^{\mathrm{lp}}(t) = -(\pi(t) - \pi(U^2))$. For $2 < W < 3$ copies of length 2 appear but $M(2) = 0$. For $3 < W < 4$ copies of length 3 enter with $M(3) = -1$, of opposite net sign.

11.5. **Local large-prime decomposition of an excursion.**

Proposition 11.54 (Local decomposition)**.** *Let $I_\nu = (u, v]$ be a completed excursion and $y = \sqrt{v}$. For $u < t \le v$,*

$$M(t) = \sum_{\substack{u<n\le t\\ P^+(n)\le y}} \mu(n) - \sum_{a<y} \mu(a)\,\Pi_a(t), \qquad \Pi_a(t) := \#\Big\{p:\ \max(y, \tfrac{u}{a}) < p \le \tfrac{t}{a}\Big\}, \tag{11.94}$$

and the bridge condition $M(v) = 0$ reads

$$\sum_{\substack{u<n\le v\\ P^+(n)\le \sqrt{v}}} \mu(n) = \sum_{a<\sqrt{v}} \mu(a)\,\Pi_a(v). \tag{11.95}$$

Proof. $M(u) = 0$. An $n \le v = y^2$ has at most one prime factor $p > y$; then $n = ap$ with $a < v/y = y < p$, so $\mu(n) = -\mu(a)$, and $(a, p) \mapsto ap$ is a bijection onto these n. □

Remark 11.55 (Primes in short intervals). The decomposition holds for every excursion, with threshold $\sqrt{v_\nu}$. The large-prime input consists of prime counts in the intervals $(u/a, t/a]$, of length at most ℓ_ν/a at height about v_ν/a. Huxley proved the prime number theorem uniformly in intervals of length $h \ge x^{7/12+\varepsilon}$ [19]; the zero-density estimates of Guth–Maynard give intervals of length $x^{17/30+o(1)}$ [10]. For Möbius sums, Matomäki–Teräväinen prove $\sum_{x<n\le x+x^\theta} \mu(n) = o(x^\theta)$ for every $\theta > 0.55$ [31]. The cited results do not provide, below these ranges, an asymptotic for Π_a uniform in a.

Lemma 11.56 (Inverse logarithms)**.** *Let C be as in Proposition 11.51. Uniformly for*

$$\exp(C(\log\log x)^2) \le z \le \sqrt{x}, \tag{11.96}$$

$$\sum_{a\le z} \frac{\mu(a)}{a\log(x/a)} = -\frac{1+o(1)}{(\log x)^2}. \tag{11.97}$$

Proof. Partial summation gives

$$\frac{\widetilde{m}(z)}{\log(x/z)} - \int_1^z \frac{\widetilde{m}(w)}{w\log^2(x/w)}\,dw. \tag{11.98}$$

Since $z \le \sqrt{x}$, the boundary term is uniformly $O(e^{-c\sqrt{\log z}}/\log x)$, hence $o(1/\log^2 x)$. Write $\log^{-2}(x/w) = g_x(w)\log^{-2} x$ with $1 \le g_x(w) \le 4$, extend $g_x \mathbf{1}_{\{w\le z\}}$ by zero, and apply dominated convergence with (11.85). □

Heuristic 11.57 (Opposite drifts)**.** Let $u \ge \sqrt{v}\exp(C(\log\log v)^2)$. Near $t \in I_\nu$ the counts Π_a grow for $a < t/\sqrt{v}$, at rate $1/(a\log(t/a))$ if primes near t/a have their expected local density uniformly in a. Over $(t, t+h]$ the large-prime part then changes by about $-h\sum_{a<t/\sqrt{v}} \mu(a)/(a\log(t/a)) = (1+o(1))h/\log^2 t$ by Lemma 11.56. Since $M(v) - M(u) = 0$, the $\sqrt{v}$-smooth part drifts at rate about $-1/\log^2 v$, independently of σ_ν. As

$$\frac{\ell_\nu}{\log^2 v_\nu} \gg \sqrt{\ell_\nu} \iff \ell_\nu \gg \log^4 v_\nu, \tag{11.99}$$

for $\ell_\nu \gg (\log v_\nu)^4$ both drift contributions exceed $\sqrt{\ell_\nu}$, and (11.95) is a balance between two opposite drifts: the local analogue of Proposition 11.51.

12. Two-point correlations and the energy of excursions

For integers $0 \le u < X$ and $1 \le h < X - u$ put

$$C_h^{(u,X]} := \sum_{u<n\le X-h} \mu(n)\mu(n+h), \qquad C_h^{[X]} := C_h^{(0,X]}. \tag{12.1}$$

For an excursion $I_\nu = (x_{0,\nu}, x_{1,\nu}]$ one has $C_h^{(\nu)} = C_h^{(x_{0,\nu},x_{1,\nu}]}$, since $\sigma_\nu^2 = 1$. Chowla's conjecture asserts $C_h^{[X]} = o(X)$ for each fixed $h \ge 1$. Matomäki, Radziwiłł and Tao [29] prove, for $K = K(X) \to \infty$, $K \le X$,

$$\sum_{h\le K} \left| \sum_{n\le X} \mu(n)\mu(n+h) \right| = o(KX). \tag{12.2}$$

Since replacing the upper limit X by $X - h$ changes the h-th sum by at most h, we shall use the triangular consequence

$$\sum_{h\le K} |C_h^{[X]}| = o(KX) \qquad (K \to \infty,\ K = o(X)). \tag{12.3}$$

Their stronger averaged Elliott form permits translated shifts of size $O(X)$ and covers μ.

12.1. **Signed lag averages and the weighted RH criterion.**

Remark 12.1 (Unweighted signed averages are degenerate). By Lemma 5.3, every excursion satisfies $\sum_h C_h^{(\nu)} = -q_\nu/2$, with $q_\nu \le \ell_\nu$. Hence $\ell_\nu^{-2} \sum_h C_h^{(\nu)} = O(\ell_\nu^{-1})$ for purely algebraic reasons: a signed average over lags of the correlations inside a bridge is small without any pseudorandomness of μ. Statements of Chowla type inside an excursion must therefore carry absolute values inside the lag average, or impose a genuinely weighted condition.

The weighted sum behaves differently. By (5.8), $Q_\nu = -\sum_h hC_h^{(\nu)}$: the weight h records at which lags the correlation responsible for the excursion energy sits. The same identity holds globally at every zero of M, and the cross terms between different excursions cancel exactly.

Proposition 12.2 (Weighted correlations at zeros of M)**.** *Let $v \ge 2$ be a zero of M. Then*

$$\sum_{n<v} M(n)^2 = -\sum_{h<v} h\, C_h^{[v]}, \qquad \sum_{h<v} h\, C_h^{[v]} = \sum_{\nu:\, x_{1,\nu}\le v} \sum_h h\, C_h^{(\nu)}. \tag{12.4}$$

Thus, in the h-weighted sum, the correlations between increments lying in different completed excursions cancel exactly; internal zero-run increments contribute nothing because $\mu(n) = 0$ there.

Proof. Apply Lemma 5.1 to the block $x_n = \mu(n)$, $1 \le n \le v$: then $S_t = M(t)$ and $S_v = M(v) = 0$, which gives the first identity. Because $M(v) = 0$, $\sum_{n\le v} M(n)^2 = \sum_{n<v} M(n)^2$. By (5.12), this equals $\sum_\nu Q_\nu$ over the completed excursions ending at or before v, and $Q_\nu = -\sum_h hC_h^{(\nu)}$ by (5.8). □

Theorem 12.3 (Weighted-correlation criterion)**.** *Using the unconditional return localization (RL) of Corollary 9.8, RH holds if and only if, for every $\varepsilon > 0$,*

$$-\sum_{h<v} h\, C_h^{[v]} \ll_\varepsilon v^{2+\varepsilon} \qquad \text{for every integer } v \text{ with } M(v) = 0. \tag{12.5}$$

Proof. If RH holds, Theorem 8.6 and (12.4) give the bound. Conversely, fix a target $\varepsilon_0 > 0$ and choose $\varepsilon > 0$ so small that $(1+\varepsilon)(2+\varepsilon) \le 2+\varepsilon_0$. Let x be large and put $N = \lfloor x \rfloor$. If $M(N) = 0$, then $\sum_{n\le x} M(n)^2 = B_2(N)$ is already evaluated at a zero. Otherwise, let $u = \max\{n \le N : M(n) = 0\}$ and let $v > N$ be the next zero. Then $(u, v]$ is the completed excursion whose open prefix contains

N, and Corollary 9.8 gives $v \le u^{1+\varepsilon} \le x^{1+\varepsilon}$ for all sufficiently large x. In either case, with $v = N$ in the zero case,

$$\sum_{n\le x} M(n)^2 \le \sum_{n<v} M(n)^2 \ll_\varepsilon v^{2+\varepsilon} \le x^{2+\varepsilon_0}. \tag{12.6}$$

Theorem 8.6 now applies. □

Proposition 12.4 (Square-root lag cutoff at zeros)**.** *Let $v \to \infty$ through integers satisfying $M(v) = 0$, and let $K = K(v) \to \infty$ with $K \le v^{1/2}$. Then*

$$\sum_{h\le K} h\, C_h^{[v]} = o(K^2 v). \tag{12.7}$$

In particular,

$$\sum_{n<v} M(n)^2 = -\sum_{\sqrt{v}<h<v} h\, C_h^{[v]} + o(v^2), \tag{12.8}$$

where the lower limit may be replaced by $\lfloor\sqrt{v}\rfloor$ without changing the statement.

Proof. By (12.3), $\sum_{h\le K} |C_h^{[v]}| = o(Kv)$. Therefore

$$\left|\sum_{h\le K} hC_h^{[v]}\right| \le K \sum_{h\le K} |C_h^{[v]}| = o(K^2 v). \tag{12.9}$$

Taking $K = \lfloor\sqrt{v}\rfloor$ and using Proposition 12.2 gives (12.8). □

Corollary 12.5 (RH after deleting all small lags)**.** *RH holds if and only if, for every $\varepsilon > 0$,*

$$-\sum_{\sqrt{v}<h<v} h\, C_h^{[v]} \ll_\varepsilon v^{2+\varepsilon} \qquad \text{for every sufficiently large zero } v \text{ of } M. \tag{12.10}$$

Thus the lags $h \le \sqrt{v}$ are unconditionally negligible at the critical v^2 scale of the weighted-correlation criterion.

Theorem 12.6 (Zero-edge lag localization)**.** *Let $v \to \infty$ through integers satisfying $M(v) = 0$. Then*

$$-\sum_{h<v} hC_h^{[v]} = \sum_{n<v} M(n)^2 = v^{1+2\Theta+o(1)}. \tag{12.11}$$

Moreover, for every fixed $0 \le \alpha < \Theta$,

$$\sum_{h\le v^\alpha} hC_h^{[v]} = o\left(\sum_{n<v} M(n)^2\right), \tag{12.12}$$

and hence

$$-\sum_{v^\alpha<h<v} hC_h^{[v]} = (1+o(1))\sum_{n<v} M(n)^2. \tag{12.13}$$

Thus the signed weighted correlation total is forced beyond every power v^α with $\alpha < \Theta$.

Proof. The first equality is Proposition 12.2. Since v is an integer zero of M, the final summand vanishes and $B_2(v) = \sum_{n\le v} M(n)^2 = \sum_{n<v} M(n)^2$; hence the asymptotic exponent follows by applying Corollary 8.11 along the subsequence of such zeros $v \to \infty$. For $K = \lfloor v^\alpha\rfloor$ with $\alpha > 0$, (12.3) gives

$$\left|\sum_{h\le K} hC_h^{[v]}\right| \le K \sum_{h\le K} |C_h^{[v]}| = o(K^2 v) = o(v^{1+2\alpha}). \tag{12.14}$$

Choose a fixed δ_0 with $0 < \delta_0 < 2(\Theta - \alpha)$. Along zeros $v \to \infty$, Corollary 8.11 gives $B_2(v) \ge v^{1+2\Theta-\delta_0}$ for all sufficiently large v, while the small-lag contribution is $o(v^{1+2\alpha})$. Their ratio is

therefore $o(v^{-2(\Theta-\alpha)+\delta_0}) = o(1)$. The case $\alpha = 0$ follows trivially from the fixed finite number of lags and $|C_h^{[v]}| \le v$. □

Remark 12.7 (Endpoint versus the universal square-root cutoff). Proposition 12.4 is uniform without knowing Θ and reaches the endpoint $K = \sqrt{v}$ at the absolute v^2 scale. Theorem 12.6 is stronger in relative terms but requires a strict power gap $\alpha < \Theta$; it does not settle the endpoint $\alpha = \Theta$. Under RH it says that every range $h \le v^{1/2-\delta}$ carries $o(1)$ of the total signed weighted correlation, while if RH fails the negligible range extends correspondingly beyond the square-root scale.

Remark 12.8 (Where the cancellation must occur). The trivial bound is $\sum_h h|C_h^{[v]}| \le \sum_h h(v-h) \le v^3/6$. RH requires the signed weighted sum to be $v^{2+o(1)}$. Absolute values cannot give this at the natural scale: even square-root cancellation $|C_h^{[v]}| \ll v^{1/2+o(1)}$ in every single lag would give only $\sum_h h|C_h^{[v]}| \ll v^{5/2+o(1)}$. Corollary 12.5 sharpens the location of the unresolved cancellation: at the v^2 scale, only lags $h > \sqrt{v}$ remain. By Proposition 12.2 the signed weighted identity itself decomposes excursion by excursion.

12.2. **Lag localization of the energy of an excursion.**

Proposition 12.9 (Energy beyond a lag cutoff). *For every excursion I_ν of length $\ell = \ell_\nu$ and every $1 \le h_0 < \ell$,*

$$\Big|Q_\nu + \sum_{h_0<h<\ell} h\, C_h^{(\nu)}\Big| \le h_0 \sum_{h\le h_0} |C_h^{(\nu)}|. \tag{12.15}$$

Consequently, if $\sum_{h\le h_0} |C_h^{(\nu)}| \le \delta h_0 \ell$, then

$$\sum_{h_0<h<\ell} h\, C_h^{(\nu)} = -Q_\nu + O(\delta h_0^2 \ell). \tag{12.16}$$

Proof. Split $Q_\nu = -\sum_{h<\ell} h C_h^{(\nu)}$ at h_0. □

Corollary 12.10 (Averaged Chowla on linear-scale intervals). *Let $(u_j, u_j + L_j]$ be intervals with $L_j \to \infty$ and $u_j \le BL_j$ for a fixed B, and let $h_j \to \infty$ with $h_j = o(L_j)$. Then*

$$\sum_{h\le h_j} |C_h^{(u_j,u_j+L_j]}| = o(h_j L_j). \tag{12.17}$$

Proof. Put $X_j = L_j - h_j$. Then $X_j \sim L_j$ and, for all large j, $u_j \le 2BX_j$. Apply the shifted form of [29, Theorem 1.6], whose hypotheses explicitly allow translation parameters $b_r \le AX_j$, to the two-point Möbius correlation. The same paper states immediately after Theorem 1.6 that the argument proving its averaged Chowla theorem also applies with λ replaced by μ. Taking one relative shift fixed and averaging the other over $1 \le h \le h_j$, with the common translation $u_j = O(X_j)$ absorbed into the allowed b_r, gives

$$\sum_{h\le h_j} \left| \sum_{1\le n\le X_j} \mu(u_j+n)\mu(u_j+n+h) \right| = o(h_j X_j) = o(h_j L_j). \tag{12.18}$$

For each $h \le h_j$, extending the inner sum from $X_j = L_j - h_j$ to $L_j - h$ adds at most $h_j - h$ terms. Hence the total triangular boundary cost is

$$\sum_{h\le h_j} (h_j - h) = O(h_j^2) = o(h_j L_j), \tag{12.19}$$

which proves the claim. □

Corollary 12.11 (Large-lag concentration in linear-scale excursions)**.** *Let $B > 0$ be fixed and let I_j be excursions with $\ell_j \to \infty$, $x_{0,j} \le B\ell_j$ and heights H_j with $H_j^3/\ell_j \to \infty$. For every $h_j \to \infty$ with $h_j^2\ell_j = o(H_j^3)$,*

$$\sum_{h_j<h<\ell_j} h\, C_h^{(I_j)} = -(1+o(1))Q_{I_j}. \tag{12.20}$$

If in addition $\Theta > 1/2$, $0 < \eta < \Theta - \frac{1}{2}$ is fixed, there is a fixed $\lambda' > 1$ with $x_{1,j} \le \lambda' x_{0,j}$, and these excursions carry Kaczorowski–Pintz witnesses $H_j \ge \frac{1}{2}x_{0,j}^{\Theta-\eta}$, then for every fixed

$$0 < \theta_{\mathrm{lag}} < \frac{3(\Theta-\eta)-1}{2} \tag{12.21}$$

one may take $h_j = x_{0,j}^{\theta_{\mathrm{lag}}}$, and hence

$$\sum_{x_{0,j}^{\theta_{\mathrm{lag}}}<h<\ell_j} h\, C_h^{(I_j)} = -(1+o(1))Q_{I_j}. \tag{12.22}$$

Proof. By Corollary 12.10, $\sum_{h\le h_j}|C_h^{(I_j)}| = \delta_j h_j \ell_j$ with $\delta_j \to 0$. Proposition 12.9 gives an error at most $\delta_j h_j^2 \ell_j = o(H_j^3) = o(Q_{I_j})$. In the witness case the hypotheses $x_{0,j} \le B\ell_j$ and $x_{1,j} \le \lambda' x_{0,j}$ give $\ell_j \asymp x_{0,j}$. Hence $H_j^3/\ell_j \gg x_{0,j}^{3\Theta-3\eta-1}$, so the displayed condition on θ_{lag} gives $h_j^2\ell_j = o(H_j^3)$. □

Remark 12.12 (Direction of the lag statement). Corollary 12.11 says that the energy lies beyond every cutoff h_j satisfying $h_j = o(H_j^{3/2}\ell_j^{-1/2})$; it does *not* say that the energy is concentrated only at lags larger than the endpoint scale $H_j^{3/2}\ell_j^{-1/2}$ itself.

12.3. **Uniform two-point cancellation in sublinear intervals.** The linear-scale restriction can be removed for mesoscopic lags by a Fourier-uniform estimate valid in every interval in the range displayed below. Theorem 1.5 of Matomäki–Teräväinen [31] gives the twisted Möbius estimate with the original logarithmic saving $A < 1/3$. Matomäki–Shao–Tao–Teräväinen record explicitly, in their discussion following equation (1.16), that [31, Remark 5.2] allows this range to be enlarged to $0 < A < 1$; see [30, discussion following (1.16)]. Thus, for every fixed $0 < A < 1$ and $\iota > 0$,

$$\sup_{\xi\in\mathbb{R}}\Big|\sum_{x<n\le x+L}\mu(n)e(\xi n)\Big| \ll_{A,\iota} \frac{L}{(\log x)^A} \qquad (x^{3/5+\iota} \le L \le x). \tag{12.23}$$

Theorem 12.13 (Uniform two-point Cesàro cancellation)**.** *Fix $0 < A < 1$ and $\iota > 0$. Uniformly for all integers u, L, R with $u^{3/5+\iota} \le L \le u$ and $1 \le R \le L/2$,*

$$\frac{1}{R}\sum_{h=1}^{R}\frac{|C_h^{(u,u+L]}|}{L-h} \ll_{A,\iota} \Big(\frac{L}{R}\Big)^{1/2}(\log u)^{-A}. \tag{12.24}$$

In particular, if $R \ge L(\log u)^{-2A+2\kappa}$ with $0 < \kappa < A$, the left side is $O((\log u)^{-\kappa})$.

Proof. Put $a_i = \mu(u+i)$ for $1 \le i \le L$ and $A_I(\xi) = \sum_i a_i e(i\xi)$. For $h > 0$ define $C_{-h}^{(u,u+L]} := \sum_{i=1}^{L-h} a_{i+h}a_i = C_h^{(u,u+L]}$ and put $C_0 = q = \sum_i a_i^2$. These are the Fourier coefficients of $|A_I(\xi)|^2$. Parseval gives the precise identity

$$\int_0^1 |A_I(\xi)|^4\, d\xi = q^2 + 2\sum_{h=1}^{L-1}|C_h^{(u,u+L]}|^2 \le \|A_I\|_\infty^2 q \le L\|A_I\|_\infty^2. \tag{12.25}$$

Hence Cauchy–Schwarz gives $\sum_{h\le R}|C_h| \le \sqrt{RL}\,\|A_I\|_\infty$, and since $L - h \ge L/2$ the left side of (12.24) is at most $2\|A_I\|_\infty/\sqrt{RL}$. Apply (12.23). □

Corollary 12.14 (Near-log-squared mesoscopic range)**.** *Fix $\iota > 0$ and $0 < \delta < 1$. Uniformly for*

$$u^{3/5+\iota} \le L \le u, \qquad \frac{L}{(\log u)^{2-\delta}} \le R \le \frac{L}{2}, \tag{12.26}$$

one has

$$\frac{1}{R}\sum_{h\le R}\frac{|C_h^{(u,u+L]}|}{L-h} \ll_{\iota,\delta} (\log u)^{-\delta/4}. \tag{12.27}$$

Proof. In Theorem 12.13 take $A = 1 - \delta/4$ and $\kappa = \delta/4$. □

Remark 12.15 (Scope of the sublinear estimate). Theorem 12.13 is a short Parseval–Cauchy–Schwarz corollary of the published Fourier estimate (12.23); it does not use the excursion condition. Its role here is to provide an arithmetic input on every adaptively selected interval, something that "almost all intervals" results cannot supply. The factor $(L/R)^{1/2}$ is inherent in this direct L^2-to-L^1 Cauchy–Schwarz step using only the displayed second-moment control of the C_h; additional information on the distribution of the correlations could in principle improve it.

Corollary 12.16 (Sublinear large-lag estimate)**.** *Fix $\iota > 0$ and $0 < \delta < 1$, and let $I = (u, u + L]$ be a completed excursion with $u^{3/5+\iota} \le L \le u$. For all sufficiently large u, put*

$$R = \left\lceil \frac{L}{(\log u)^{2-\delta}} \right\rceil. \tag{12.28}$$

For sufficiently large u, $(\log u)^{2-\delta} \ge 3$, and hence $1 \le R \le L/2$; therefore the preceding two-point estimate applies. Then

$$\sum_{R<h<L} hC_h^{(I)} = -Q_I + O_{\iota,\delta}\left(\frac{L^3}{(\log u)^{4-7\delta/4}}\right). \tag{12.29}$$

Consequently, if $Q_I L^{-3}(\log u)^{4-7\delta/4} \to \infty$, then the left side is $-(1 + o(1))Q_I$.

Proof. Corollary 12.14 gives $\sum_{h\le R}|C_h^{(I)}| \ll RL(\log u)^{-\delta/4}$. Hence

$$\left|\sum_{h\le R} hC_h^{(I)}\right| \le R\sum_{h\le R}|C_h^{(I)}| \ll R^2 L(\log u)^{-\delta/4}, \tag{12.30}$$

which is the error in (12.29) after inserting the chosen value of R into $Q_I = -\sum_{h<L} hC_h^{(I)}$. □

Remark 12.17 (Connection with wide peaks). The quantitative form of the short-interval estimate in [31, Theorem 1.1] gives, after the blocking argument of Lemma 9.42, $\omega_u \ll_\varepsilon (\log u)^{-1/3+\varepsilon} + u^{-c}$ for some $c > 0$ (depending on the two fixed short-interval exponents used there). Thus, in the regime $H/h_u \to \infty$ of Proposition 9.43,

$$Q_I \gg_\varepsilon H^3(\log u)^{1/3-\varepsilon}. \tag{12.31}$$

Combining this with Corollary 12.16, a sufficient condition for large-lag concentration is

$$\left(\frac{H}{L}\right)^3 (\log u)^{13/3-7\delta/4-\varepsilon} \longrightarrow \infty. \tag{12.32}$$

For linearly localized Kaczorowski–Pintz excursions satisfying $v \le \lambda u$ with a fixed $\lambda < 2$, this machinery applies when $\Theta > 3/5$ (after choosing the loss parameters so that $\Theta - \eta > 3/5 + \iota$); the remaining obstruction is that the excursion may be much longer than its height.

12.4. **Fejér reduction and the weaker local-variance target.**

Remark 12.18 (Excursion geometry alone does not give cancellation). For the tent excursion $a_i = +1$ $(i \le k)$, $a_i = -1$ $(k < i \le 2k)$, which satisfies the bridge and one-sign constraints of a primitive excursion, $C_h = 2k - 3h$ for $1 \le h < k$. Hence for $R \to \infty$, $R = o(k)$,

$$\frac{1}{R}\sum_{h\le R}\frac{|C_h|}{2k-h} \longrightarrow 1, \qquad \sum_{h<R}(R-h)C_h \asymp kR^2, \tag{12.33}$$

whereas $RL \asymp kR$. Arithmetic input is indispensable.

Lemma 12.19 (Finite Fejér identity, classical)**.** *For a real sequence $(a_i)_{1\le i\le L}$, extended by 0 outside $[1, L]$, put*

$$q = \sum_{i=1}^{L} a_i^2, \qquad C_h = \sum_{i=1}^{L-h} a_i a_{i+h}, \qquad F_R := \sum_{h=1}^{R-1}(R-h)C_h. \tag{12.34}$$

Then for $1 \le R \le L$,

$$Rq + 2F_R = \sum_{m\in\mathbb{Z}}\left(\sum_{j=1}^{R} a_{m+j}\right)^2. \tag{12.35}$$

Proof. Expand the square. Each diagonal term occurs R times, and a pair at lag h, $1 \le h < R$, occurs $2(R-h)$ times. □

Proposition 12.20 (Bridge–Fejér identity and interior variance)**.** *Let $I = (u, u+L]$ be a completed excursion, put $a_i = \mu(u+i)$ and $S_t = \sum_{i=1}^{t} a_i = M(u+t)$ for $0 \le t \le L$. Then $S_0 = S_L = 0$, and for every $1 \le R < L$,*

$$Rq + 2F_R = 2Q_I - 2\sum_{t=1}^{L-1-R} S_t S_{t+R}. \tag{12.36}$$

Moreover

$$Rq + 2F_R = E_{\mathrm{bd}}(R) + \sum_{m=0}^{L-R}\big(M(u+m+R) - M(u+m)\big)^2, \tag{12.37}$$

where

$$E_{\mathrm{bd}}(R) = \sum_{t=1}^{R-1} S_t^2 + \sum_{t=L-R+1}^{L-1} S_t^2 = O(R^3). \tag{12.38}$$

Consequently, if $R = o(\sqrt{L})$, then

$$F_R = o(RL) \quad\Longleftrightarrow\quad \sum_{m=0}^{L-R}\big(M(u+m+R) - M(u+m)\big)^2 = Rq + o(RL). \tag{12.39}$$

If in addition $L \ge u^{1/2+\varepsilon}$ and $L \le u$, then

$$q = \left(\frac{6}{\pi^2} + o(1)\right)L, \tag{12.40}$$

and the right side of (12.39) *is equivalently*

$$\sum_{m=0}^{L-R}\big(M(u+m+R) - M(u+m)\big)^2 = \left(\frac{6}{\pi^2} + o(1)\right)RL. \tag{12.41}$$

Proof. Extend S_t by 0 outside $0 \le t \le L$. Since $\sum_{j=1}^{R} a_{m+j} = S_{m+R} - S_m$, (12.35) gives

$$Rq + 2F_R = \sum_m (S_{m+R} - S_m)^2 = 2\sum_m S_m^2 - 2\sum_m S_m S_{m+R}, \tag{12.42}$$

The natural range of the last sum is $0 \le m \le L-R$, but the terms $m=0$ and $m=L-R$ vanish because $S_0 = S_L = 0$, leaving $1 \le t \le L-1-R$; this is (12.36). Splitting the same sum into $m<0$, $0 \le m \le L-R$, and $m > L-R$ gives (12.37): for $-R < m < 0$ the contribution is S_{m+R}^2, for $L-R < m < L$ it is S_m^2, and the endpoint terms vanish because $S_0 = S_L = 0$. Since $|a_i| \le 1$ and the path returns to zero, $|S_t| \le t$ and $|S_{L-t}| \le t$, whence $E_{\mathrm{bd}}(R) \le 2\sum_{t<R} t^2 = O(R^3)$. Thus $E_{\mathrm{bd}}(R) = o(RL)$ when $R = o(\sqrt{L})$, proving the equivalence. Finally the classical estimate $\sum_{n\le x} \mu(n)^2 = (6/\pi^2)x + O(\sqrt{x})$ gives $q = (6/\pi^2)L + O(\sqrt{u+L})$. Under $u^{1/2+\varepsilon} \le L \le u$,

$$\frac{\sqrt{u+L}}{L} \le \frac{\sqrt{2u}}{u^{1/2+\varepsilon}} \ll u^{-\varepsilon}, \tag{12.43}$$

so the error is uniform and (12.40) follows. □

Remark 12.21 (Fejér density at frequency zero). With $A_I(\xi) = \sum_{i=1}^{L} a_i e(i\xi)$ and

$$K_R(\xi) = \frac{1}{R}\left|\sum_{j=1}^{R} e(j\xi)\right|^2, \tag{12.44}$$

one has

$$q + \frac{2F_R}{R} = \int_0^1 |A_I(\xi)|^2 K_R(\xi)\, d\xi. \tag{12.45}$$

Thus $F_R = o(RL)$ says that the Fejér average of the local spectral density at frequency 0, at resolution $1/R$, equals $q + o(L)$. It is not a statement that $|A_I(\xi)|^2$ is spectrally flat on the whole circle.

Remark 12.22 (Three distinct targets). The following conditions must not be conflated.

First, absolute averaged Chowla, $\sum_{h<R} |C_h| = o(RL)$, implies only $|F_R| \le R\sum_{h<R}|C_h| = o(R^2L)$, not the sharper $F_R = o(RL)$ when $R \to \infty$. Conversely, $F_R = o(RL)$ allows cancellation between the C_h and does not control their absolute values.

Second, F_R does not by itself control the small-lag contribution to the excursion energy. If

$$V_R := \sum_{h<R} hC_h, \tag{12.46}$$

then exactly

$$F_R + V_R = R\sum_{h<R} C_h. \tag{12.47}$$

Thus a Fejér estimate alone does not imply $V_R = o(Q_I)$.

Third, Proposition 12.20 requires both endpoints of the block to be zeros of M. For a block beginning at a zero but ending before the return, additional boundary terms remain.

Remark 12.23 (Why the variance target may go below 3/5). For the Fejér problem one does not need the stronger absolute estimate $\sum |C_h|$. By Proposition 12.20, on a completed excursion and for $R = o(\sqrt{L})$ it is enough to prove the diagonal local-variance law

$$\sum_{m=0}^{L-R} \big(M(u+m+R) - M(u+m)\big)^2 = Rq + o(RL). \tag{12.48}$$

This is strictly weaker than absolute averaged Chowla and is therefore a plausible target below the Fourier-uniform threshold $L \ge u^{3/5+\iota}$. The all-interval U^2/Fourier estimates currently used above do not, by themselves, supply this asymptotic for every adaptively selected block in the range $L \ge u^{1/2+\varepsilon}$. Results holding for almost all starting points cannot be substituted, because excursion endpoints are selected by M itself.

No fixed-prefix formula for the increments in (12.41) is asserted here. In particular, the identity $\sum_{d\le x} \mu(d)\lfloor x/d \rfloor = 1$ introduces lower-scale terms such as $M(x/2)$ when one truncates at $x/2$; omitting those terms would give an invalid reduction.

13. Extreme values in stochastic models of excursions

The counts of Section 9.1 bound the number of excursions in a family, not their heights. They acquire a second function when combined with a tail hypothesis on a normalized excursion statistic: they fix the scale of the family maximum, with no independence assumption.

Let $(\mathfrak{X}, \mathcal{G}, \Pr)$ be a probability space carrying a random sequence of disjoint integer intervals $I_k = (x_{0,k}, x_{1,k}]$ with $x_{1,k} \le x_{0,k+1}$, and real random variables Z_k (for instance a model of the Möbius walk with $Z_k = H_k/\sqrt{\ell_k}$). A family $\mathcal{F}$ is given by a rule selecting indices from $(x_{0,k}, x_{1,k})_k$, such as $\mathcal{F}_{\mathfrak{h}}$; let $k_{\mathcal{F}}(1) < k_{\mathcal{F}}(2) < \cdots$ enumerate its members, and put

$$Z_{\max,\mathcal{F}}(X) := \max\{Z_k : k \in \mathcal{F},\ x_{0,k} \le X\} \qquad (\max \emptyset = -\infty). \tag{13.1}$$

The lemmas of Section 9.1 hold for every realization.

Hypothesis 13.1 (SG_{ϖ})**.** There are $C, \kappa_0, \varpi > 0$ such that for every $j \ge 1$ and $z \ge 0$

$$\Pr(k_{\mathcal{F}}(j) \text{ exists and } Z_{k_{\mathcal{F}}(j)} > z) \le C e^{-\kappa_0 z^{\varpi}}. \tag{13.2}$$

For $\varpi = 2$ this is a uniform sub-Gaussian tail, denoted (SG).

Theorem 13.2 (Extreme scale from a count)**.** *Assume* (SG_{ϖ}) *for* $\mathcal{F}$ *and suppose that* $E_{\mathcal{F}}(X) \le \mathcal{N}(X)$ *for every realization, with* $\mathcal{N}(X) \ge 1$ *deterministic. Then*

$$\Pr(Z_{\max,\mathcal{F}}(X) > z) \le \mathcal{N}(X)\, C e^{-\kappa_0 z^{\varpi}}. \tag{13.3}$$

Consequently, for every $\delta > 0$*,*

$$\Pr\Big(Z_{\max,\mathcal{F}}(X) > \Big(\frac{1+\delta}{\kappa_0} \log \mathcal{N}(X)\Big)^{1/\varpi}\Big) \le C\, \mathcal{N}(X)^{-\delta}, \tag{13.4}$$

and $Z_{\max,\mathcal{F}}(X) = O_{\Pr}\big((\log \mathcal{N}(X))^{1/\varpi}\big)$ *as* $\mathcal{N}(X) \to \infty$*.*

Proof. The members of $\mathcal{F}$ starting at most X are the first $E_{\mathcal{F}}(X) \le \mathcal{N}(X)$ members of the enumeration, so $\{Z_{\max,\mathcal{F}}(X) > z\} \subset \bigcup_{1 \le j \le \lfloor \mathcal{N}(X) \rfloor} \{k_{\mathcal{F}}(j) \text{ exists},\ Z_{k_{\mathcal{F}}(j)} > z\}$, and (13.3) is the union bound. Substituting $\kappa_0 z^{\varpi} = (1+\delta)\log \mathcal{N}(X)$ gives (13.4). □

Corollary 13.3 (Triple-logarithmic scale on the square family)**.** *Assume (SG) for the square-exceptional family* $\mathcal{F}_{x^2}$ *of the model, and write* $Z_{\max,\square} := Z_{\max,\mathcal{F}_{x^2}}$*. Then, for* $X \ge e^e$ *and every* $\delta > 0$*,*

$$\Pr(Z_{\max,\square}(X) > z) \le 4C(\log\log X) e^{-\kappa_0 z^2}. \tag{13.5}$$

For all sufficiently large X*,*

$$\Pr\Big(Z_{\max,\square}(X) > \sqrt{\tfrac{1+\delta}{\kappa_0} \log\log\log X}\Big) \le 4C(\log\log X)^{-\delta}. \tag{13.6}$$

Thus, for all sufficiently large X*,* $Z_{\max,\square}(X) = O_{\Pr}(\sqrt{\log\log\log X})$*. The same holds for the family* $\mathcal{F}_{cx^2}$ *of Lemma 9.2 with a constant depending on* c*.*

Proof. Use (9.4) in (13.3) and note that $e^{-\kappa_0 z^2} = (\log\log X)^{-(1+\delta)}$ at the stated z. □

The union bound gives convergence in probability; a Borel–Cantelli argument along a sequence adapted to $\mathcal{N}$ gives an almost sure bound with the same constant.

Theorem 13.4 (Almost sure extreme scale)**.** *Assume* (SG_{ϖ}) *for* $\mathcal{F}$ *and* $E_{\mathcal{F}}(X) \le \mathcal{N}(X)$ *for every realization, where* $\mathcal{N}$ *is continuous, nondecreasing and unbounded. Then almost surely*

$$\limsup_{X \to \infty} \frac{Z_{\max,\mathcal{F}}(X)}{(\log \mathcal{N}(X))^{1/\varpi}} \le \kappa_0^{-1/\varpi}. \tag{13.7}$$

Proof. Fix $\delta > 0$. For $j \geq j_0$ let $X_j := \inf\{X : \mathcal{N}(X) \geq 2^j\}$. By continuity, monotonicity and unboundedness, after increasing j_0 if necessary one has $\mathcal{N}(X_j) = 2^j$. Put $z_j = \big((1+\delta)(j+1)\log 2/\kappa_0\big)^{1/\varpi}$ and $A_j = \{Z_{\max,\mathcal{F}}(X_{j+1}) > z_j\}$. By (13.3), $\Pr(A_j) \leq 2^{j+1} C e^{-(1+\delta)(j+1)\log 2} = C 2^{-(j+1)\delta}$, which is summable, so almost surely A_j occurs only finitely often. For $X_j \leq X \leq X_{j+1}$, monotonicity of $Z_{\max,\mathcal{F}}$ in X gives $Z_{\max,\mathcal{F}}(X) \leq z_j$ outside A_j, while $\log \mathcal{N}(X) \geq j \log 2$. Hence the lim sup is at most $\lim_j \big((1+\delta)(j+1)/(\kappa_0 j)\big)^{1/\varpi} = \big((1+\delta)/\kappa_0\big)^{1/\varpi}$. Let $\delta \downarrow 0$ along a countable sequence. □

With $\mathcal{N}_\square(X) = 2 + \log_2(\log X/\log 2)$, a continuous increasing majorant of the integer counting function, Theorem 13.4 gives under (SG)

$$\limsup_{X\to\infty} \frac{Z_{\max,\square}(X)}{\sqrt{\log\log\log X}} \leq \kappa_0^{-1/2} \quad \text{almost surely.} \tag{13.8}$$

Applied to the three counts available for any sequence of disjoint intervals (all excursions: at most $X+1$; Lemma 9.4; Lemma 9.1), Theorems 13.2 and 13.4 give three levels:

family	members starting $\leq X$	extreme scale under (SG)
all excursions	$\leq X+1$	$\sqrt{\log X}$
$x_1 > \lambda x_0$	$O(\log X)$	$\sqrt{\log\log X}$
$x_1 > c x_0^2$ (or x_0^A)	$O(\log\log X)$	$\sqrt{\log\log\log X}$

(13.9)

Each level of lacunarity removes one logarithm from the extreme scale. Under (SG_ϖ) the last column reads $(\log X)^{1/\varpi}$, $(\log\log X)^{1/\varpi}$, $(\log\log\log X)^{1/\varpi}$.

Remark 13.5 (Scope). For M itself the square-exceptional family is finite by Corollary 9.8, so Theorems 13.2–13.4 concern stochastic models of the Möbius walk, in which the pathwise counts of Section 9.1 persist while no analogue of the Kaczorowski–Pintz localization is available. Hypothesis (SG) is an assumption on the model; the theorems state exactly what the counts contribute to its extremes.

14. A one-frequency model for the net area curve

Heuristic 14.1 (Scale and period)**.** Motivated by (6.1), model $M(x) \approx \sqrt{x}\, F(\log x)$ with F almost periodic, and retain only the lowest zero γ_1 (RH does not imply that its coefficient dominates). Within an excursion of length $\asymp x$ this gives $H_\nu \asymp \sqrt{x}$ and $\sqrt{H_\nu}\ell_\nu \asymp x^{5/4}$, so $\mathcal{A}$ oscillates at scale

$$|\mathcal{A}(x)| \asymp x^{5/4}, \tag{14.1}$$

the scale of Corollary 7.4, with period $2\pi/\gamma_1$ in $\log x$, i.e. a factor $e^{2\pi/\gamma_1} \approx 1.5597$ in x. In this model consecutive zeros of M are separated by the factor $e^{\pi/\gamma_1} = \lambda_*$, the sharp threshold of the interval-counting comparison used here in Corollary 9.14(iii): a model excursion is exactly λ_*-long.

15. Numerical data

The following census covers all $n \leq N$, where $N = 455\,052\,511$; it contains 87 385 excursions and records $(u_\nu, \tau_\nu, v_\nu, H_\nu, \sigma_\nu)$ and the curves $\mathcal{A}_+, \mathcal{A}_-, \mathcal{A}$:

$$\mathcal{A}_+(N) = 6.3449 \times 10^9, \quad \mathcal{A}_-(N) = -7.0877 \times 10^9, \quad \mathcal{A}(N) = -7.4279 \times 10^8, \tag{15.1}$$

$$\max_\nu \frac{H_\nu}{\sqrt{\ell_\nu}} = 2.0743, \qquad \max_\nu \frac{\mathfrak{b}_\nu}{\ell_\nu^{3/2}} = 1.0214. \tag{15.2}$$

Since $|M(x)| \leq 0.571\sqrt{x}$ for $33 \leq x \leq 10^{16}$ [18], and $|M(n)| \leq \sqrt{n}$ for $n < 33$ with equality only at $n = 1$, the constant $C_0 = 1$ is admissible in Lemma 7.1, which gives $\mathcal{A}_+(N) + |\mathcal{A}_-(N)| \leq \frac{4}{5} N^{5/4} = 5.317 \times 10^{10}$ against the observed 1.3433×10^{10}, a ratio of 3.96. Heuristic 8.5 with $L = \sqrt{N}$ predicts $\#\{\ell_\nu > \sqrt{N}\} \asymp N^{1/4} \approx 146$. At this range $N^{0.05} \approx 2.7$, so factors x^ε are not separable from

constants. These numerical values are reported from the stated census and are not independently reproducible from the manuscript alone without the underlying census file and regeneration code. Reproducing them requires the excursion file, the area-integration convention of Proposition 4.2, and the terminal convention of Definition 3.6.

16. Conclusions

16.1. **What the paper does.** The paper replaces the single function M by the sequence of its excursions, and asks which statements about M can be read, and which can be localized, excursion by excursion. Three facts make this possible. First, an excursion is a bridge: it starts and ends at 0, so its quadratic mass is a lag-weighted sum of Möbius correlations internal to it (Lemma 5.1), and its height and length bound its linear and quadratic masses sharply (Proposition 5.6). Second, excursions are disjoint intervals, and disjoint intervals that grow by a factor λ are few: at most $\log X/\log\lambda + O(1)$ of them start below X (Lemma 9.4). Third, the oscillation theorems of Kaczorowski–Pintz and Pintz give, respectively, the number and the exponent of the large values of M. Every main result is obtained by combining two of these three facts.

16.2. **Horizon localization, area and arithmetic selection.** The horizon-localization theorem (Theorem 9.12) shows that RH is decided by the values of M on the part of each excursion below any horizon crossed by fewer than $(\gamma_1/\pi)\log X$ excursions. Its smallest linear instance obtained by the counting comparison is the horizon λx_0 with $\lambda > e^{\pi/\gamma_1}$ (Corollaries 9.14 and 9.15), says that the large values of M that decide RH can be sought below 1.25 times the starting point of their excursion; the threshold is exactly where the upper count of long intervals and the lower count of witnesses meet, and Proposition 9.33 shows that the counting mechanism cannot go below it. The first-arrival audit sharpens what an arithmetic improvement would have to achieve: after exact accounting of the zero runs, the sufficient record-return coefficient is $\lambda_* - 1$ (Corollary 9.38), with the positive excess E''' measuring the additional retrogression cost.

The exact area exponent (Corollary 8.13) identifies the growth of the skeleton area as $x^{1+\Theta/2+o(1)}$. The upper bound is a concavity argument; the lower bound needs the inequality $B_1 \le 2(m^*)^{1/2}\mathcal{A}_{\mathrm{abs}}(v^*)$, the exponent of B_1 from Pintz's theorem and the return localization of Corollary 9.8. Together these give that the area bound $x^{5/4+\varepsilon}$ is equivalent to RH.

The analysis of the localized window (Section 11) shows that the RH-equivalent part of an excursion is an explicit linear function of μ on the prefix up to χu, $\chi > \frac{1}{2}e^{\pi/\gamma_1}$; it decomposes this function into scale channels, proves a channel-descent inequality with an explicit geometric loss, and splits M into a smooth part and a large-prime part which, in the range of the split considered, have size $(1+o(1))t/\log^2 t$ and opposite signs (Proposition 11.51).

The weighted correlation identity at zeros, combined with averaged Chowla and with Pintz's exponent (Theorem 12.6), locates the signed weighted correlation total representing $\sum_{n<v} M(n)^2$ at lags larger than v^α for every $\alpha < \Theta$. On the reflection fibre (Section 10), divisor equations reduce the 2^J orientations compatible with the profile M^2 to at most $2N$, while Theorem 10.4 shows that a fixed-shift correlation cannot be controlled uniformly by $o(N)$ divisor rows.

16.3. **Structural identities.** The skeleton and its area, energy and inflection formulas (Sections 3–4), the bridge identity $Q_\nu = -\sum_h hC_h^{(\nu)}$ with its sharp bounds, the four counting lemmas for disjoint intervals (Lemmas 9.1–9.4), the dipole identity (Proposition 6.1) and the positivity of the excursion blocks $D_I(\varsigma)$ each follow in a few lines from the definitions. Their role is structural: they turn statements about M into statements about a sequence of disjoint bridges, to which counting and the oscillation theorems can be applied.

16.4. **Equivalent criteria and transported estimates.** The L^1, L^2, energy and correlation criteria (Proposition 8.1, Theorems 8.6 and 8.9) are equivalent to Littlewood's criterion; here they are derived from the excursion decomposition, and the energy criterion is squeezed between the

L^1 mass and the maximum. The exponents of $\sum |M|$ and $\sum M^2$ are immediate consequences of Pintz's theorem. The linear window identity follows from $\mu * 1 = \delta$, and the bilinear identity is structurally analogous to the $k = 2$ Heath–Brown/Vaughan truncation pattern. The results on correlations in excursion blocks transport published short-interval and averaged-Chowla theorems to intervals selected by M; the additional ingredient specific to the present formulation is that the selected intervals are completed excursions.

16.5. **Limits of the method and open directions.** Several negative results proved here delimit the new excursion mechanisms. Pointwise channel descent loses a fixed factor per step (Remark 11.11); termwise contraction of the exact kernel gives nothing beyond the unit-step cone (Proposition 11.23); the relaxed class retaining only future-window divisor constraints admits ballistic paths (Proposition 11.19); and, most importantly for the linear horizon, Proposition 9.33 shows that the geometric inputs alone cannot force the positional surplus needed below λ_*. A successful improvement must therefore use arithmetic information that is sensitive to the location of the witness inside its excursion.

The first new route is the logarithmic-surplus programme created by Proposition 9.32. The constant $\lambda_* = e^{\pi/\gamma_1}$ is the balance point between the Kaczorowski–Pintz witness density $d_1 = \gamma_1/\pi$ and the logarithmic budget of disjoint excursions. For a target $1 < \lambda < \lambda_*$, the exact missing amount is $\log(\lambda_*/\lambda)$. Thus one does not need an unspecified improvement: one needs an arithmetic mechanism that forces that fixed mean surplus on the bad witness family.

The second new route is the first-arrival record cost developed in the deductive audit. Proposition 9.36 and Lemma 9.37 remove the squarefree-density loss and give the sufficient positional estimate

$$\frac{t_H}{u} \le 1 + \frac{\mathcal{V}(u,H)}{u} + o(1). \tag{16.1}$$

Consequently the relevant coefficient is exactly $\lambda_* - 1$, rather than $(6/\pi^2)(\lambda_* - 1)$: a bound $\mathcal{V} \le (\lambda_* - 1 - \varepsilon_*)u$ on an unbounded Kaczorowski–Pintz first-arrival sequence would place those witnesses inside a strictly shorter linear horizon. The exact identity (9.91) also identifies the obstruction: in the linear regime $\mathcal{V}/u = h/u + E'''/u + o(1)$. Hence the missing estimate must control the retrogression excess E''' independently, or control $h + E'''$ with its signs intact; further algebraic expansion of the same total cost does not by itself weaken the positional problem.

A third, closely related route removes the first-arrival drift before this positive excess is introduced. Proposition 9.40 centers each record block and converts its position into a signed weighted correlation energy. It gives $h \le 3\mathcal{Q}^\circ + \frac{3}{2}H$ and therefore a second explicit target,

$$\mathcal{Q}^\circ \le \left(\frac{\lambda_* - 1}{3} - \varepsilon_*\right) u. \tag{16.2}$$

This route is again specific to the record structure of the excursion. Its present obstruction is also precise: the adaptive record blocks can have mean length $u^{1-\alpha}$ when a Kaczorowski–Pintz first arrival has height u^α, so the existing all-interval Fourier input used elsewhere in the paper does not reach the block scale. These two first-arrival routes—direct control of E''' and centered record correlations—are therefore the principal nonclassical directions opened by the present results for pushing the horizon below λ_*.

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